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# Image discrimination robustness of classical image quality metrics: an analysis on MAE, MSE, ERGAS, LMSE, SSIM, SAM, UIQI, PSNR, NIQE, PIQE, SNR, VIF, FSIM, GMSD, and NCC

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**Abstract**
Classical Image Quality Metrics (IQMs) are widely used in fields like remote sensing and computer vision for quantitative image discrimination. They face significant robustness challenges that compromise reliability in critical applications. These issues are evident when perceptually distinct images, especially after enhancements such as Spline-based non-linear Contrast Sketching, yield identical IQM scores, hindering effective differentiation and raising concerns about the trustworthiness of automated systems. To address these limitations, this study first analyzes the image discrimination robustness of 15 conventional IQMs on a dataset of 12 *Test Image*s, establishing baselines and failure points. Furthermore, this study presents and evaluates the *Localized IQM Calculation Method* (*Localized-IQM*), a potential enhancement to existing approaches. This method enhances IQM robustness by incorporating local image characteristics and statistical variations, moving beyond less discriminative global evaluations. Experimental findings demonstrate that *Localized-IQM* yields substantially more robust and reliable outcomes for challenging image discrimination problems compared to the direct global application of conventional IQMs. Its enhanced performance is consistent across diverse image types and common alteration levels. Consequently, the proposed method offers a significantly improved, dependable framework for image quality assessment, promising to advance image analysis accuracy and reliability in fields requiring precise image differentiation.

**Keywords** Image quality metrics · MAE, MSE, ERGAS, LMSE, SSIM, SAM, UIQI, PSNR, NIQE, PIQE, SNR, VIF, FSIM, GMSD, NCC · Contrast sketching · Colony-based search algorithm · Localized-IQM

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## 1 Introduction

Image Quality Metrics (IQMs) (Alaei et al. 2024; Almen et al. 1996; Guo et al. 2023) are indispensable instruments within numerous technologically advanced sectors, including remote sensing, computer vision, photogrammetry, and medical imaging, which underpin modern industrial information production. The deployment of IQMs is paramount as they provide an objective framework for evaluating intrinsic properties of imaging systems and processed imagery, such as resolution, noise, and artifacts. Rigorous application of IQMs is essential for sensor calibration, validation of imaging performance, and enhancement of processing algorithms, ensuring the fidelity and reliability of visual data crucial for computational modeling and decision-making. The need for objective IQMs emerged due to the limitations of subjective assessment, which is time-consuming, costly, inconsistent, and influenced by individual neuropsychological variability. IQMs offer a standardized, quantifiable, and replicable methodology for measuring image quality, facilitating reliable comparisons (Li et al. 2022; Shu et al. 2025). In critical industries like medical imaging or remote sensing, IQMs enable quality control via numerical thresholds, ensuring data meet required standards (Yang et al. 2019).

The advantages of IQMs are considerable, impacting various stages of image analysis. For researchers developing algorithms or compression techniques, IQMs serve as invaluable tools for objective evaluation, allowing systematic comparison against existing methods and performance tuning according to specific requirements. In scenarios where manual evaluation is impractical, IQMs enable automated, efficient quality assessment. Furthermore, IQMs are vital in research and standardization efforts, establishing benchmarks and comparative frameworks that promote consistency and facilitate technological advancements. IQMs assess the perceived quality or fidelity of a test image, often relative to a reference. Depending on reference availability, IQMs are categorized as *full-reference* (FR), *reduced-reference* (RR), or *no-reference* (NR) . The landscape of IQMs is diverse, encompassing methods like histogram-based metrics, pixel correspondence measures, and advanced approaches using Convolutional Neural Networks (CNNs) or Generative Adversarial Networks (GANs), alongside spatial, frequency-domain, and color-specific techniques. This diversity reflects the multifaceted challenge of quantifying image quality (Li et al. 2022; Kim et al. 2020; Miura et al. 2025).

Despite their utility, existing IQMs possess significant limitations (Zhang et al. 2025). A primary deficiency is their frequent failure to comprehensively encapsulate human visual perception. Human perception is influenced by context, individual preferences (e.g., for vibrant or muted colors), experience, and semantic understanding—aspects that IQMs, as mathematical constructs, may not adequately address. Consequently, assessments can diverge from genuine human appraisal, and artifacts highly noticeable to humans might be poorly reflected in IQM scores; a high score does not always equate to a visually pleasing image. Many IQMs are tailored to specific degradations (e.g., blur, noise, compression artifacts), potentially overlooking other critical issues like geometric distortions, color inaccuracies, or the preservation of fine details. Their scope can disregard aesthetic appeal or artistic value, crucial in certain contexts. IQM performance can also be content-dependent, yielding inconsistent results across diverse image types, such as natural scenes versus medical imagery or images with complex patterns. Moreover, IQMs typically evaluate images in isolation, ignoring the application context where perceived quality is paramount. For instance, an image acceptable for a web thumbnail might be unsuitable for high-resolution medical diagnosis. User intent or image purpose is generally not considered, meaning an IQM might rate an image as lower quality even if it is fit for its intended use (Sun et al. 2025; Yoo et al. 2025). This contextual blindness means IQM scores often misalign with subjective human judgments. Some metrics might exhibit biases towards particular image characteristics or be biased by the specific datasets used in their development, hindering generalization. The computational cost of some advanced IQMs can limit their real-time applicability. Crucially, different IQMs often produce divergent quality rankings for the same images, and no individual metric can be considered universally optimal; rather, the appropriate choice and interpretation depend on the specific application and distortion types.

A significant issue with many IQMs is their poor image discrimination power: the inability to assign unique scores to perceptually distinct images, particularly against a common reference. As a result, visibly different processed image versions can yield identical IQM values, demonstrating an insufficient capacity to distinguish quality variations. Compounding this problem, comprehensive studies thoroughly examining this critical limitation are notably scarce. Moreover, an IQM's over-sensitivity to minor, perceptually irrelevant changes, or its insensitivity to significant visual alterations, undermines accurate quality assessment. These combined factors can lead to misinterpreting IQM values, yielding misleading conclusions that risk derailing research or fostering suboptimal processing decisions.

Current IQMs exhibit several critical limitations: a divergence from human perception, specificity to certain artifacts, content and context dependency, inherent biases, and notably, poor image discrimination. These shortcomings collectively underscore the urgent need for more robust and perceptually aligned IQM methodologies. Precisely understanding these drawbacks is vital to prevent scientific studies from yielding paradoxical results or depending on flawed quality assessments. The observation that different images can produce identical IQM scores highlights a fundamental weakness, demanding methods that more reliably characterize image fidelity. Therefore, future research must prioritize innovating IQMs that correlate more strongly with human perception across diverse content and distortions, while also possessing markedly enhanced discriminatory power and stability. Developing advanced metrics, perhaps by incorporating uncertainty quantification or sophisticated models of the human visual system, is essential. Such advancements are imperative for IQMs to effectively support the rigorous demands of evolving image-dependent technologies and to ensure more reliable and meaningful quality assessments across all pertinent scientific and industrial domains.

This paper proposes a novel method, the *Localized IQM Calculation Method* (*Localized-IQM*), for the more robust calculation of IQM values. *Localized-IQM* is relatively more robust compared to conventional IQM calculation techniques, and it takes into account local statistical fluctuations within the relevant images.

This paper investigates the drawbacks encountered when employing 15 classical image quality metrics—namely, *Mean Absolute Error* (MAE) (Hodson 2022), *Mean Squared Error* (MSE) (Koller et al. 2022; Wang et al. 2021), *Erreur Relative Globale Adimensionnelle de Synthèse* (ERGAS) (Gao et al. 2014; Renza et al. 2013), *Laplacian Mean Squared Error* (LMSE) (Na 2004, 2004), *Structural Similarity Index Measure*(SSIM) (Boubechal et al. 2019; Lin et al. 2022), (Salmeron Gomez et al. 2020) , *Spectral Angle Mapper* (SAM) (Chen et al. 2012; Lee et al. 2022; Rao and Guha 2018), *Universal Image Quality Index* (UIQI) (Wang and Bovik 2002), *Peak Signal-to-Noise Ratio* (PSNR) (Martini 2025; Shang et al. 2019; Winkler and Mohandas 2008), *Natural Image Quality Evaluator* (NIQE) (Wu et al. 2021), *Perception based Image Quality Evaluator* (PIQE) (Pandev et al. 2020), *Signal-to-Noise Ratio* (SNR) (Chen et al. 2016; Moore et al. 2015; Tian et al. 2023), *Visual Information Fidelity* (VIF) (Kuo et al. 2016; Wu et al. 2013) *Feature Similarity Index* (FSIM) (Zhang et al. 2011), *Gradient Magnitude Similarity Deviation* (GMSD) (Li et al. 2020; Xue et al. 2014), and *Normalized Cross-Correlation* (NCC) (Chen et al. 2021; Duong et al. 2019; Lee et al. 2024), which are among the most frequently used and widely accepted by the image processing community—specifically in the context of *image discrimination problems*.

The primary contributions of this paper to the literature are as follows:

- An empirical demonstration that many widely used IQMs exhibit limited efficacy in complex image discrimination tasks, despite their prevalent adoption.
- Elucidation of the challenges and often questionable performance of most IQMs in one-to-many quality comparison scenarios, contrasting with their more established utility in direct (one-to-one) assessments.
- The introduction and validation of a novel computational method that yields IQM values with demonstrably and significantly enhanced robustness compared to conventional techniques.

The rest of this paper is organized as follows: In Sect. 2, Image Quality Assessment Methods are introduced. In Sect. 3, Localized IQM Calculation Method is presented. In Sect. 4, Experiments are discussed, and in Sect. 5, Results are provided. Finally, in Sect. 6, Conclusion is given.

## 2 Image quality assessment methods

Quantifying visual information quality is a complex yet crucial endeavor for advancing image processing and user experience (Alaei et al. 2024; Guo et al. 2023; Shu et al. 2025). IQMs have been developed, from pixel-error summations to advanced models emulating human vision, categorized by reliance on a reference image (*full*, *reduced*, or *no-reference*). This section presents the mathematical details of fifteen prominent IQMs that have significantly shaped the field, continue to be instrumental in various analytical contexts, and are employed in the experiments conducted in this paper.

The selected IQMs illustrate the field's evolution, from simple error calculations to advanced *no-reference* paradigms. This paper explores fifteen such methods: MAE, MSE, ERGAS, LMSE, SSIM, SAM, UIQI, PSNR, NIQE, PIQE, SNR, VIF, FSIM, GMSD, and NCC, which were selected for this study's experiments due to their widespread use, rapid execution, and no pre-setting requirements. Understanding these unique and vital tools, whose principles and utility are discussed herein, is key for researchers.

MAE averages the absolute pixel differences between a reference ($I$) and distorted ($K$) image (M×N); lower MAE means better quality (Eq. 1).

$$MAE = \frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} |I(i,j) - K(i,j)| \tag{1}$$

It is simple, less outlier-sensitive than MSE, but often correlates poorly with perceived quality, treating all pixel errors equally.

MSE averages squared pixel differences between reference (*I*) and distorted (*K*) images (M×N); lower MSE suggests better quality (Eq. 2).

$$MSE = \frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} (I(i,j) - K(i,j))^2 \tag{2}$$

Simple and mathematically convenient, MSE heavily penalizes large errors but correlates poorly with perception due to outlier sensitivity and neglect of error context.

ERGAS (Erreur Relative Globale Adimensionnelle de Synthèse) is an index used to assess the spectral quality of pansharpened images by quantifying the overall spectral distortion introduced during the image fusion process. Lower ERGAS values indicate a higher quality for the fused image, with an ideal value being 0. The mathematical formula for ERGAS is expressed by using Eq. (3):

$$ERGAS = 100\frac{h}{l}\sqrt{\frac{1}{N}\sum_{k=1}^{N}\left(\frac{RMSE(B_k)}{\mu_k}\right)^2} \tag{3}$$

In this formula, the term *h* represents the spatial resolution of the high-resolution panchromatic (PAN) image, while *l* denotes the spatial resolution of the lower-resolution multispectral (MS) image. The variable *N* signifies the number of spectral bands in the MS image that are being evaluated. $RMSE(B_k)$ is the Root Mean Square Error calculated for the *k*-th spectral band; this error quantifies the difference between the pixel values of the fused image ($I_{fused_k}$) and the reference image ($I_{ref_k}$) over *P* total pixels for that band, according to the formula : $RMSE(B_k) = \sqrt{\frac{1}{P}\sum_{i=1}^{P}(I_{fused_k}(i) - I_{ref_k}(i))^2}$. Furthermore, $\mu_k$ represents the mean pixel value of the *k*-th band within the original reference MS image, and it is calculated as : $\mu_k = \frac{1}{P}\sum_{i=1}^{P} I_{ref_k}(i)$. The constant 100 in the ERGAS equation serves as a scaling factor, and the ERGAS index itself is a dimensionless quantity. Finally, the ratio $\frac{h}{l}$ represents the proportional difference in spatial resolution between the panchromatic and multispectral images.

LMSE refines MSE by applying a Laplacian filter, which highlights rapid intensity changes, to both reference (*I*) and distorted (*K*) images before computing MSE (Eqs. 4–5). This approach emphasizes errors in high-detail regions, aligning with human visual system (HVS) sensitivity. Common 2D Laplacian kernels ($L_1$ for 4-connectivity, $L_2$ for 8-connectivity) include:

$$L_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \text{or} \quad L_2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

If $I'(i,j) = (L * I)(i,j)$ and $K'(i,j) = (L * K)(i,j)$ are the Laplacian-filtered images, LMSE is defined as:

$$LMSE = \frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} (I'(i,j) - K'(i,j))^2 \tag{4}$$

This can also be expressed using convolution:

$$LMSE = \frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} ((L * I)(i,j) - (L * K)(i,j))^2 \tag{5}$$

LMSE offers better perceptual relevance than standard MSE for distortions affecting edges and fine details, and it is sensitive to blurring. However, it remains an error-summation metric, its performance is dependent on the chosen Laplacian kernel, it can amplify noise, and it treats all differences in the Laplacian domain equally.

SSIM is a perceptual Image Quality Assessment (IQA) metric that quantifies image degradation based on the human visual system's adaptation for extracting structural information. It assesses similarity between two image windows, *x* (reference) and *y* (distorted), by considering luminance, contrast, and structural components (Eq. 6).

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{6}$$

Here, $\mu_x, \mu_y$ are local means; $\sigma_x^2, \sigma_y^2$ are local variances; $\sigma_{xy}$ is local covariance; and $C_1 = (k_1 L)^2, C_2 = (k_2 L)^2$ are small constants to prevent instability (where $L$ is the dynamic range of pixel values, and typically $k_1 = 0.01, k_2 = 0.03$). The overall image SSIM is usually the mean of SSIM values from all local windows. A key advantage of SSIM is its generally superior correlation with human perception of image quality compared to metrics like MSE, as it explicitly accounts for structural information. Nevertheless, SSIM's computation is more complex, it can sometimes be misled by distortions where structure is preserved but perceptual quality is altered (or vice-versa), and it requires a *reference image*.

SAM is primarily used in hyperspectral image analysis to determine spectral similarity between two spectra, such as a pixel's spectrum and a reference, or corresponding spectra from two different hyperspectral images. It treats each spectrum as a vector in an $N_b$-dimensional space (where $N_b$ is the number of spectral bands) and calculates the angle between these vectors (Eq. 7); a smaller angle indicates greater similarity.

$$SAM(\mathrm{v}_1, \mathrm{v}_2) = \arccos\left(\frac{\mathrm{v}_1 \cdot \mathrm{v}_2}{\|\mathrm{v}_1\| \|\mathrm{v}_2\|}\right) = \arccos\left(\frac{\sum_{k=1}^{N_b} v_{1k} v_{2k}}{\sqrt{\sum_{k=1}^{N_b} v_{1k}^2}\sqrt{\sum_{k=1}^{N_b} v_{2k}^2}}\right) \tag{7}$$

When applied to images, SAM is often reported as the average angle computed for all corresponding pixel spectra. SAM's main advantages include its insensitivity to illumination changes (multiplicative scaling), making it useful for material identification and assessing spectral fidelity in hyperspectral data. However, SAM primarily focuses on the shape of the spectra (vector direction) and does not directly consider magnitude differences if spectra are collinear but different in length. Its application is mainly for spectral data; it's not directly a spatial quality metric for grayscale or RGB images unless applied band-wise and aggregated, and it requires a reference.

UIQI, or Q-index, a precursor to SSIM, models image distortion through loss of correlation, luminance distortion, and contrast distortion (Eq. 8). It locally compares reference (*x*) and distorted (*y*) image windows, averaging scores for overall quality.

$$Q = \frac{\sigma_{xy}}{\sigma_x \sigma_y} \cdot \frac{2\bar{x}\bar{y}}{(\bar{x})^2 + (\bar{y})^2} \cdot \frac{2\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \tag{8}$$

Here, $\bar{x}, \bar{y}$ are local means, $\sigma_x, \sigma_y$ are local standard deviations, and $\sigma_{xy}$ local covariance; stability constants are often added. UIQI offers improved perceptual correlation over MSE/PSNR by considering these three factors. However, it can be less robust than SSIM, sensitive to mean shifts if not handled carefully locally, and is a *full-reference* metric.

PSNR quantifies the ratio of a signal's maximum possible power to the power of corrupting noise, typically derived from the MSE between reference and distorted images (Eq. 9). A higher PSNR value generally indicates better image quality.

$$PSNR = 10 \log_{10} \left( \frac{MAX_I^2}{MSE} \right) \tag{9}$$

$MAX_I$ represents the maximum possible pixel value of the image (e.g., 255 for an 8-bit image). PSNR is widely used due to its calculation simplicity. Its main disadvantage is its poor correlation with perceived visual quality, as it treats all error types equally and can be disproportionately affected by small MSE changes, especially at low MSE values.

NIQE is a *no-reference* (blind) IQA model. It assesses quality by measuring an image's statistical deviation from *quality aware* Natural Scene Statistics (NSS) typically found in pristine images; a lower NIQE score indicates better perceptual quality. NIQE models pristine image NSS features with a multivariate Gaussian (MVG) model $(\mathrm{v}_p, \mathbf{\Sigma}_p)$ and quantifies quality as the distance (Eq. 10) to the test image's NSS feature model $(\mathrm{v}_d, \mathbf{\Sigma}_d)$. Features often include Mean Subtracted Contrast Normalized (MSCN) coefficients and their products.

$$D(\mathrm{v}_p, \mathrm{v}_d, \mathbf{\Sigma}_p, \mathbf{\Sigma}_d) = \sqrt{\left( (\mathrm{v}_p - \mathrm{v}_d)^T \left( \frac{\mathbf{\Sigma}_p + \mathbf{\Sigma}_d}{2} \right)^{-1} (\mathrm{v}_p - \mathrm{v}_d) \right)} \tag{10}$$

NIQE's key advantage is its *no-reference* capability, making it highly applicable where pristine originals are unavailable, and it generally correlates well with human perception for certain distortions. However, its performance depends on the training corpus of natural images, it can be sensitive to image content deviating from typical natural scenes, and it is computationally more intensive than simple *full-reference* metrics.

PIQE is another *no-reference* IQA algorithm operating in the spatial domain. It estimates block-wise distortion levels and computes an overall quality score based on local image activity and noticeable artifacts, without requiring training on human-rated images or prior knowledge of distortions; lower PIQE scores indicate better quality. PIQE's stages typically include:

- Dividing the input image into non-overlapping blocks.
- Identifying high spatially active (assumed less distorted) and low spatially active blocks.
- Estimating distortion in potentially distorted blocks using features like MSCN statistics and indicators of noise or blockiness.
- Pooling these local distortion estimates, often weighted by local activity and artifact noticeability, for a final quality score.

PIQE's primary advantages are being a *no-reference* metric and not requiring prior training on human opinion scores, enhancing its general applicability. It is also relatively fast. However, its accuracy may not always match *full-reference* or some opinion-aware blind metrics, and performance can vary with image content and artifact types.

SNR compares the desired signal level to background noise. In image processing, it is the ratio of the original *reference image* power ($P_{signal}$) to the error or noise power ($P_{noise}$), often represented by Mean Squared Error (MSE), as defined in Eq. (11). A higher SNR generally indicates better image quality.

$$SNR = 10 \log_{10} \left( \frac{P_{signal}}{P_{noise}} \right) \tag{11}$$

Signal power $P_{signal}$ is often calculated with Eq. (12), and noise power $P_{noise}$ is typically the MSE (Eq. 13).

$$P_{signal} = \frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} I(i,j)^2 \tag{12}$$

$$P_{noise} = \frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} (I(i,j) - K(i,j))^2 = MSE \tag{13}$$

Thus, a common formulation for SNR is given in Eq. (14):

$$SNR = 10 \log_{10} \left( \frac{\sum_{i=0}^{M-1} \sum_{j=0}^{N-1} I(i,j)^2}{\sum_{i=0}^{M-1} \sum_{j=0}^{N-1} (I(i,j) - K(i,j))^2} \right) \tag{14}$$

SNR is easy to calculate, widely used, and objectively characterizes noise levels. However, it often correlates poorly with perceived visual quality as it treats all errors equally and may not accurately reflect localized distortions.

VIF is a *full-reference* image quality metric that quantifies information shared between a *reference image* (*C*) and a distorted image (*D*). It models the reference as a natural scene output and the Human Visual System (HVS) as a channel extracting this information; higher VIF values (often 0–1, but can exceed 1) suggest better fidelity. VIF (Eq. 15) is expressed as a ratio of two mutual information quantities summed over different wavelet subbands $j$, comparing information from the perceived test image ($F^j$) versus the perceived *reference image* ($E^j$), given natural scene statistics ($s_j^N$).

$$VIF = \frac{\sum_{j\in\text{subbands}} I(C^j; F^j|s_j^N)}{\sum_{j\in\text{subbands}} I(C^j; E^j|s_j^N)} \tag{15}$$

VIF generally shows a strong correlation with human subjective judgments, being based on sophisticated models of natural scene statistics and the HVS. Its disadvantages include being computationally intensive and requiring a *reference image*.

FSIM is a *full-reference* metric evaluating image quality based on the similarity of low-level features—primarily Phase Congruency (PC, measuring feature significance) and Gradient Magnitude (GM, measuring intensity change strength)—extracted from reference ($I_1$) and distorted ($I_2$) images. Local similarities for PC ($S_{PC}$, Eq. (16)) and GM ($S_{GM}$, Eq. 17), using stability constants $T_1, T_2$, are combined into $S_L(x) = [S_{PC}(x)]^\alpha \cdot [S_{GM}(x)]^\beta$ (typically $\alpha = \beta = 1$).

$$S_{PC}(x) = \frac{2PC_1(x)PC_2(x) + T_1}{PC_1(x)^2 + PC_2(x)^2 + T_1} \tag{16}$$

$$S_{GM}(x) = \frac{2GM_1(x)GM_2(x) + T_2}{GM_1(x)^2 + GM_2(x)^2 + T_2} \tag{17}$$

The FSIM index (Eq. 18) is then a weighted average of $S_L(x)$ over the image, using $PC_m(x) = \max(PC_1(x), PC_2(x))$ as the weighting function.

$$FSIM = \frac{\sum_{x\in\Omega} S_L(x) \cdot PC_m(x)}{\sum_{x\in\Omega} PC_m(x)} \tag{18}$$

FSIM correlates well with human visual perception by utilizing perceptually important features and capturing structural distortions. However, it is more computationally demanding than basic metrics, requires a *reference image*, and can be slightly affected by parameter choices, though defaults usually perform well.

GMSD is a *full-reference* IQM that assesses image quality by computing a pixel-wise Gradient Magnitude Similarity (GMS) map between a reference ($I_r$) and a distorted ($I_d$) image, using their gradient magnitudes ($m_r(i), m_d(i)$). The GMS map (Eq. 19), incorporating a stability constant $c$, measures local structural similarity.

$$GMS(i) = \frac{2m_r(i)m_d(i) + c}{m_r(i)^2 + m_d(i)^2 + c} \tag{19}$$

The GMSD (Eq. 20) is the standard deviation of this GMS map, where $N$ is the total number of pixels and $\overline{GMS}$ is the mean of the GMS map; a lower GMSD value indicates higher image quality.

$$GMSD = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(GMS(i) - \overline{GMS})^2} \tag{20}$$

GMSD is computationally efficient and performs well in predicting subjective image quality, especially for distortions affecting image gradients. Nevertheless, its reliance on gradient information might make it less sensitive to distortions not significantly altering gradients (e.g., subtle color shifts, uniform blurring), and it requires a *reference image*.

NCC measures similarity between two images ($I_1, I_2$) by normalizing their cross-correlation by the product of their standard deviations (Eq. 21). NCC values range from -1 (perfect negative correlation) to +1 (perfect positive correlation), with 0 indicating no correlation; $\bar{I}_1$ and $\bar{I}_2$ are the mean pixel values of the images.

$$NCC = \frac{\sum_{i=0}^{M-1}\sum_{j=0}^{N-1}(I_1(i,j) - \bar{I}_1)(I_2(i,j) - \bar{I}_2)}{\sqrt{\left(\sum_{i=0}^{M-1}\sum_{j=0}^{N-1}(I_1(i,j) - \bar{I}_1)^2\right)\left(\sum_{i=0}^{M-1}\sum_{j=0}^{N-1}(I_2(i,j) - \bar{I}_2)^2\right)}} \tag{21}$$

NCC's advantages include invariance to linear changes in brightness and contrast and robustness in template matching. However, as a global IQM, NCC may not strongly correlate with human perception of overall image quality as it doesn't model HVS characteristics. It is also sensitive to non-linear intensity variations, geometric distortions, and may not effectively capture localized distortions when used for comparing entire images for quality.

Image quality evaluation employs two main paradigms: qualitative (subjective) and quantitative (objective) (Almen et al. 1996; Shu et al. 2025). Qualitative assessment uses human judgment (e.g., Mean Opinion Score tests) for direct measures of perceived quality, vital for ground-truth data and critical evaluations where human insight is paramount, capturing nuances objective metrics may miss. Quantitative assessment uses IQMs for automated scoring. These are extensively applied in algorithm benchmarking, real-time monitoring, and large-scale testing due to their repeatability, efficiency, and cost-effectiveness. However, achieving high correlation between IQM scores and human perception remains a central research challenge. A variety of classical IQMs have been developed, ranging from simple pixel-difference calculations to more sophisticated models reflecting human vision or image statistics. The choice of a classical IQM typically depends on the expected distortion type, *reference image* availability, and computational constraints.

Table 1 summarizes common classical IQMs—their purpose, pros, and cons—to clarify their applications and limitations. These enduring metrics remain relevant as benchmarks due to their simplicity and clarity, despite newer methods.

Deep learning has introduced powerful IQA models achieving state-of-the-art correlation with human scores. However, these models' reliance on extensive training data presents practical challenges; adapting them to new data or distortions often requires time-consuming retraining, which can be infeasible. Consequently, despite their high accuracy, deployment may be impractical under certain constraints. In contrast, classical IQMs remain important due to inherent advantages such as simpler structures, straightforward implementation, and minimal computational needs, making them suitable for real-time applications or use on low-power, resource-constrained devices. Within the deep learning domain, approaches

**Table 1** Overview of common classical IQMs

| Metric | Primary purpose | Advantages | Disadvantages |
|---|---|---|---|
| MAE | To measure the average absolute difference between pixel values of reference and distorted images | Simple to calculate/interpret | Poor perceptual correlation |
| | | Less sensitive to outliers than MSE | Treats all pixel differences equally |
| | | Computationally inexpensive | Requires *reference image* |
| MSE | To measure the average squared difference between pixel values, indicating overall pixel-wise fidelity error | Simple, widely understood | Poor perceptual correlation |
| | | Mathematically convenient | Sensitive to outliers |
| | | Penalizes larger errors more | Ignores structural aspects |
| | | | Requires *reference image* |
| ERGAS | To assess the overall quality of fused images (e.g., pansharpened multispectral images) considering both spectral and spatial fidelity | Specifically designed for fused image quality | Requires *reference image* |
| | | Considers spectral and spatial aspects | More complex than basic metrics |
| | | Provides a global dimensionless error | Primarily for remote sensing |
| LMSE | To quantify error by emphasizing differences in edge and texture regions via Laplacian filtering prior to MSE calculation | Better perceptual relevance than MSE for edge distortions | Still an error-summation metric |
| | | Simple modification of MSE | Filter-dependent performance |
| | | Sensitive to blurring | Can amplify noise |
| SSIM | To measure structural similarity considering luminance, contrast, and structure, aiming for better perceptual relevance than pixel-wise errors | Better perceptual correlation than MSE/PSNR | More complex than MSE/PSNR |
| | | Accounts for structure | Can be fooled by some distortions |
| | | Widely adopted perceptual metric | Requires *reference image* |
| SAM | To determine spectral similarity between two spectra (e.g., in hyperspectral images) by calculating the angle between them | Insensitive to illumination changes | Primarily for spectral data |
| | | Useful for spectral fidelity/material ID | Not a direct spatial quality metric |
| | | Measures spectral shape similarity | Requires *reference image*/spectrum |
| UIQI | To model image distortion as a combination of loss of correlation, luminance distortion, and contrast distortion | Better perceptual correlation than MSE/PSNR | Less robust than SSIM |
| | | Considers correlation, luminance, contrast | Sensitive to mean shifts |
| | | Influential early perceptual metric | Requires *reference image* |
| PSNR | To measure the ratio between the maximum possible power of a signal and the power of corrupting noise, derived from MSE | Widely used benchmark | Poor perceptual correlation |
| | | Easy to calculate | Treats all errors equally |
| | | Logarithmic scale (dB) | Sensitive MSE changes |
| | | | Requires *reference image* |
| NIQE | To blindly assess image quality by comparing statistical features of a test image to those of natural, pristine images | *No-reference* (blind) | Performance depends on training corpus |
| | | Based on natural scene statistics | Sensitive to non-naturalistic content |
| | | Good perceptual correlation for some distortions | Computationally intensive |

**Table 1** (continued)

| Metric | Primary purpose | Advantages | Disadvantages |
|---|---|---|---|
| PIQE | To blindly assess image quality by estimating block-wise distortion in the spatial domain without prior training on human scores | *No-reference* (blind) | May not be as accurate as FR or some opinion-aware NR metrics |
| | | No training on human scores needed | Performance can vary with content/artifacts |
| | | Relatively fast for a blind IQA | |
| SNR | To measure the ratio of the power of the original signal to the power of the noise, indicating signal strength relative to noise | Simple, widely understood | Poor perceptual correlation |
| | | Objective noise level measure | Treats all errors (noise) equally |
| | | Fundamental concept | Different definitions exist |
| | | | Requires *reference image* |
| VIF | To quantify the information shared between a reference and a distorted image, based on HVS models and natural scene statistics | Strong perceptual correlation | Computationally complex |
| | | Based on information theory, HVS models | Requires *reference image* |
| | | Principled approach | Model assumptions may not always hold |
| FSIM | To measure image quality based on low-level features (phase congruency, gradient magnitude) that are important for human visual perception | Good perceptual correlation | Computationally intensive |
| | | Uses perceptually significant features | Requires *reference image* |
| | | Robust to structural distortions | Parameter dependent |
| GMSD | To evaluate image distortion based on the pixel-wise gradient magnitude similarity map, with lower deviation indicating better quality | Computationally efficient | Less sensitive to non-gradient distortions |
| | | Good performance for gradient-affecting distortions | Requires *reference image* |
| | | Clear interpretation | |
| NCC | To measure the similarity between two images based on normalized cross-correlation, often used in template matching | Invariant to linear brightness/contrast changes | Limited perceptual correlation for IQA |
| | | Robust for matching | Sensitive to non-linear changes, rotation, scaling |
| | | Bounded output | Requires *reference image* |

such as CNN-based models (Kang et al. 2014), NIMA (Talebi and Milanfar 2018), and DeepSim (Gao et al. 2017) have significantly advanced no-reference and perceptual image quality assessment by providing robust frameworks for evaluating visual fidelity. More recently, Transformer-based methods (You and Korhonen 2021) have further extended this progress by introducing attention mechanisms that enhance adaptability and accuracy in image quality assessment.

Most classical *full-reference* IQMs require no training or complex tuning, offering immediate, *plug-and-play* application unlike many deep learning models. They also serve as vital historical benchmarks and offer mathematical transparency, facilitating easier result interpretation compared to some *black-box* deep learning systems. While all IQA methods have distinct pros and cons, the straightforward applicability and low overhead of classical IQMs ensure their continued relevance, especially for quick assessments when resources for training complex models are limited.

## 3 Localized IQM calculation method

This paper introduces a novel approach, the *Localized IQM Calculation Method* (*Localized-IQM*), which enables a relatively more robust calculation of IQMs utilized in solving image discrimination problems. *Localized-IQM* considers the uncertainties stemming from numerical fluctuations in the local features of the compared images when calculating IQM values. In this method, image planes are partitioned into sub-regions using a [k × k] dimensional virtual grid during the IQM value calculation process, where k = 64 is used in this paper. Subsequently, the corresponding IQM value for each grid cell is calculated individually. The standard deviation of the dataset, composed of the local IQM values thus obtained, is utilized as a factor in weighting these said local IQM values. Ultimately, *Localized-IQM* produces the weighted average of these local IQM values as the final IQM value.

By incorporating the standard deviation value into its calculations, *Localized-IQM* offers a more robust and statistically consistent evaluation compared to traditional IQM methods. Although the method possesses certain limitations, its intuitive weighting mechanism, ease of implementation, and capability to manage heterogeneous data reliability render *Localized-IQM* an attractive option for diverse applications requiring nuanced similarity assessments. While *Localized-IQM* does not define a new type of IQM per se, it provides significant advantages through the integration of its uncertainty-based weighting concept into the calculation processes of existing IQMs. The fundamental contribution of the method lies in systematically addressing the heterogeneous uncertainties frequently encountered in extracted image features or associated subjective rating data, thereby enhancing the robustness and perceptual consistency of quality assessments. This nuanced approach, compared to methods that overlook such variabilities, facilitates a more refined interpretation of image quality, consequently elevating the reliability of traditional image quality metrics.

A key advantage of *Localized-IQM* becomes particularly evident in scenarios where an IQM synthesizes its final score from a composite of multiple local image features, each potentially exhibiting distinct levels of reliability or susceptibility to noise. For instance, textural descriptors extracted from poorly illuminated or highly complex regions may possess greater uncertainty compared to features derived from well-defined, homogeneous areas; similarly, features obtained from different spectral channels can exhibit varying noise profiles. *Localized-IQM* modulates the contribution of each local IQM value to the overall result in inverse proportion to its corresponding standard deviation value, thereby naturally limiting the impact of noisier local features on the aggregate quality assessment.

In multifaceted quality analysis scenarios, such as when evaluating images across different spectral channels (e.g., RGB, multi-spectral) or distinct spatial regions where the reliability of quality assessment can significantly vary, *Localized-IQM* allows for a more effective and meaningful amalgamation of these disparate quality indicators. In this approach, the weighting coefficients that determine the contribution of quality metrics derived from channels or regions characterized by high local uncertainty to the overall result are reduced. This meticulous aggregation process ensures that the consolidated IQM score, by preventing disproportionate distortion from less reliable partial assessments, more accurately reflects the overall perceptually dominant information.

Figure 1 illustrates the *Localized-IQM* process utilized in the calculation of *full-reference* IQM values (Alaei et al. 2024; Guo et al. 2023; Shu et al. 2025). Figure 2, on the other hand, presents the *Localized-IQM* process applied in the calculation of *no-reference* (blind)

IQM values (Almen et al. 1996; Li et al. 2022; Yang et al. 2019). In Figs. 1 and 2, $x$ denotes the reference image and $y$ denotes the deformed image by using Eq. 22. The *out* value is the calculated *Localized-IQM* value. The *computeIQM* function denotes the calculation of the numerical value for the relevant classical IQM. Uint8 denotes the transformation to an integer-valued *Galois field* over the closed interval [0, 255], applicable for Uint8 images. The *var* function denotes the calculation of the *variance* for the respective value.

## 4 Experiments

In this section, MAE, MSE, ERGAS, LMSE, SSIM, SAM, UIQI, PSNR, NIQE, PIQE, SNR, VIF, FSIM, GMSD, and NCC were experimentally examined to determine their robustness in image discrimination problems. The robustness of an IQM is defined by its ability to not respond with significant amplitude variations to small changes in the relevant image. An IQM should not be overly sensitive to low-amplitude noise occurring in the relevant image and numerical turbulence in pixel values due to thermal behaviours of the image capture sensor. In the conducted experiments, 12 reference Test Images with [512×512] pixels in 2D size and 8-bit RGB were used.

In the experiments conducted in this study, the relevant test images were deformed using the *Cubic Splines based Contrast Enhancement* (CSCE) method. The parameters defining the CSCE polynomial (Durrleman and Simon 1989; Enjilela et al. 2019; Yin et al. 2009)

```
Input: Image matrices x, y
Output: out: IQM value
Function out = computeIQMValue(x, y)
    x ← double(x)
    y ← double(y)
    bSize ← 64
    nBlk ← 512/bSize
    errVals ← []
    for i ← 1 to nBlk do
        for j ← 1 to nBlk do
            rs ← (i − 1) · bSize + 1
            re ← i · bSize
            cs ← (j − 1) · bSize + 1
            ce ← j · bSize
            X ← uint8(x[rs:re, cs:ce, :])
            Y ← uint8(y[rs:re, cs:ce, :])
            err ← computeIQM(X, Y)
            errVals ← append(errVals, err)
        end
    end
    varErr ← var(errVals)
    out = 1/nBlk² · sqrt( Σ errVals∘2 / varErr )
    return out
end
```

$$out = \frac{1}{nBlk^2} \cdot \sqrt{\frac{\sum errVals^{\circ 2}}{varErr}}$$

**Fig. 1** The pseudo-code for the computation process of *full-referenceLocalized-IQM* values

```
Input: Image matrix x
Output: out: IQM value
Function out=computeNoReferenceIQM(x)
    x ← double(x)
    bSize ← 64
    nBlk ← 512/bSize
    errVals ← []
    for i ← 1 to nBlk do
        for j ← 1 to nBlk do
            rs ← (i − 1) · bSize + 1
            re ← i · bSize
            cs ← (j − 1) · bSize + 1
            ce ← j · bSize
            y ← uint8(x[rs..re, cs..ce, :])
            err ← computeIQM(y)
            errVals ← append(errVals, err)
        end
    end
    varErr ← var(errVals)
    out = 1/nBlk^2 · sqrt(∑ errVals^∘2 / varErr)
    return out
```

**Fig. 2** The pseudo-code for the computation process of *no-referenceLocalized-IQM* values (i.e., Localized-NIQE and Localized-PIQE values)

were optimized using the *Colony-Based Search Algorithm* (CSA) (Civicioglu and Besdok 2024a) such that the numerical difference between the classical IQM value calculated for the reference and deformed images, and a target empirical IQM value, (i.e., $IQM_0(x, y)$) for the specific IQM, was less than $10^{-2}$. The discrete nature of images renders the CSCE optimization a multimodal problem. Consequently, by employing CSA solutions with different initial colonies, it was possible to synthesize three visually distinct images for each test image, all possessing the same classical IQM value. The structural parameter values used for CSA are as follows: the lower search bound is set to 0 and the upper search bound is set to 255. The optimization process was continued for 1000 epochs. The colony size of CSA was chosen as 10.

In the experiments performed in this section, the number of control points utilized to define the CSCE polynomial is set to 10. Therefore, the optimization problem has focused on determining the positions of the relevant control points that enable obtaining the desired empirical IQM value between the reference and deformed images. The relevant control points enable the definition of the CSCE polynomial, and the related polynomial is used in the generation of the deformed image. The analytical definition of this optimization problem is provided in Eq. (22).

$$\underbrace{\arg\min}_{x_k} \sum |IQM(x, y) - IQM_0| \tag{22}$$

Here, *x* is the *reference image* and *y* is the image deformed using CSCE. In order to calculate the numerical values of the polynomial coefficients of CSCE, the $x_k$ positions were optimized with the help of CSA using Eq. (22).The $IQM_0$ values used in Eq. (22) for the deformed images are presented in Table 2. The NIQE and PIQE values presented in Table 2 are the calculated NIQE and PIQE scores for the *reference image*.

Upon collective analysis of the experimental results, the IQM values of the deformed images were observed to have a numerical precision of $10^{-2}$ corresponding to a 99.935% confidence interval for *normal distribution*. This indicates that the respective deformed images were deformed very successfully following the employed CSA-based evolutionary optimization process. In the experiments performed in this paper, the respective Image Quality Metric (IQM) was considered to exhibit *weak robustness* when the absolute difference between the classical *IQM*(*x*, *y*) value and its predetermined numerical value, $IQM_0(x, y)$, fell below $10^{-2}$.

### 4.1 Cubic splines based contrast enhancement

Image contrast manipulation aims to remap the intensity values of pixels to enhance visual distinctions or achieve specific aesthetic effects. Spline interpolation provides a powerful and flexible method for defining a smooth, non-linear mapping function based on a discrete set of user-defined control points. Let these *N* control points be denoted as $(x_k, y_k)$ for $k = 0, \ldots, N-1$, where $x_k$ represents the original input intensity levels and $y_k$ represents the desired target output intensity levels for those inputs. Typically, the input intensity values $x_k$ are uniformly distributed across the normalized intensity range, such as [0, 1], ensuring coverage of the entire tonal spectrum. A spline function, *S*(*x*), is then constructed such that it passes through these specified control points, satisfying the condition $S(x_k) = y_k$ for all *k*.

Cubic splines are frequently employed for this purpose due to their inherent smoothness, which is achieved by ensuring continuity of the function itself, as well as its first and second derivatives, at each interior control point, often referred to as a knot. Between any two consecutive knots $x_k$ and $x_{k+1}$, the spline *S*(*x*) is represented by a distinct cubic polynomial, denoted as $S_k(x)$. This polynomial for the *k*-th interval $[x_k, x_{k+1}]$ can be expressed by using Eq. (23).

$$S_k(x) = a_k(x - x_k)^3 + b_k(x - x_k)^2 + c_k(x - x_k) + d_k \tag{23}$$

The coefficients $a_k, b_k, c_k, d_k$ for each segment are determined by satisfying several critical conditions. Firstly, the interpolation condition dictates that $S_k(x_k) = y_k$, which directly implies that $d_k = y_k$ for each segment. Secondly, at each interior knot $x_k$ (for $k = 1, \ldots, N-2$), the values of the adjacent polynomial segments, their first derivatives, and their second derivatives must match to ensure overall smoothness of the composite curve. For instance, the continuity of the first derivative requires $S'_{k-1}(x_k) = S'_k(x_k)$, ensuring a smooth transition in slope. Similarly, for the second derivative, the continuity condition is $S''_{k-1}(x_k) = S''_k(x_k)$, ensuring a smooth transition in curvature.

These continuity constraints, combined with the interpolation requirements and suitable boundary conditions (e.g., *natural* splines where $S''(x_0) = 0$ and $S''(x_{N-1}) = 0$), form a system of linear equations. Solving this system typically involves first determining the values of the second derivatives $S''(x_k)$ at each knot. Once these second derivative values

**Table 2** Baseline values of $(IQM_0)$ used in the experiments

| IQMs | Test Images | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | #1 | #2 | #3 | #4 | #5 | #6 | #7 | #8 | #9 | #10 | #11 | #12 |
| MAE | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 | 20 |
| MSE | 200 | 200 | 200 | 200 | 200 | 200 | 200 | 200 | 200 | 200 | 200 | 200 |
| ERGAS | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |
| LMSE | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| SSIM | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 |
| SAM | 45 | 45 | 45 | 45 | 45 | 45 | 45 | 45 | 45 | 45 | 45 | 45 |
| UIQI | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 |
| PSNR | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 |
| NIQE | 3.56 | 3.32 | 2.93 | 4.38 | 2.53 | 2.79 | 2.76 | 3.05 | 4.22 | 5.19 | 5.26 | 3.76 |
| PIQE | 26.49 | 47.31 | 30.36 | 46.91 | 25.42 | 31.65 | 25.15 | 32.8 | 27.02 | 23.05 | 16.51 | 26.88 |
| SNR | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 |
| VIF | 0.5 | 0.5 | 0.5 | 0.5 | 0.5 | 0.5 | 0.5 | 0.5 | 0.5 | 0.5 | 0.5 | 0.5 |
| FSIM | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 |
| GMSD | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 |
| NCC | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 |

are known, the remaining coefficients $a_k, b_k,$ and $c_k$ for each cubic polynomial segment can then be readily computed. After the entire spline function *S*(*x*) is fully defined by these piecewise polynomials, it effectively serves as a continuous lookup table or a direct intensity mapping function.

For processing an RGB image, represented as *I*, let *I*(*u*, *v*) be the vector of intensity values at pixel coordinates (*u*, *v*), with individual components $I_R(u,v), I_G(u,v)$, and $I_B(u,v)$ for the red, green, and blue channels, respectively. The spline transformation *S*(*x*) is then applied independently to each color channel $c \in \{R, G, B\}$ for every pixel (*u*, *v*) in the image. The new intensity value $I'_c(u,v)$ for a given channel is thus determined by using Eq. (24).

$$I'_c(u,v) = S(I_c(u,v)) \tag{24}$$

Prior to applying the spline function, input pixel intensities $I_c(u,v)$ are usually normalized to the domain over which *S*(*x*) is defined, typically the interval [0, 1]. The output values $I'_c(u,v)$ obtained from the spline function are subsequently clamped to this same range, [0, 1], to ensure they represent valid intensity levels before any potential denormalization back to the original image data type (e.g., 8-bit integers). This comprehensive process allows for highly flexible and nuanced adjustments to global image contrast by enabling users to interactively define the precise shape of the intensity mapping curve through the strategic placement of the $y_k$ control point values.

### 4.2 Colony-based search algorithm (CSA)

The CSA is an evolutionary-based global minimizer specifically engineered for single-objective numerical problems, accommodating both bounded and unbounded real-valued variables (Civicioglu and Besdok 2024a). In its current form, CSA primarily focuses on refining patterns within the clan matrix, which are initially sampled stochastically from a larger parental collective, designated as the colony matrix. Characterized by its non-recursive, iterative, and inherently robust nature, CSA operates through a structured framework comprising initialization, selection, numerical evolution, and updating phases. A critical feature of the algorithm is the seamless integration of its mutation mechanism with a random crossover process, a synergistic operation collectively referred to as morphogenesis (Civicioglu and Besdok 2018, 2019, 2021, 2022, 2023, 2024b). A comprehensive, step-by-step exposition of CSA's algorithmic design is provided in the subsequent sections.

The CSA's primary population, referred to as the Colony Matrix (*p*0), is initialized as a random solution vector. Its dimensions are determined by multiplying the size of the Clan Matrix (*p*) by *T* times. Both the initial Colony Matrix (*p*0) and the objective function values of its patterns (*fitp*0) are established using the methodology outlined in Eq. (25).

$$\begin{cases} p0_{i=1:T\cdot N, j=1:D} \sim U(low_j, up_j) \\ fitp0 = ObjFun(p0) \end{cases} \tag{25}$$

Here, *ObjFun* represents the objective function, while *low* and *up* denote the lower and upper bounds of the search space, respectively. For generating the random numbers essential across all experimental methods in this study, the reliable Mersenne Twister uniform pseudo-random number generator is employed. Within CSA, the initial values for *moment* and *initindex* are set to $moment_{initindex} = 0$ and $initindex = \{1 : N\}$, respectively.

Equation (26) then describes how the clan (*p*) and its corresponding objective function values (*fitp*) are expressed.

$$[p;\ fitp] = [p0(index, :);\ fitp0(index)] \tag{26}$$

In this context, $index = rperm(T \cdot N; N)|(initindex \sim= index)$, and the *initindex* is subsequently updated to *index*. The *rperm* denotes *position permuting* operator on vectors in *Discrete Mathematics*.

To control the amplitude of the direction vector during the optimization process, CSA utilizes a scale factor, or *scale*. This factor is dynamically generated using the mechanism described in Eq. (27):

$$scale = \begin{cases} (rand\,(N; c) - 0.50) \oslash (rand\,(N, c) - 0.50) & If\ rand() < rand() \\ sign\,(rand\,(N; 1) - 0.50) \circ \Phi & else \end{cases} \tag{27}$$

Here, $\Phi \sim LevyDist\,(\alpha, \beta, N, 1)$ signifies a scaling vector, and $\oslash$ represents the *Hadamard division* operator. The $\circ$ operator denotes *Hadamard multiplication*. The *rand* function generates pseudo random numbers using a uniform distribution U(0,1) each time it is called. The individual elements of $\Phi$ follow a Levy Distribution with $\alpha$ and $\beta$ as its location and shape parameters. This inclusion of $\Phi$ is crucial as it imbues CSA with Levy flight capability. Within the CSA framework, Levy flights provide the essential exploration capacity needed to effectively navigate and overcome local optima, thereby enabling the discovery of diverse high-quality solutions within complex search spaces. By incorporating sporadic long jumps, Levy flights significantly strengthen the global search process, enhancing the algorithm's robustness and markedly reducing the likelihood of premature stagnation.

The Levy distribution itself is a probability distribution renowned for its heavy-tailed nature. This characteristic means it exhibits a greater propensity to generate extreme values compared to the more common normal distribution. Its behavior is precisely characterized by its probability density function (PDF), as specified in Eq. (28).

$$f(x; \alpha, \beta) = \frac{\beta}{2\pi} \exp\left(-\frac{\beta}{2\,(x-\alpha)}\right)\left(-\frac{1}{(x-\alpha)^{3/2}}\right) \tag{28}$$

In this equation, *x* denotes the random variable, $\alpha$ is the location parameter, and $\beta$ is the scale parameter of the distribution. In CSA, the $\Phi$ utilized in Eq. (27) is specifically generated using the formulation presented in Eq. (29):

$$\Phi = \beta \cdot (rand() + 1) \oslash \omega^{\frac{1}{\alpha}}\ \mid\ \omega \sim \Gamma\,(\alpha; \kappa)\,,\ \kappa \sim \{2 : 5\} \tag{29}$$

Here, $\Gamma\,(\alpha; \kappa)$ denotes the Gamma distribution with parameters $\alpha$ and $\kappa$.

The value of *c*, which is used in Eq. (27), is calculated according to the logic presented in Eq. (30):

$$If\ rand() < rand()\ ,\ then\ c = 1\ \ else\ \ c = D \tag{30}$$

The binary-valued mutation control matrix, *m*, of CSA is systematically produced using the steps outlined in Eq. 31. This matrix plays a critical role in guiding the mutation process.

$$\begin{cases} m = 0_{N \times D} \\ for\ \ j = 1\ \ to\ \ N \\ ind = rperm(D) \\ k_{index} = abs(randi([0,1]) - rand()^{\bullet randi([1,10])}) \\ e = ind\,(1 : \lceil[\![ k_{index} \cdot D ]\!]\rceil) \\ m(k,e) := 1 \\ end \end{cases} \tag{31}$$

In this context, • denotes the piecewise-power operator. Following the creation of the mutation control matrix, CSA utilizes Eq. (32) to generate the evolutionary direction, *dx*. This vector dictates the path of potential improvement for the solutions.

$$dx = \begin{cases} p[v2] - p[v1] & v = 1 \\ p[v1] - p & v = 2 \\ p\,[index0\,[randi\,(1, \lceil[\![ N/5 ]\!]\rceil)]\,,:] - p & v = 3 \end{cases} \tag{32}$$

Here, $[v1, v2] = [rperm(N;N), rperm(N;N)]$ ensures that $v1 \neq v2$, and $index0 = sortindex(fitp,'ascend')\ \wedge\ \ v \sim U\{1,2,3\}$.

The morphogenesis matrix, *px*, which represents the new generation of solutions in CSA, is then generated using the formula presented in Eq. (33). This step combines the current patterns with the calculated evolutionary direction and scale factor.

$$px = p + scale \circ m \circ dx + s \tag{33}$$

Here, $s = (rand(N;1) - 0.50) \circ rand(N;1)^{\bullet randi([2;10])}$.

A crucial aspect of any optimization algorithm is handling solutions that fall outside the defined search boundaries. In CSA, Eq. (34) is employed to update individuals that exceed these search limits, effectively bringing them back within the feasible region.

$$\begin{array}{l} for\ k = 1\ to\ N \\ for\ l = 1\ to\ D \\ If\ px(k,l) < low(l),\ px(k,l) = low(l) + rand()^{randi([1;5])} \cdot (up(l) - low(l)) \\ If\ px(k,l) > up(l),\ px(k,l) = up(l) + rand()^{randi([1;5])} \cdot (low(l) - up(l)) \\ end \\ end \end{array} \tag{34}$$

Following the generation and boundary correction of the new patterns, their objective function values (*fitpx*) are computed. This evaluation step, shown in Eq. (35), is fundamental for assessing the quality of the new solutions.

$$fitpx = ObjFun(px) \tag{35}$$

Next, Eq. (36) is used to identify the indices (*ind*) of the *px* patterns that offer superior solutions compared to their corresponding patterns in the current clan. This step pinpoints which new solutions are improvements.

$$ind \leftarrow fitpx_{\{1:N\}} < fitp_{\{1:N\}} \;\mid\; ind \in \{1:N\} \tag{36}$$

In the concluding stage of each iteration, CSA strategically modifies the associated Colony patterns by incorporating these improved Clan patterns, as precisely outlined in Eq. (37). This ensures that the population evolves towards better solutions.

$$\begin{cases} [p(ind,:), fitp(ind)] = [px(ind,:), fitpx(ind)] \\ [p0(index,:), fit0(index)] = [p, fitp] \\ indbest \leftarrow fitp0_{indbest} = \min(fitp0) \\ [gbest, gmin] = [p0(indbest,:), fit0(indbest)] \end{cases} \tag{37}$$

Finally, at the end of the current iteration, the numerical value of *moment* is revised according to the formula presented in Eq. (38), contributing to the algorithm's dynamic behavior.

$$moment = (abs(randi([0,1];N,1)) - m) \circ dx \tag{38}$$

In summary, CSA is a stochastic, iterative, and non-recursive evolutionary search method. It is specifically developed for solving single-objective, real-valued numerical problems, and the version presented here is engineered as a global minimizer. A key distinguishing feature of CSA is its unique scaling method, which effectively regulates the amplitude of its direction vectors, contributing to its robust search capabilities.

Table 15 presents a summary table designed to help readers quickly interpret the comparative results, providing the robustness ranking of Image Quality Metrics (IQMs) in discrimination tasks (lower $\rightarrow$ higher).

### 4.3 Limitations and future research

The proposed Localized-IQM framework substantially improves discrimination by incorporating local statistical variations, yet certain boundaries of its applicability should be acknowledged. In cases where degradations are largely global and uniform, such as heavy blur or strong additive noise, the relative advantage of localized partitioning may be reduced. Similarly, for highly homogeneous image regions, limited variability across tiles can lessen the contribution of local weighting. The method also introduces a modest computational overhead compared to global IQMs, although this remains a constant-factor increase that does not preclude real-time deployment on modern CPUs and GPUs. As Localized-IQM operates on top of classical metrics, it may still inherit some baseline sensitivities from the underlying IQM formulation.

Despite these considerations, the framework provides a robust foundation for future extensions. Broader validation across application-specific domains such as medical imaging, environmental monitoring, or surveillance remains an important next step, as distortion types and perceptual demands differ across these contexts. Another promising direction is the integration of adaptive grid sizing or perceptual weighting to ensure stronger alignment with human visual sensitivity. Moreover, combining Localized-IQM with models of the human visual system or incorporating uncertainty quantification could enhance both interpretability and robustness. Overall, these extensions can strengthen the method's applicability while preserving its demonstrated advantages in discriminative image quality assessment.

## 5 Results

In the experiments conducted in this paper, three different image datasets were utilized: *Animal Images Dataset* (Github 2023), *Human Images Dataset* (Hub 2024), and *Earth Observation Images Dataset* (NVIDIA 2019). Each image dataset contains 4 different *Test Images*. In the conducted experiments, for each *Test Image*, 3 new images that are visually different from each other but achieve the same classical-IQM values were computed. In the experiments performed, a total of 540 deformed images were used for all IQMs. These images, generated using CSA and Eq. (22) to achieve identical classical-IQM values to the *reference image*, are visualized in Figs. 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 14. For the images presented in Figs. 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 14, classical IQMs produce completely identical IQM values between images with different appearances and the *reference image*. However, the proposed *Localized-IQM* values in this paper can detect local differences between the corresponding images with different appearances and the *reference image*. Therefore, *Localized-IQM* can easily distinguish images that have identical semantic content but different visual appearances. The *Localized-IQM* values are provided in Tables 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 14.

The experimental results for the *Animal Images Dataset*, presented in Figs. 3, 4, 5 and 6, reveal a significant inconsistency between human perception and the values provided by classical IQMs. All classical IQMs employed in the experiments showed limited robustness when evaluating the deformed images. Notably, metrics such as MAE, ERGAS, LMSE, PSNR, NIQE, SNR, PIQU, FSM, GMSD, and NCC produced identical IQM values at a $10^{-2}$ precision level, even for images that were visually extremely different. This finding clearly demonstrates that classical IQMs fail to adequately discriminate images for the test images within this dataset.

Upon examining the experimental results for the *Human Images Dataset*, showcased in Figs. 7, 8, 9 and 10, a clear discrepancy between quantitative and qualitative image quality assessment outcomes is observable. All classical IQMs used in these experiments showed limited ability to discriminate the deformed images. The results obtained for all test images in this dataset revealed that classical IQMs readily produced misleading values at a $10^{-2}$ precision level, even for images exhibiting striking visual differences.

Experiments conducted within the *Earth Observations Images Dataset*, with results presented in Figs. 11, 12, 13 and 14, unequivocally demonstrate the weakness of classical IQMs in image discrimination performance. For all test images in this test set, identical classical IQM values were obtained at a $10^{-2}$ precision level, even with synthetic images that were visually extremely different. This situation clearly indicates that the success level of classical IQMs in image discrimination is insufficient in the context of the test images within this particular image dataset.

The proposed *Localized-IQM* values are presented in Tables 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 14. In these tables, the results indicated in bold font represent the Localized-IQM values generated for the respective classical IQM. As is evident from all tables, these val-

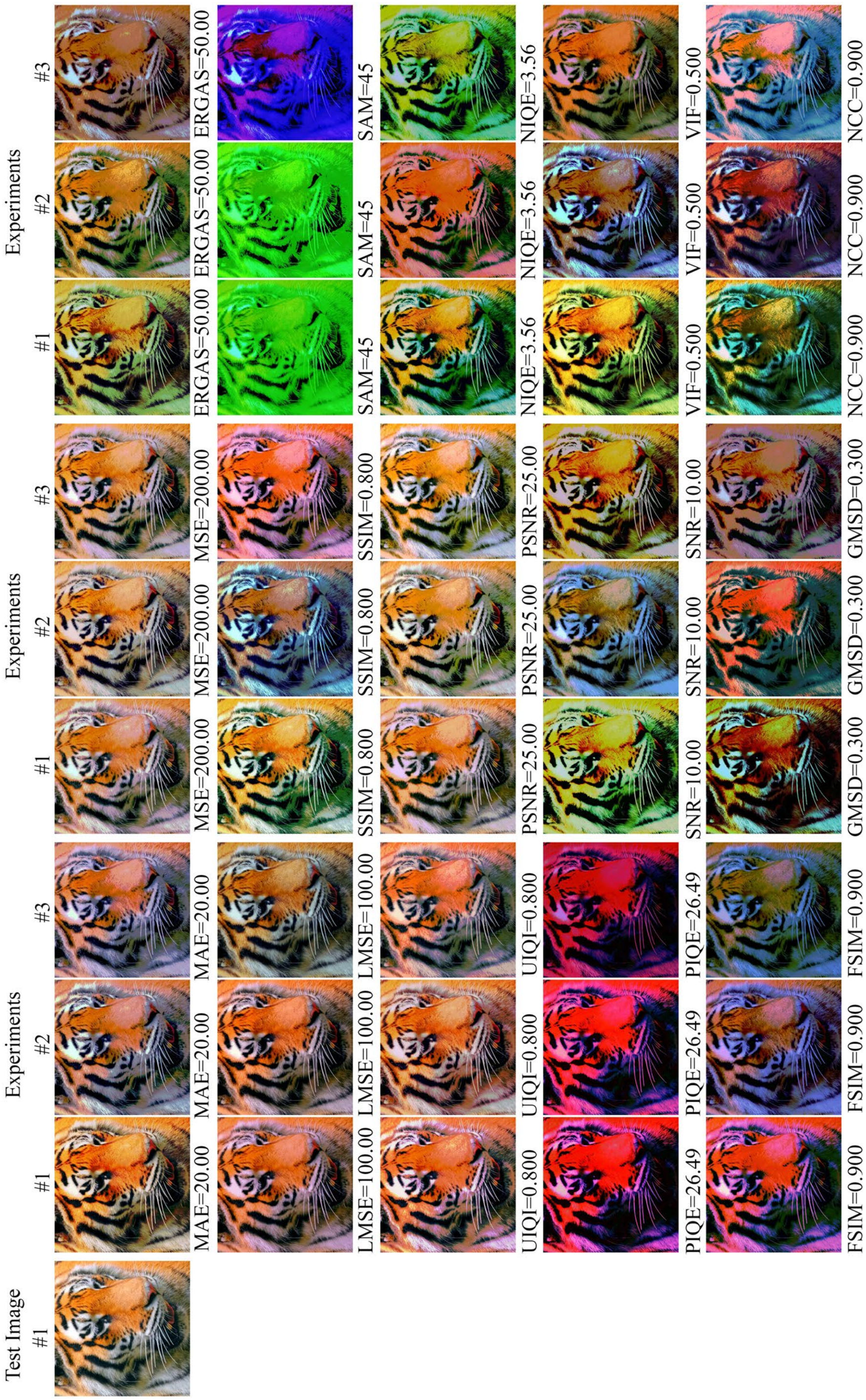


**Fig. 3** The experimental results for the Test Image #1 of *Animal Images Dataset*. Please use the zoom function on your computer to see more details

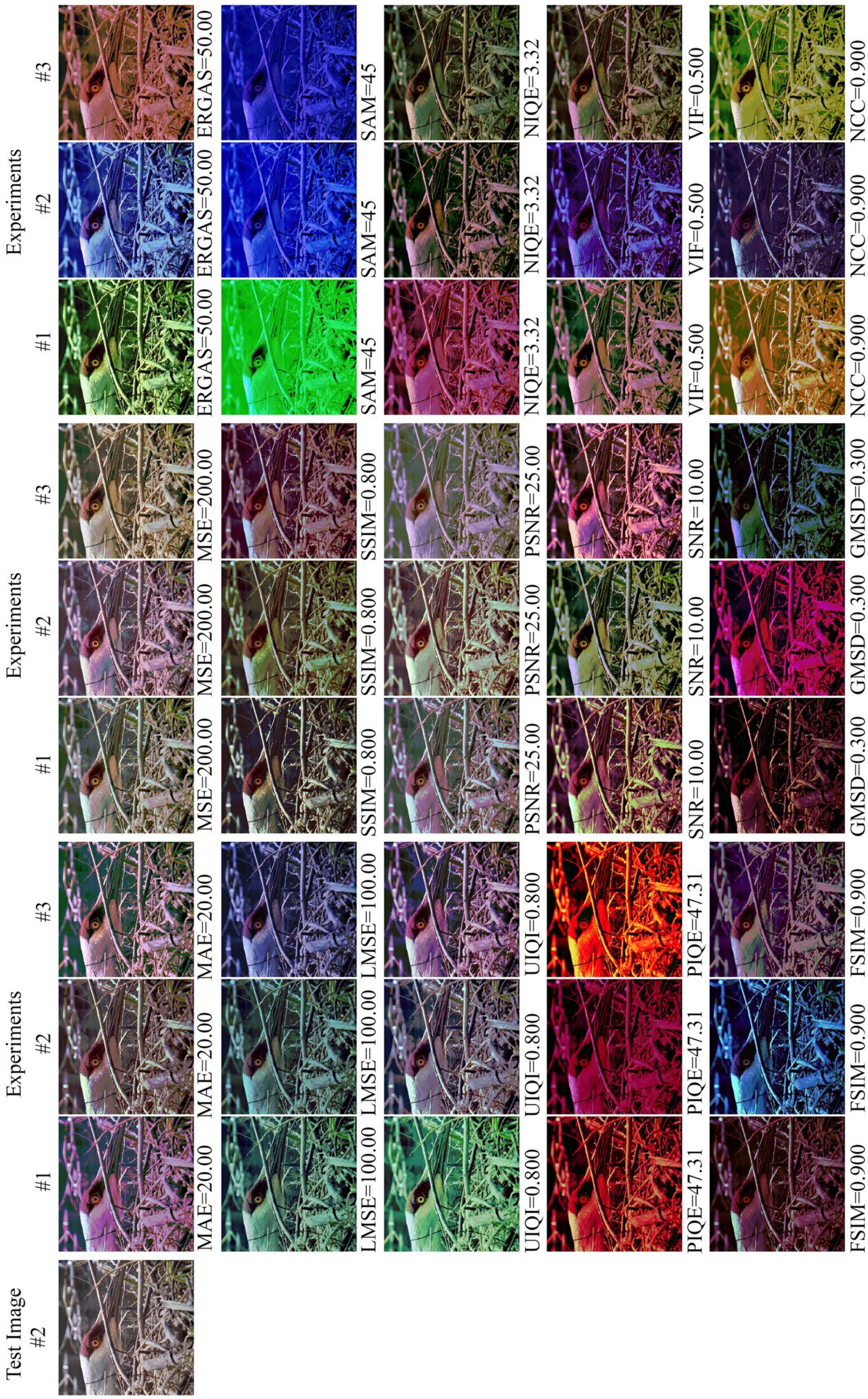


**Fig. 4** The experimental results for the Test Image #2 of *Animal Images Dataset*. Please use the zoom function on your computer to see more details

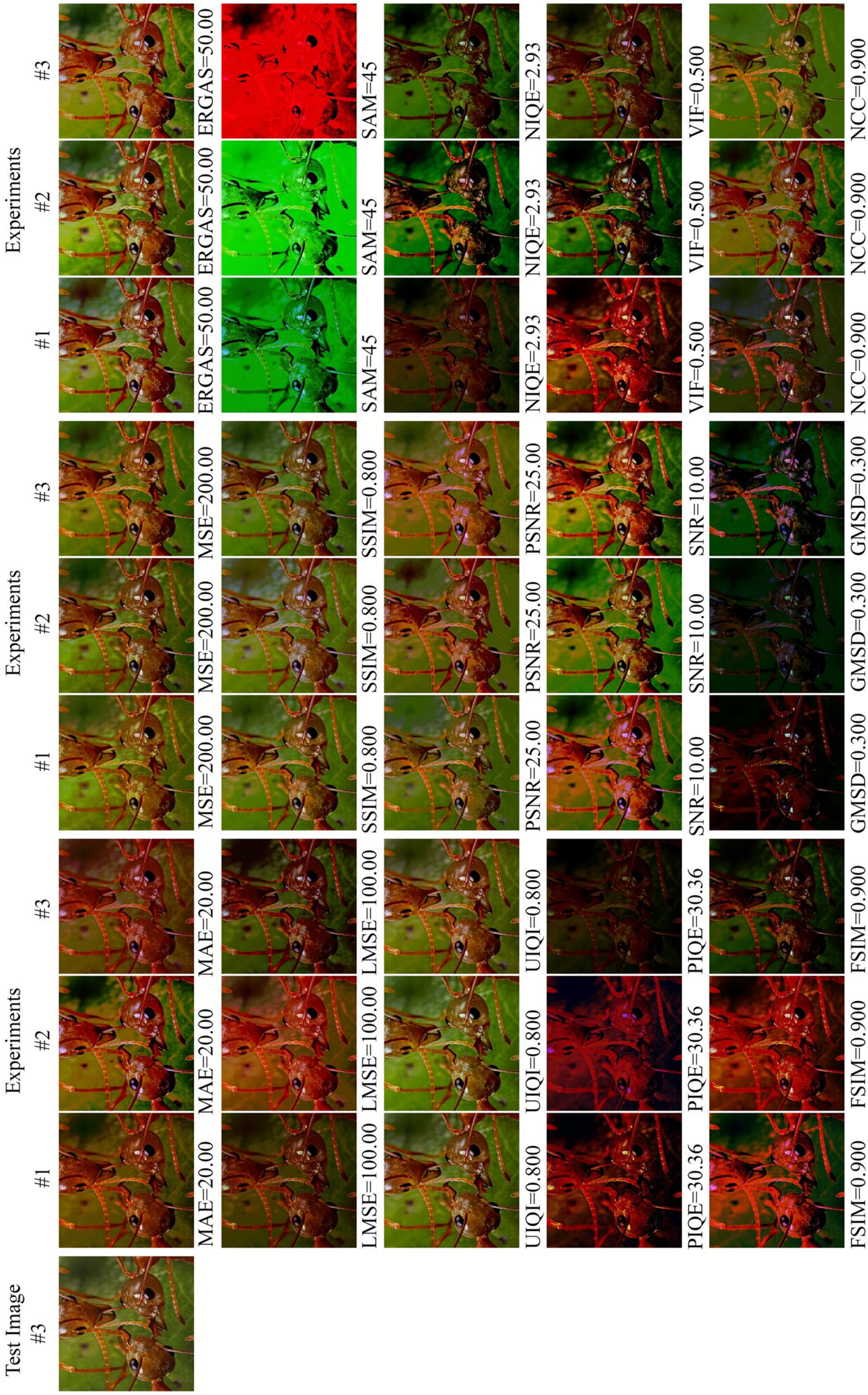


**Fig. 5** The experimental results for the Test Image #3 of *Animal Images Dataset*. Please use the zoom function on your computer to see more details

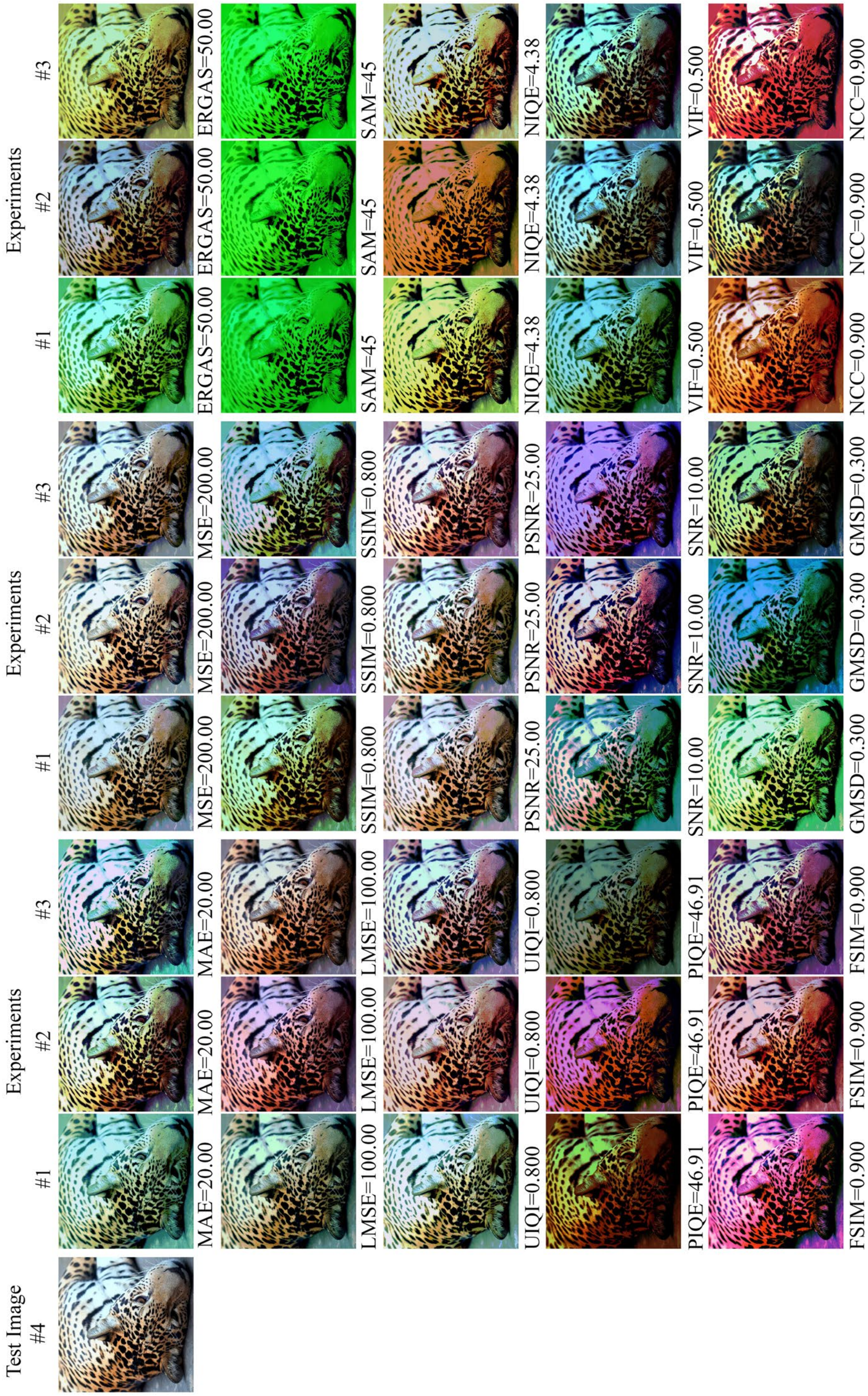


**Fig. 6** The experimental results for the Test Image #4 of *Animal Images Dataset*. Please use the zoom function on your computer to see more details

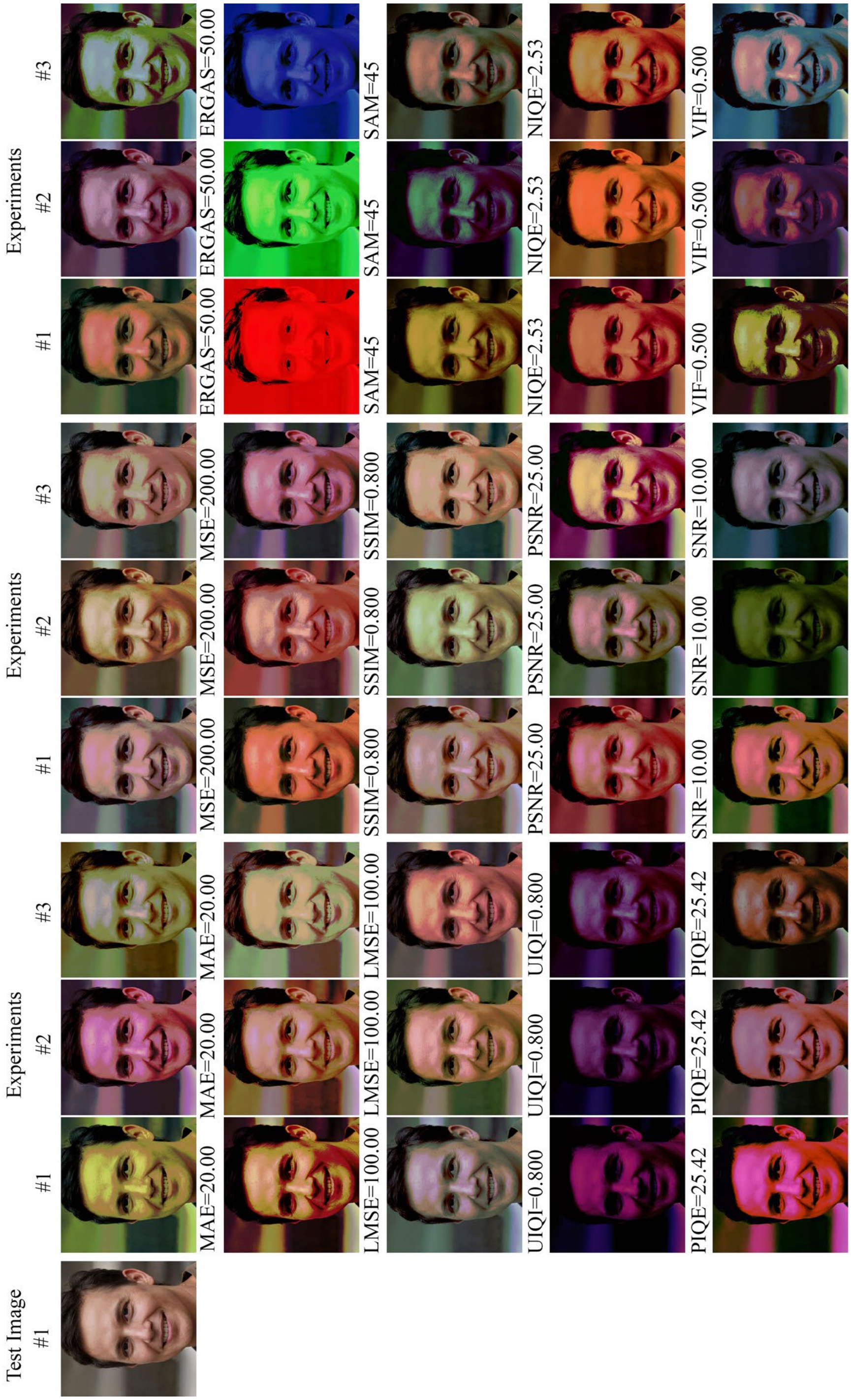


**Fig. 7** The experimental results for the Test Image #1 of *Human Images Dataset*. Please use the zoom function on your computer to see more details

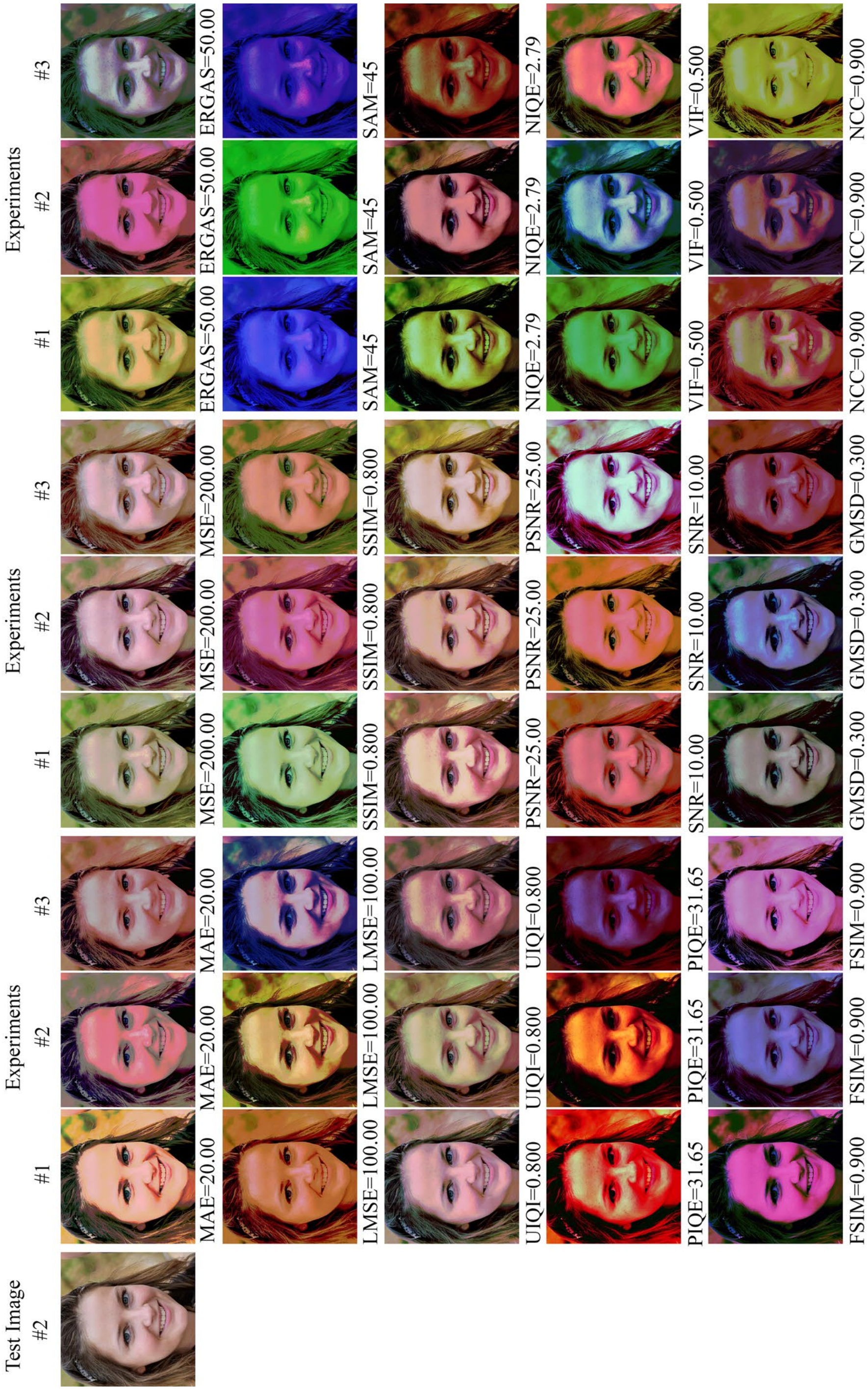


**Fig. 8** The experimental results for the Test Image #2 of *Human Images Dataset*. Please use the zoom function on your computer to see more details

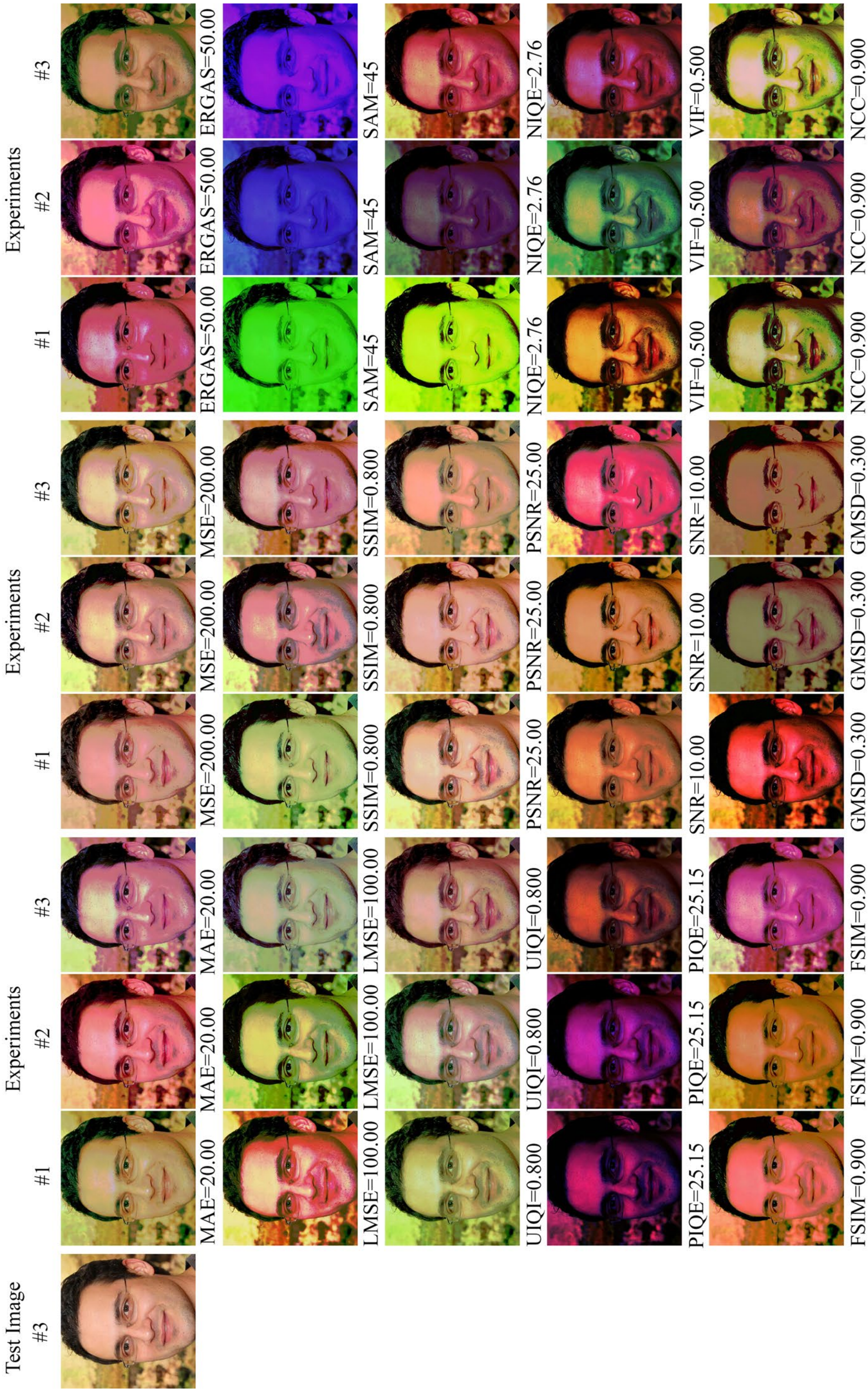


**Fig. 9** The experimental results for the Test Image #3 of *Human Images Dataset*. Please use the zoom function on your computer to see more details

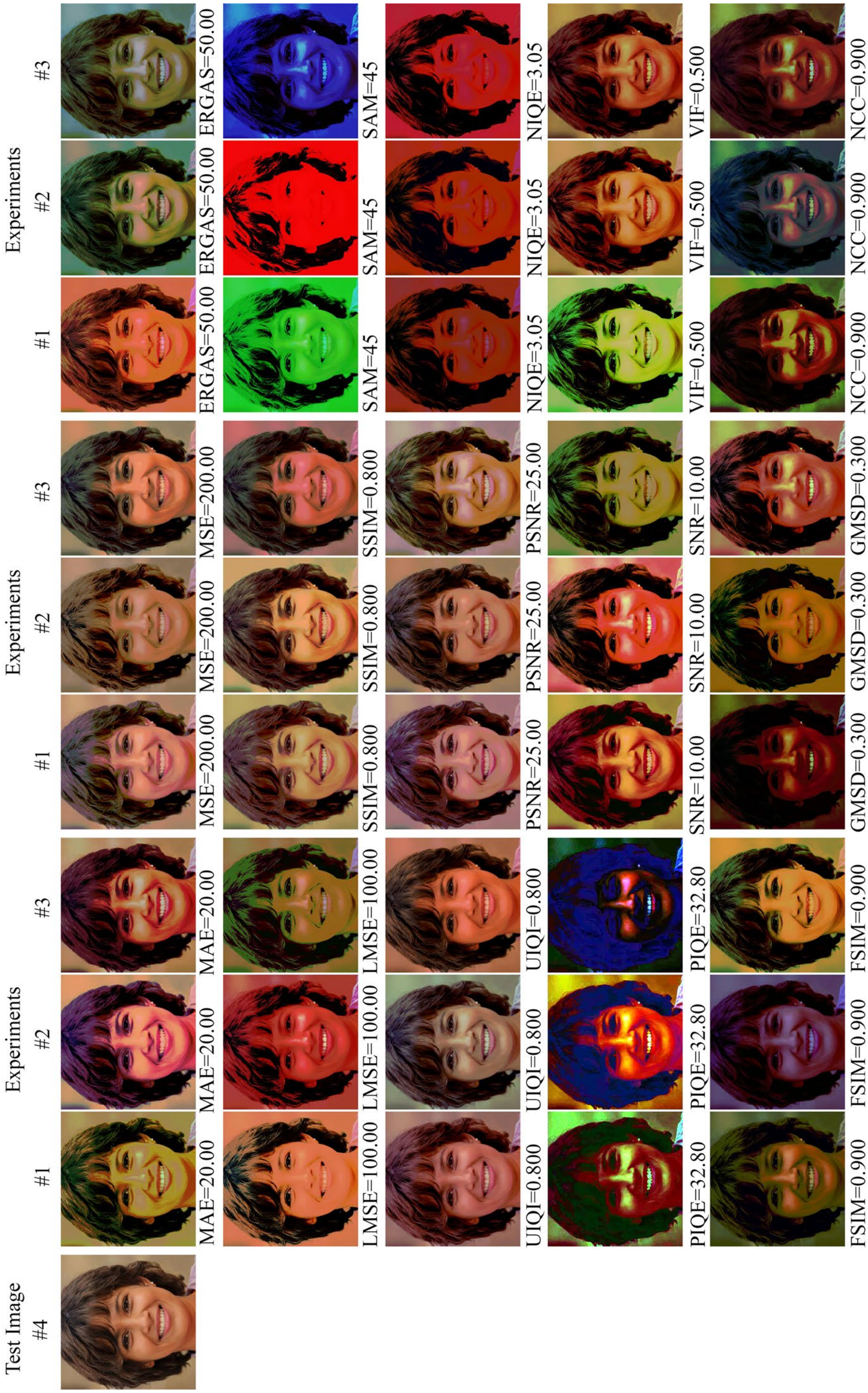


**Fig. 10** The experimental results for the Test Image #4 of *Human Images Dataset*. Only the response images from PIQE were enhanced for better visualization. Please use the zoom function on your computer to see more details

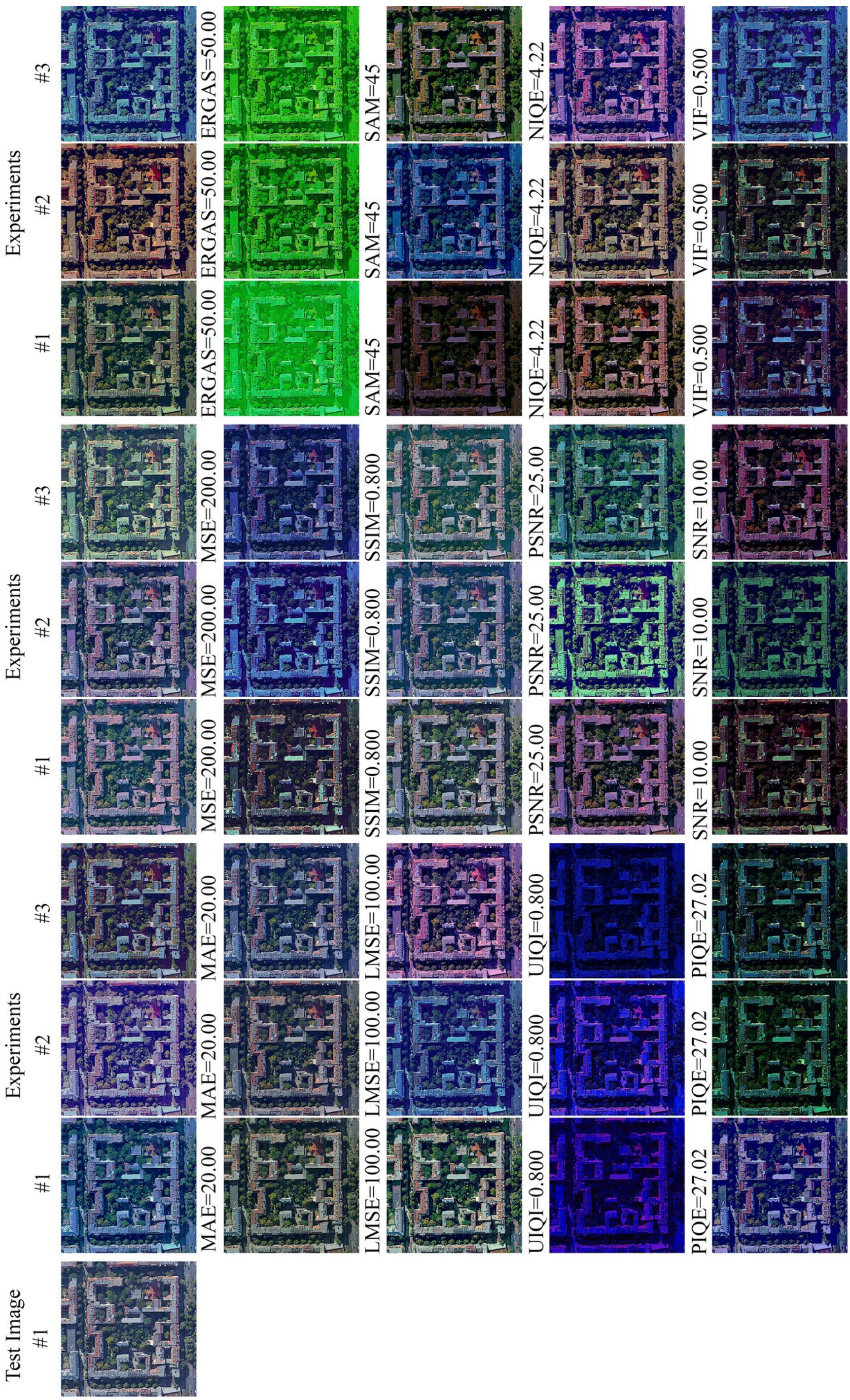


**Fig. 11** The experimental results for the Test Image #1 of *Earth Observation Images Dataset*. Please use the zoom function on your computer to see more details

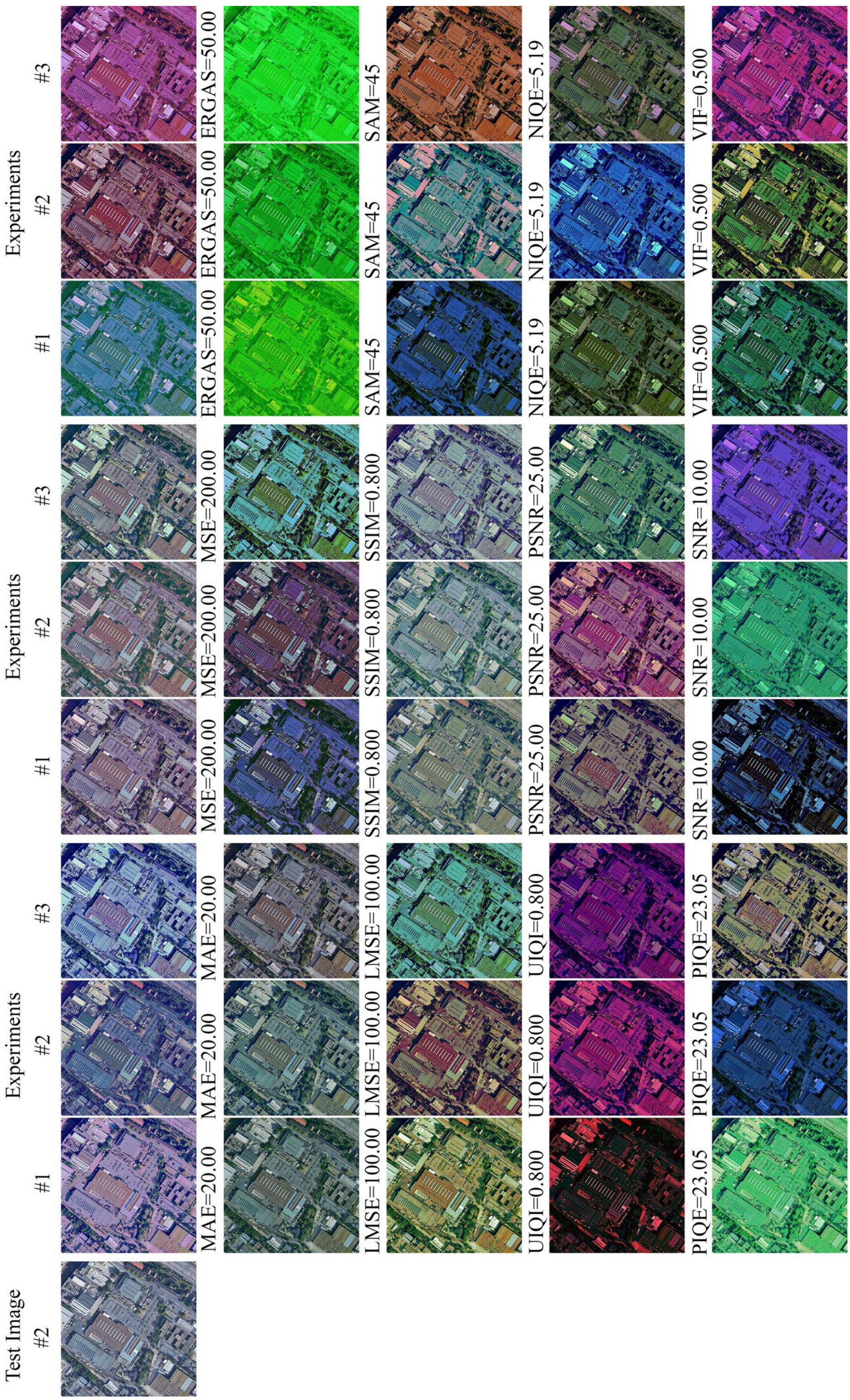


**Fig. 12** The experimental results for the Test Image #2 of *Earth Observation Images Dataset*. Please use the zoom function on your computer to see more details

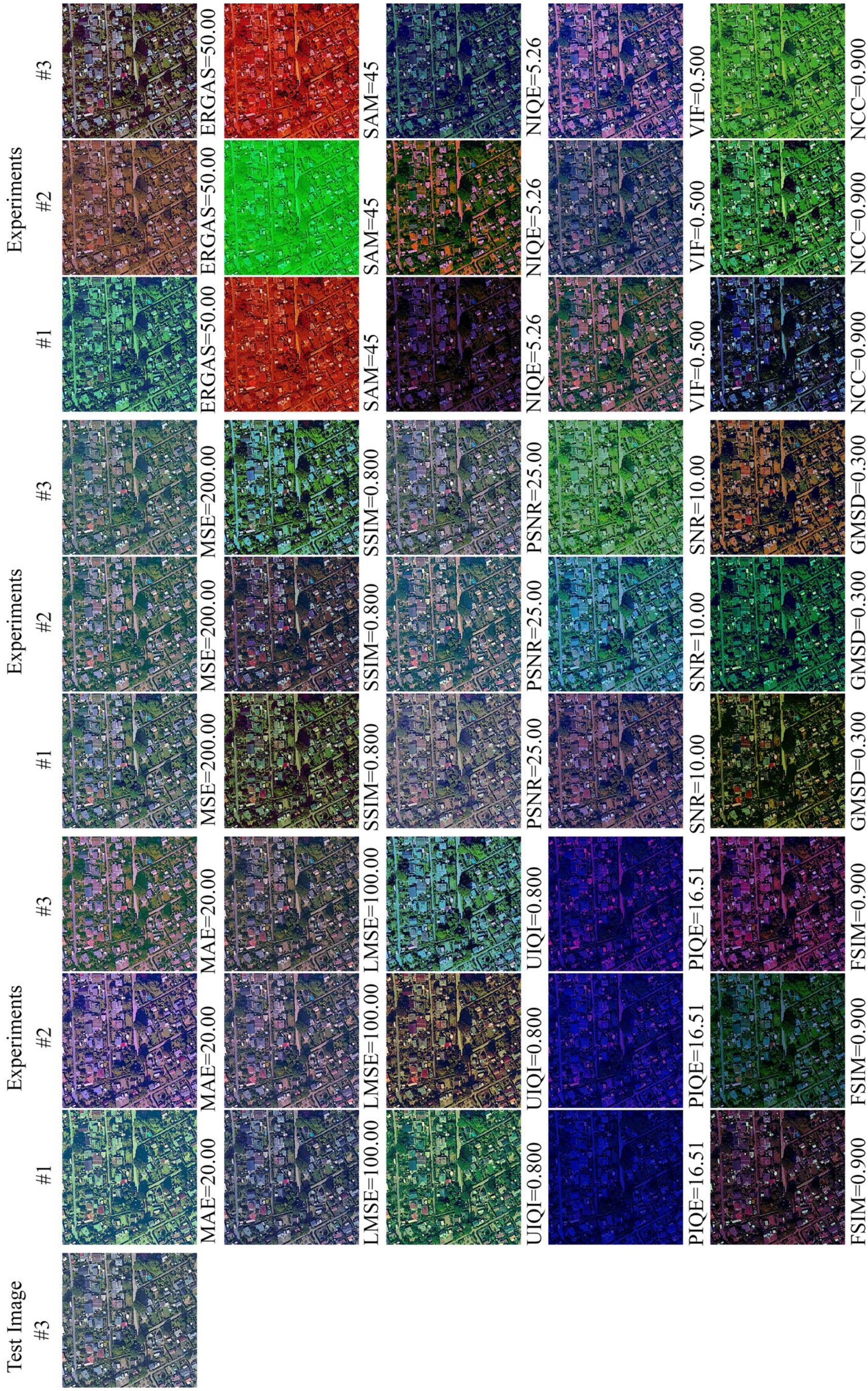


**Fig. 13** The experimental results for the Test Image #3 of *Earth Observation Images Dataset*. Please use the zoom function on your computer to see more details

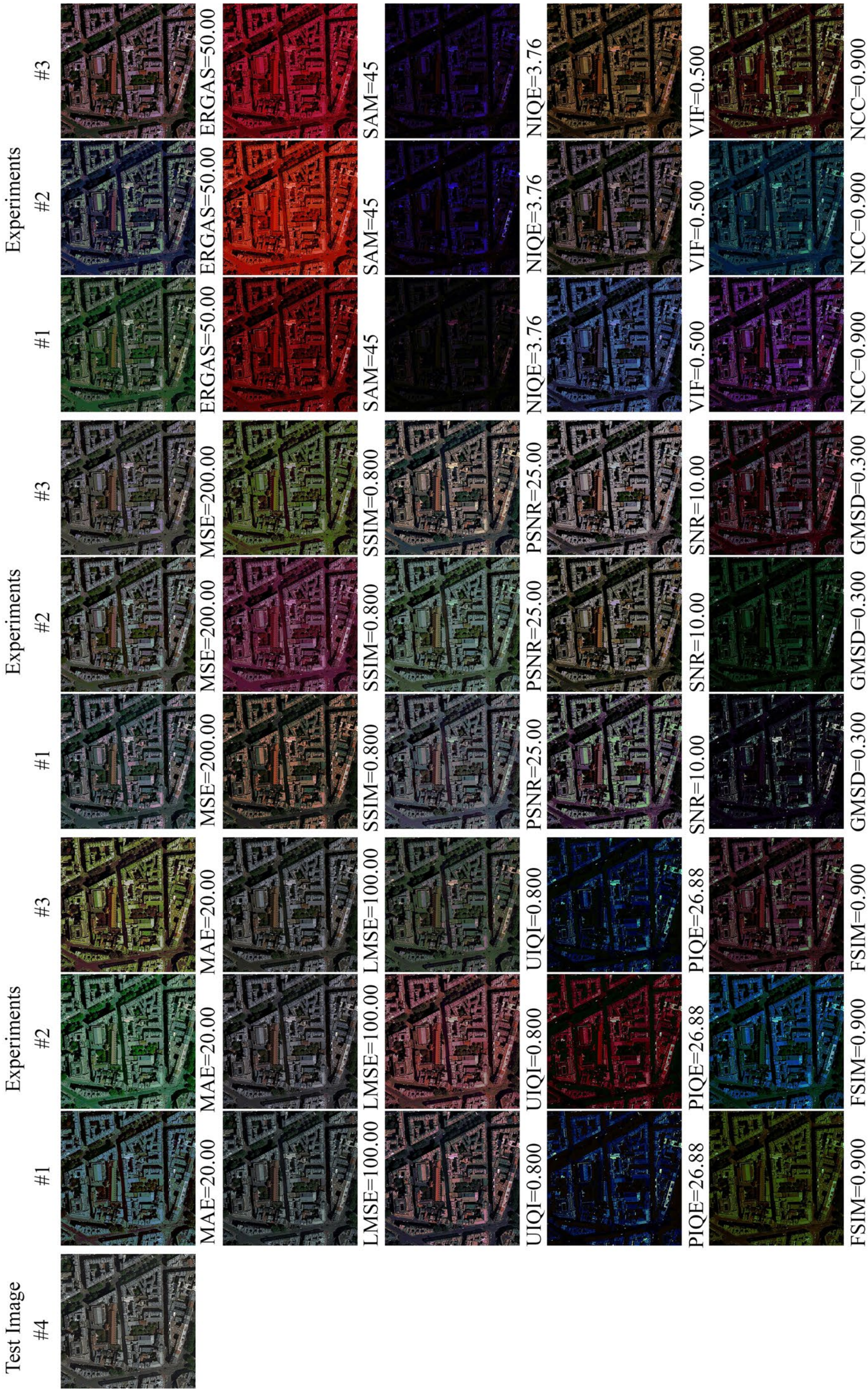


**Fig. 14** The experimental results for the Test Image #4 of *Earth Observation Images Dataset*. Please use the zoom function on your computer to see more details

**Table 3** *Localized-IQM* values obtained from experiments performed using Test Image #1 from the Animal Image Dataset

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| | | 20.0000 | 200.0000 | 50.0000 | 100.0000 | 0.8000 | 45.0000 | 0.8000 | 25.0000 | – | – | 10.0000 | 0.5000 | 0.9000 | 0.3000 | 0.9000 |
| MAE | #1 | **0.4906** | 0.2433 | 0.4519 | 0.2098 | 0.3418 | 0.5304 | 0.6523 | 1.3811 | 847.5678 | 0.2968 | 0.6540 | 0.4058 | 3.4980 | 0.3052 | 1.5246 |
| MAE | #1 | **1.0356** | 0.6507 | 0.3853 | 0.2864 | 0.8179 | 0.4407 | 1.3673 | 2.1519 | 441.8626 | 0.3526 | 0.6483 | 1.0339 | 7.4451 | 0.2526 | 7.6220 |
| MAE | #2 | **1.1265** | 0.6761 | 0.3767 | 0.2886 | 0.8162 | 0.2676 | 1.4355 | 4.1082 | 575.9005 | 0.3697 | 0.9956 | 1.2299 | 4.8269 | 0.3259 | 5.1254 |
| MAE | #3 | **1.1631** | 0.7434 | 0.4312 | 0.3161 | 0.5635 | 0.3097 | 1.2072 | 3.5913 | 619.7125 | 0.3585 | 1.0928 | 1.3225 | 4.9317 | 0.4421 | 6.1241 |
| MSE | #1 | 0.5923 | **0.3266** | 0.3345 | 0.3214 | 1.3372 | 0.3137 | 1.7154 | 2.2117 | 655.8316 | 0.3415 | 1.5421 | 1.2721 | 12.0641 | 0.3841 | 7.2229 |
| MSE | #2 | 0.7133 | **0.3556** | 0.2681 | 0.3094 | 1.3166 | 0.2198 | 1.8139 | 1.9589 | 501.9107 | 0.3669 | 0.7960 | 1.7284 | 6.3637 | 0.3988 | 7.0631 |
| MSE | #3 | 0.8980 | **0.4336** | 0.2802 | 0.2730 | 1.2797 | 0.2232 | 2.1691 | 2.2095 | 498.8953 | 0.3361 | 0.8597 | 0.9951 | 5.5366 | 0.3477 | 9.4455 |
| ERGAS | #1 | 1.2158 | 0.5425 | **0.5844** | 0.2484 | 0.6028 | 0.4392 | 1.3983 | 2.7799 | 459.9079 | 0.3722 | 0.7487 | 1.1659 | 4.8825 | 0.4616 | 7.5185 |
| ERGAS | #2 | 0.9138 | 0.5171 | **0.4522** | 0.2848 | 0.6697 | 0.3228 | 1.1081 | 2.4416 | 582.3030 | 0.4037 | 0.6655 | 0.7558 | 4.2325 | 0.4507 | 4.4916 |
| ERGAS | #3 | 0.5039 | 0.3179 | **0.6222** | 0.2647 | 0.7685 | 0.2797 | 1.1717 | 0.8862 | 584.8145 | 0.3647 | 1.3789 | 1.0076 | 3.8198 | 0.6156 | 3.4460 |
| LMSE | #1 | 1.4858 | 0.6519 | 0.3978 | **0.2188** | 1.0448 | 0.4618 | 1.6139 | 3.3507 | 574.0045 | 0.3336 | 1.0919 | 1.9148 | 15.1080 | 0.1902 | 20.1245 |
| LMSE | #2 | 1.8695 | 1.6399 | 0.3714 | **0.1897** | 1.0557 | 0.3469 | 1.1117 | 2.8227 | 545.9696 | 0.3522 | 0.7335 | 1.3298 | 5.5804 | 0.1848 | 13.7670 |
| LMSE | #3 | 1.2029 | 0.6439 | 0.4803 | **0.2153** | 1.4627 | 0.2439 | 1.4787 | 3.3618 | 554.4253 | 0.3496 | 1.0183 | 1.8368 | 10.8410 | 0.1725 | 22.0306 |
| SSIM | #1 | 0.5957 | 0.3038 | 0.2948 | 0.2752 | **0.5973** | 0.2623 | 0.9640 | 1.7032 | 497.3296 | 0.3903 | 0.4721 | 1.0877 | 2.9861 | 0.3798 | 3.7413 |
| SSIM | #2 | 1.1964 | 0.5642 | 0.4598 | 0.2600 | **0.3353** | 0.3725 | 1.4800 | 3.1742 | 574.9617 | 0.4140 | 0.8110 | 0.8910 | 5.3072 | 0.4857 | 4.4843 |
| SSIM | #3 | 1.4658 | 0.8499 | 0.3792 | 0.2842 | **0.5978** | 0.4012 | 1.1027 | 3.2309 | 602.5192 | 0.3435 | 0.9348 | 0.9902 | 4.4579 | 0.3279 | 3.6584 |
| SAM | #1 | 0.7013 | 0.4224 | 0.8771 | 0.1758 | 0.3257 | **1.9324** | 0.7315 | 1.0382 | 677.0960 | 0.3289 | 0.3854 | 0.6591 | 2.6480 | 0.5921 | 2.9105 |
| SAM | #2 | 0.6891 | 0.4288 | 0.8351 | 0.1810 | 0.3168 | **2.2833** | 0.4684 | 0.9346 | 714.3552 | 0.3268 | 0.3620 | 0.4994 | 1.6575 | 0.7855 | 3.2994 |
| SAM | #3 | 0.7427 | 0.5505 | 0.4691 | 0.2052 | 0.1287 | **0.9364** | 0.4423 | 0.8927 | 760.2247 | 0.3218 | 0.2132 | 0.5612 | 1.7311 | 0.7273 | 2.1770 |
| UIQI | #1 | 1.0824 | 0.6900 | 0.4232 | 0.2459 | 0.6983 | 0.4509 | **1.8638** | 2.8531 | 550.5976 | 0.3695 | 0.7811 | 0.8826 | 6.6194 | 0.4778 | 6.2939 |
| UIQI | #2 | 1.2412 | 0.6286 | 0.3408 | 0.3084 | 0.6960 | 0.2652 | **1.5376** | 3.1202 | 535.0966 | 0.3630 | 0.6775 | 1.9856 | 6.5678 | 0.2690 | 4.4192 |
| UIQI | #3 | 0.8941 | 0.5004 | 0.4870 | 0.3118 | 0.8281 | 0.4305 | **1.6163** | 1.7789 | 603.1164 | 0.3560 | 1.0463 | 1.2860 | 9.6276 | 0.4394 | 8.6953 |
| PSNR | #1 | 0.9197 | 0.4504 | 0.2381 | 0.2963 | 1.1886 | 0.2365 | 1.1974 | **2.8724** | 505.3604 | 0.3839 | 0.8133 | 1.0408 | 8.1238 | 0.4941 | 3.4624 |
| PSNR | #2 | 0.5153 | 0.3135 | 0.2325 | 0.2403 | 1.1238 | 0.2211 | 1.1020 | **1.5493** | 591.5266 | 0.3660 | 0.5662 | 0.8822 | 3.6393 | 0.3867 | 4.7538 |
| PSNR | #3 | 1.1567 | 0.6832 | 0.2829 | 0.2660 | 0.9963 | 0.2827 | 1.1753 | **3.3632** | 600.2663 | 0.3496 | 0.7303 | 1.4777 | 13.0807 | 0.2984 | 6.6034 |
| NIQE | #1 | 0.5750 | 0.3772 | 0.3852 | 0.2259 | 0.3181 | 0.3091 | 0.4648 | 0.9538 | **469.1237** | 0.3808 | 0.3447 | 0.4722 | 2.6973 | 0.2608 | 0.8463 |

**Table 3** (continued)

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| NIQE | #2 | 0.8965 | 0.5769 | 0.4964 | 0.2338 | 0.3391 | 0.3493 | 0.5570 | 1.7698 | **478.1001** | 0.3100 | 0.4934 | 0.4393 | 2.2677 | 0.4606 | 1.0515 |
| NIQE | #3 | 0.9331 | 0.6466 | 0.5303 | 0.2180 | 0.3759 | 0.4687 | 0.5896 | 2.2139 | **440.3989** | 0.3566 | 0.5245 | 0.5526 | 3.3147 | 0.3072 | 1.0375 |
| PIQE | #1 | 0.8586 | 0.6201 | 0.5236 | 0.2152 | 0.2591 | 0.5112 | 0.3232 | 1.2999 | 593.2118 | **0.3050** | 0.3304 | 0.3214 | 1.1800 | 0.5408 | 0.9733 |
| PIQE | #2 | 0.9796 | 0.6364 | 0.5452 | 0.2133 | 0.2511 | 0.4920 | 0.4291 | 1.5361 | 557.5049 | **0.3133** | 0.3719 | 0.4415 | 1.4736 | 0.4819 | 1.0978 |
| PIQE | #3 | 0.7390 | 0.4688 | 0.7427 | 0.2064 | 0.2399 | 0.5677 | 0.3593 | 1.0618 | 698.0475 | **0.2870** | 0.3538 | 0.4550 | 1.2402 | 0.7658 | 1.3564 |
| SNR | #1 | 0.8276 | 0.5643 | 0.3603 | 0.2343 | 0.3982 | 0.3858 | 0.4907 | 1.8413 | 450.8119 | 0.3508 | **0.4466** | 0.4824 | 2.1247 | 0.5016 | 1.2484 |
| SNR | #2 | 0.8370 | 0.4084 | 0.4593 | 0.2430 | 0.4094 | 0.6202 | 1.0993 | 2.2541 | 565.6280 | 0.3631 | **0.9544** | 0.7734 | 4.0976 | 0.2684 | 4.7700 |
| SNR | #3 | 0.8211 | 0.5006 | 0.4493 | 0.2385 | 0.5113 | 0.3415 | 0.6291 | 1.8523 | 467.3485 | 0.3790 | **0.5025** | 0.5760 | 2.2360 | 0.3607 | 2.8997 |
| VIF | #1 | 0.6714 | 0.4317 | 0.5798 | 0.2208 | 0.4009 | 0.6348 | 0.7192 | 1.0440 | 424.9089 | 0.3522 | 0.4711 | **0.6108** | 3.3613 | 0.2237 | 5.9688 |
| VIF | #2 | 1.0494 | 0.5811 | 0.4220 | 0.2128 | 0.3809 | 0.3257 | 0.6548 | 2.8369 | 487.6923 | 0.4284 | 0.5895 | **0.5828** | 2.8506 | 0.3715 | 4.0808 |
| VIF | #3 | 0.8406 | 0.4793 | 0.6306 | 0.1970 | 0.5365 | 0.5657 | 0.9145 | 1.5768 | 552.6804 | 0.3306 | 0.5678 | **0.8828** | 3.9277 | 0.2679 | 6.1867 |
| FSIM | #1 | 0.8951 | 0.6585 | 0.4577 | 0.2280 | 0.3357 | 0.5078 | 0.4766 | 1.2170 | 556.3074 | 0.3110 | 0.4439 | 0.4760 | **2.5250** | 0.3115 | 1.4470 |
| FSIM | #2 | 0.9118 | 0.5186 | 0.5261 | 0.2242 | 0.3354 | 0.3550 | 0.7492 | 2.4385 | 495.1349 | 0.3537 | 0.7114 | 0.6466 | **2.4064** | 0.2442 | 4.4061 |
| FSIM | #3 | 0.6693 | 0.3708 | 0.6627 | 0.2323 | 0.3701 | 0.5193 | 0.8089 | 1.4731 | 541.8485 | 0.3368 | 0.9247 | 0.7680 | **3.3704** | 0.3038 | 3.2417 |
| GMSD | #1 | 0.8375 | 0.6240 | 0.4790 | 0.2332 | 0.3513 | 0.4142 | 0.3474 | 1.4061 | 463.4918 | 0.3630 | 0.3529 | 0.3558 | 1.3979 | **0.3713** | 1.5127 |
| GMSD | #2 | 0.6641 | 0.4062 | 0.5522 | 0.2131 | 0.2974 | 0.5522 | 0.6432 | 1.0608 | 678.7662 | 0.3611 | 0.4102 | 0.6358 | 2.0827 | **0.4751** | 2.4128 |
| GMSD | #3 | 0.7503 | 0.4303 | 0.5533 | 0.2628 | 0.6016 | 0.2577 | 0.7298 | 1.4827 | 526.2806 | 0.3572 | 0.8128 | 0.8164 | 2.2581 | **0.5836** | 1.5561 |
| NCC | #1 | 0.7607 | 0.4486 | 0.5528 | 0.2147 | 0.4578 | 0.6019 | 0.8093 | 1.5522 | 608.7158 | 0.3805 | 0.4800 | 0.4593 | 2.6969 | 0.4814 | **4.0904** |
| NCC | #2 | 0.7245 | 0.5051 | 0.6183 | 0.2412 | 0.3331 | 0.5121 | 0.7512 | 1.0798 | 622.8823 | 0.3534 | 0.4824 | 0.5434 | 2.8956 | 0.5445 | **3.4276** |
| NCC | #3 | 0.7556 | 0.5013 | 0.4411 | 0.2283 | 0.2310 | 0.4581 | 1.3715 | 1.7651 | 646.4764 | 0.3453 | 0.4855 | 0.7628 | 7.8888 | 0.3935 | **3.7451** |

Bold values indicates the ability to distinguish between different images with the same content for the corresponding Normalized (Localized)-IQM values

**Table 4** *Localized-IQM* values obtained from experiments performed using Test Image #2 from the Animal Image Dataset

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| | | 20.0000 | 200.0000 | 50.0000 | 100.0000 | 0.8000 | 45.0000 | 0.8000 | 25.0000 | – | – | 10.0000 | 0.5000 | 0.9000 | 0.3000 | 0.9000 |
| MAE | #1 | **1.2767** | 0.8110 | 0.7210 | 0.2751 | 0.6307 | 0.6180 | 1.0707 | 3.0721 | 718.7118 | 0.3999 | 1.1684 | 0.8595 | 7.6497 | 0.3840 | 5.1160 |
| MAE | #2 | **0.6901** | 0.4018 | 0.7548 | 0.2604 | 1.6779 | 0.3242 | 1.6222 | 1.8198 | 788.0552 | 0.4761 | 1.2901 | 1.0328 | 4.5890 | 0.4170 | 5.2114 |
| MAE | #3 | **1.8168** | 1.1443 | 0.5727 | 0.2643 | 0.5471 | 0.4687 | 1.2715 | 4.4523 | 720.0538 | 0.4212 | 0.9424 | 1.4648 | 7.7437 | 0.5805 | 6.8628 |
| MSE | #1 | 0.8863 | **0.4655** | 0.4310 | 0.2760 | 1.2642 | 0.3333 | 2.0439 | 2.7646 | 717.7399 | 0.3941 | 1.1442 | 1.2284 | 9.9299 | 0.4929 | 10.6115 |
| MSE | #2 | 0.8324 | **0.4934** | 0.7785 | 0.2900 | 1.2675 | 0.4590 | 2.1929 | 2.6023 | 767.6004 | 0.4497 | 1.8682 | 1.4075 | 13.3122 | 0.6071 | 8.5517 |
| MSE | #3 | 1.0879 | **0.6804** | 0.5542 | 0.2849 | 0.9755 | 0.4911 | 2.0072 | 3.9274 | 749.7893 | 0.4493 | 1.2288 | 1.3427 | 11.3472 | 0.5185 | 16.0092 |
| ERGAS | #1 | 2.5292 | 1.3651 | **0.6792** | 0.2646 | 0.5809 | 0.8530 | 1.1122 | 7.0895 | 707.4227 | 0.4594 | 0.7840 | 1.3421 | 7.0999 | 0.3244 | 10.1590 |
| ERGAS | #2 | 1.8601 | 1.2784 | **0.6587** | 0.2824 | 0.3296 | 0.7801 | 1.0590 | 3.9388 | 688.8434 | 0.4610 | 0.6812 | 1.3130 | 8.8910 | 0.4002 | 12.3915 |
| ERGAS | #3 | 1.5178 | 0.7347 | **1.1704** | 0.2560 | 0.6007 | 0.7957 | 1.4824 | 3.4573 | 688.3063 | 0.4572 | 1.5579 | 0.8862 | 5.6230 | 0.6365 | 7.6699 |
| LMSE | #1 | 2.3528 | 1.8364 | 0.7588 | **0.2658** | 0.5386 | 0.6443 | 1.8287 | 3.6190 | 760.6371 | 0.4479 | 1.0999 | 1.7617 | 10.2519 | 0.2956 | 35.6109 |
| LMSE | #2 | 2.0102 | 0.7720 | 1.0711 | **0.2824** | 0.9228 | 0.5067 | 1.5054 | 5.2679 | 802.9915 | 0.4293 | 1.3283 | 1.9523 | 12.0870 | 0.2323 | 19.5201 |
| LMSE | #3 | 2.6627 | 1.1934 | 0.8773 | **0.2633** | 0.5886 | 1.0187 | 1.4759 | 9.8473 | 708.9691 | 0.4382 | 0.9148 | 1.6168 | 11.8798 | 0.2866 | 21.1506 |
| SSIM | #1 | 1.2643 | 0.7537 | 0.7214 | 0.2495 | **1.3937** | 0.4840 | 0.9425 | 2.8074 | 702.6964 | 0.4680 | 0.7633 | 1.0347 | 7.6150 | 0.5423 | 5.7342 |
| SSIM | #2 | 1.1247 | 0.5113 | 0.9513 | 0.2534 | **0.7842** | 0.5095 | 1.3637 | 2.6218 | 750.3055 | 0.4981 | 1.1404 | 1.1856 | 9.2959 | 0.5814 | 4.9986 |
| SSIM | #3 | 1.7313 | 0.9827 | 0.9110 | 0.2654 | **0.7455** | 0.5788 | 1.6647 | 2.5529 | 552.2380 | 0.5062 | 1.1381 | 1.0901 | 5.7354 | 0.6674 | 9.3441 |
| SAM | #1 | 1.7800 | 1.1492 | 0.8725 | 0.2364 | 0.1752 | **2.1693** | 0.7645 | 2.2882 | 782.3362 | 0.4405 | 0.3055 | 1.3972 | 2.5111 | 1.0712 | 3.4951 |
| SAM | #2 | 1.8122 | 1.1946 | 0.8585 | 0.2350 | 0.1669 | **1.6472** | 0.7040 | 2.5832 | 788.4151 | 0.3950 | 0.3359 | 1.1987 | 8.2189 | 0.5335 | 4.3812 |
| SAM | #3 | 1.4524 | 0.7212 | 1.3712 | 0.2413 | 0.1818 | **2.0164** | 0.7593 | 2.6101 | 808.2048 | 0.3974 | 0.5865 | 1.0815 | 5.5092 | 0.8891 | 7.0434 |
| UIQI | #1 | 1.5754 | 0.9386 | 0.6935 | 0.2508 | 0.4539 | 0.7482 | **1.5375** | 3.9661 | 742.0261 | 0.4552 | 0.8915 | 1.0131 | 7.3226 | 0.5003 | 9.4831 |
| UIQI | #2 | 1.0399 | 0.6402 | 0.6009 | 0.2648 | 0.7935 | 0.3876 | **1.5568** | 2.8850 | 557.4568 | 0.5153 | 1.0659 | 0.7182 | 6.7995 | 0.6626 | 7.2289 |
| UIQI | #3 | 0.9688 | 0.9215 | 0.5380 | 0.2751 | 0.9544 | 0.7184 | **1.4770** | 2.3593 | 626.9728 | 0.4209 | 0.8002 | 1.1834 | 17.5144 | 0.2830 | 9.3106 |
| PSNR | #1 | 1.1623 | 0.5339 | 1.0922 | 0.2553 | 0.9647 | 0.7216 | 2.6701 | **2.9605** | 752.2926 | 0.3971 | 2.8638 | 1.3295 | 8.1680 | 0.4746 | 8.5087 |
| PSNR | #2 | 1.2527 | 0.7263 | 0.6881 | 0.2811 | 1.2640 | 0.4188 | 3.2848 | **3.7088** | 589.4351 | 0.4358 | 1.7190 | 1.1978 | 11.5140 | 0.4825 | 13.6980 |
| PSNR | #3 | 0.8282 | 0.5667 | 0.6720 | 0.2541 | 0.9330 | 0.4211 | 1.3747 | **2.6264** | 801.1156 | 0.3831 | 1.6237 | 1.7495 | 7.4358 | 0.4395 | 9.0244 |
| NIQE | #1 | 2.3659 | 0.9047 | 0.9969 | 0.2526 | 0.6253 | 0.9991 | 1.0815 | 5.0593 | **606.9757** | 0.4421 | 1.0314 | 0.5559 | 3.9891 | 0.4278 | 5.2219 |
| NIQE | #2 | 1.9978 | 1.2152 | 0.6563 | 0.2470 | 0.4708 | 0.5072 | 0.5624 | 3.8470 | **688.6138** | 0.4692 | 0.4908 | 0.7971 | 3.5289 | 0.2377 | 5.4736 |

**Table 4** (continued)

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| NIQE | #3 | 1.6441 | 0.8267 | 1.0512 | 0.2412 | 0.7122 | 0.7451 | 0.9818 | 2.7964 | **666.7569** | 0.4370 | 0.8334 | 0.9768 | 4.8175 | 0.2422 | 11.1902 |
| PIQE | #1 | 1.4317 | 0.7560 | 1.1139 | 0.2353 | 0.4039 | 1.3007 | 0.5738 | 3.0908 | 687.2511 | **0.4451** | 0.5518 | 0.6780 | 3.5646 | 0.3472 | 1.9482 |
| PIQE | #2 | 1.2331 | 0.5753 | 1.7477 | 0.2340 | 0.4880 | 1.6199 | 0.6879 | 2.2437 | 666.6254 | **0.3959** | 0.7294 | 0.7025 | 2.2472 | 0.6131 | 0.7856 |
| PIQE | #3 | 0.9303 | 0.5671 | 1.3700 | 0.2371 | 0.3698 | 1.8238 | 0.5093 | 1.2209 | 656.8098 | **0.4047** | 0.4669 | 0.6400 | 4.5534 | 0.3767 | 1.8521 |
| SNR | #1 | 2.1889 | 1.3357 | 0.7119 | 0.2351 | 0.7421 | 0.8146 | 1.2433 | 3.7044 | 636.8865 | 0.4135 | **0.8695** | 0.5453 | 7.4441 | 0.1887 | 7.6207 |
| SNR | #2 | 1.2399 | 0.9686 | 0.5600 | 0.2514 | 0.5591 | 0.5517 | 1.0496 | 2.3834 | 551.8622 | 0.4688 | **0.6823** | 0.5479 | 4.9456 | 0.3288 | 7.0699 |
| SNR | #3 | 2.7549 | 2.0351 | 0.6703 | 0.2425 | 0.6819 | 0.7065 | 1.2380 | 4.8749 | 688.7638 | 0.4344 | **0.8307** | 0.7734 | 8.0655 | 0.1913 | 10.8367 |
| VIF | #1 | 1.8227 | 1.1280 | 0.6101 | 0.2575 | 0.5195 | 0.5142 | 1.0429 | 3.7415 | 599.2614 | 0.4604 | 0.7084 | **0.7197** | 4.4270 | 0.3882 | 6.0217 |
| VIF | #2 | 1.4753 | 0.7290 | 1.2101 | 0.2454 | 0.5831 | 1.8107 | 1.2124 | 3.0969 | 711.4317 | 0.4339 | 0.9084 | **1.2172** | 5.0519 | 0.5331 | 6.3951 |
| VIF | #3 | 1.8644 | 0.8925 | 1.0113 | 0.2295 | 0.7078 | 0.8132 | 1.2248 | 3.3698 | 694.6770 | 0.3995 | 0.8873 | **1.0126** | 7.4881 | 0.1827 | 13.4677 |
| FSIM | #1 | 1.4938 | 0.8152 | 1.1104 | 0.2436 | 0.6349 | 0.7138 | 0.7345 | 2.3024 | 684.0307 | 0.4250 | 0.6278 | 1.2109 | **3.9761** | 0.4011 | 8.4333 |
| FSIM | #2 | 2.5900 | 1.0635 | 0.9421 | 0.2487 | 0.4502 | 1.2104 | 0.7851 | 6.0909 | 717.0207 | 0.4995 | 0.6781 | 0.8807 | **3.6064** | 0.5570 | 10.8514 |
| FSIM | #3 | 1.7714 | 0.9104 | 0.9414 | 0.2615 | 0.7562 | 0.4958 | 1.1960 | 2.7070 | 608.6830 | 0.5393 | 0.9252 | 1.1569 | **3.1319** | 0.4901 | 6.8410 |
| GMSD | #1 | 1.2599 | 0.6939 | 1.1580 | 0.2342 | 0.4537 | 1.0419 | 0.4223 | 1.7417 | 859.2590 | 0.5395 | 0.4135 | 0.9829 | 2.2237 | **0.9483** | 4.5826 |
| GMSD | #2 | 1.5840 | 0.6025 | 1.6081 | 0.2496 | 0.5737 | 1.7816 | 0.8142 | 4.6519 | 653.2077 | 0.4914 | 0.7830 | 0.6189 | 2.3848 | **0.6964** | 2.1872 |
| GMSD | #3 | 1.5182 | 0.8402 | 1.0786 | 0.2335 | 0.6200 | 0.9562 | 0.4801 | 2.1860 | 757.1016 | 0.4877 | 0.5067 | 1.0627 | 2.4733 | **0.8203** | 5.7789 |
| NCC | #1 | 1.0308 | 0.6741 | 0.8306 | 0.2353 | 0.3673 | 1.0178 | 0.8003 | 2.1313 | 907.2933 | 0.3963 | 0.7689 | 0.7659 | 7.3474 | 0.4265 | **4.1658** |
| NCC | #2 | 0.8254 | 0.4902 | 1.2317 | 0.2457 | 0.6186 | 0.9928 | 0.8912 | 1.2714 | 827.4750 | 0.4783 | 0.7345 | 0.7312 | 3.3543 | 1.0381 | **2.7688** |
| NCC | #3 | 2.6745 | 0.8464 | 0.8674 | 0.2602 | 0.3697 | 0.9149 | 0.9517 | 4.2619 | 653.2172 | 0.4873 | 0.8307 | 0.6302 | 4.0002 | 0.7471 | **4.0329** |

Bold values indicates the ability to distinguish between different images with the same content for the corresponding Normalized (Localized)-IQM values

**Table 5** *Localized-IQM* values obtained from experiments performed using Test Image #3 from the Animal Image Dataset

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| | | 20.0000 | 200.0000 | 50.0000 | 100.0000 | 0.8000 | 45.0000 | 0.8000 | 25.0000 | – | – | 10.0000 | 0.5000 | 0.9000 | 0.3000 | 0.9000 |
| MAE | #1 | **0.7175** | 0.4408 | 1.0112 | 0.1485 | 1.5952 | 0.7177 | 1.0865 | 1.7927 | 759.0379 | 0.2549 | 0.8336 | 0.8685 | 14.6930 | 0.3352 | 0.5485 |
| MAE | #2 | **0.6766** | 0.4152 | 0.7118 | 0.1597 | 0.8263 | 0.6599 | 0.5502 | 2.0427 | 721.2796 | 0.3049 | 0.5310 | 0.3779 | 6.0122 | 0.3681 | 0.5187 |
| MAE | #3 | **1.0614** | 0.6625 | 0.5676 | 0.1589 | 1.1126 | 0.4898 | 0.8157 | 3.0225 | 771.7723 | 0.2829 | 0.7966 | 0.5288 | 7.6463 | 0.3538 | 2.2911 |
| MSE | #1 | 0.8845 | **0.3569** | 0.6110 | 0.1565 | 2.9349 | 0.5285 | 1.0996 | 2.3337 | 698.6678 | 0.3005 | 1.3480 | 0.5132 | 5.9518 | 0.4886 | 2.4350 |
| MSE | #2 | 0.7358 | **0.4540** | 0.4061 | 0.1603 | 1.9464 | 0.3441 | 1.0241 | 2.4473 | 766.4906 | 0.2893 | 0.7477 | 0.5187 | 10.6559 | 0.3092 | 2.4874 |
| MSE | #3 | 0.8734 | **0.6022** | 0.5421 | 0.1595 | 2.5692 | 0.4507 | 0.9815 | 2.4037 | 583.9645 | 0.2781 | 0.8682 | 0.7531 | 9.1015 | 0.3232 | 2.5752 |
| ERGAS | #1 | 1.3391 | 0.5643 | **0.5690** | 0.1626 | 1.8789 | 0.3985 | 0.9866 | 2.4332 | 623.6821 | 0.3384 | 1.0121 | 0.3052 | 5.2901 | 0.4032 | 2.3718 |
| ERGAS | #2 | 0.5039 | 0.2671 | **0.4624** | 0.1599 | 3.1152 | 0.4406 | 1.0055 | 1.6099 | 640.3078 | 0.3208 | 1.3294 | 0.3359 | 5.2688 | 0.5415 | 2.4183 |
| ERGAS | #3 | 1.2297 | 0.8474 | **0.3940** | 0.1584 | 2.3228 | 0.3597 | 0.9697 | 3.8977 | 647.3065 | 0.3040 | 0.9445 | 0.2899 | 5.9847 | 0.3348 | 2.4552 |
| LMSE | #1 | 0.9216 | 0.5001 | 1.1227 | **0.1539** | 1.5696 | 0.5862 | 0.8171 | 1.8140 | 711.0657 | 0.2766 | 0.8768 | 0.7835 | 6.1745 | 0.3926 | 1.2532 |
| LMSE | #2 | 0.9127 | 0.5376 | 0.5625 | **0.1632** | 1.6205 | 0.8175 | 0.6486 | 2.1060 | 702.8823 | 0.2982 | 1.0321 | 0.3890 | 9.3618 | 0.4771 | 1.9367 |
| LMSE | #3 | 1.4295 | 0.9662 | 0.7802 | **0.1566** | 0.6525 | 0.3546 | 0.6102 | 2.6836 | 814.2331 | 0.3246 | 0.5119 | 0.4969 | 3.5769 | 0.3923 | 1.9362 |
| SSIM | #1 | 1.5128 | 0.8808 | 0.3553 | 0.1618 | **3.1838** | 0.4193 | 1.0172 | 4.0866 | 602.8953 | 0.2755 | 1.0695 | 0.4524 | 8.4961 | 0.3012 | 2.3970 |
| SSIM | #2 | 0.7121 | 0.3532 | 0.6291 | 0.1634 | **1.7872** | 0.5389 | 1.1392 | 1.7791 | 662.5024 | 0.2734 | 1.0664 | 0.4107 | 10.5850 | 0.3269 | 2.4045 |
| SSIM | #3 | 1.2267 | 0.7375 | 0.3512 | 0.1618 | **3.0344** | 0.4499 | 1.0654 | 1.7127 | 647.2703 | 0.2616 | 1.2357 | 0.5955 | 13.3860 | 0.2276 | 2.3668 |
| SAM | #1 | 0.9942 | 0.4813 | 1.2946 | 0.1557 | 0.3371 | **1.2348** | 0.2802 | 1.6472 | 591.4216 | 0.3255 | 0.4316 | 0.5113 | 3.6123 | 0.5124 | 0.4180 |
| SAM | #2 | 1.2557 | 0.9862 | 0.7894 | 0.1591 | 0.3587 | **1.2310** | 0.2708 | 1.4782 | 556.9049 | 0.3278 | 0.1243 | 0.5105 | 1.7257 | 1.1095 | 0.5043 |
| SAM | #3 | 0.7351 | 0.4717 | 1.4476 | 0.1530 | 0.4849 | **0.7073** | 0.2087 | 0.9616 | 1202.7762 | 0.3183 | 0.1266 | 0.1723 | 0.8033 | 0.5732 | 0.4122 |
| UIQI | #1 | 0.7806 | 0.4433 | 0.3100 | 0.1709 | 3.4730 | 0.3435 | **1.1375** | 2.5007 | 680.7271 | 0.2761 | 0.8236 | 0.7505 | 18.7971 | 0.1778 | 2.4272 |
| UIQI | #2 | 0.9486 | 0.5541 | 0.3200 | 0.1705 | 3.2700 | 0.4617 | **1.1471** | 4.9561 | 653.8123 | 0.2580 | 0.9586 | 0.6906 | 11.8889 | 0.2047 | 2.4222 |
| UIQI | #3 | 0.7333 | 0.4245 | 0.3085 | 0.1699 | 3.7185 | 0.3664 | **1.1618** | 2.5509 | 690.3938 | 0.2812 | 0.8476 | 0.7096 | 20.9652 | 0.1848 | 2.4136 |
| PSNR | #1 | 1.4745 | 0.7327 | 0.3598 | 0.1637 | 2.4470 | 0.4586 | 0.5490 | **4.3246** | 682.8686 | 0.2943 | 0.9739 | 0.5116 | 3.6920 | 0.4739 | 1.1145 |
| PSNR | #2 | 0.8276 | 0.4772 | 0.4053 | 0.1589 | 2.5165 | 0.3620 | 0.5720 | **2.1161** | 595.7827 | 0.3385 | 1.0424 | 0.3227 | 3.9749 | 0.5444 | 1.7836 |
| PSNR | #3 | 0.5530 | 0.2959 | 0.5264 | 0.1530 | 3.1554 | 0.5826 | 0.7741 | **1.4525** | 619.5411 | 0.2890 | 1.6338 | 0.3980 | 5.8552 | 0.3642 | 1.8216 |
| NIQE | #1 | 0.9003 | 0.4775 | 1.2114 | 0.1473 | 0.5011 | 0.4130 | 0.7252 | 1.2672 | **750.5413** | 0.2633 | 0.5769 | 0.8233 | 2.0547 | 0.4474 | 1.0695 |
| NIQE | #2 | 1.1126 | 0.6901 | 0.8683 | 0.1489 | 0.3321 | 0.4806 | 0.4034 | 1.3808 | **659.9002** | 0.3562 | 0.3095 | 0.5042 | 2.2541 | 0.3155 | 0.3215 |

**Table 5** (continued)

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| NIQE | #3 | 0.9407 | 0.4796 | 0.9951 | 0.1498 | 0.5807 | 0.5972 | 0.6381 | 1.4762 | **687.2609** | 0.3003 | 0.5580 | 0.4168 | 3.8995 | 0.3306 | 1.0828 |
| PIQE | #1 | 0.8159 | 0.5694 | 1.1017 | 0.1474 | 0.2586 | 0.5080 | 0.3411 | 1.5458 | 825.8371 | **0.2734** | 0.2838 | 0.3624 | 2.5544 | 0.3605 | 0.3608 |
| PIQE | #2 | 0.8301 | 0.4805 | 0.8950 | 0.1460 | 0.2139 | 0.4981 | 0.3583 | 0.9302 | 740.8802 | **0.2599** | 0.3282 | 0.3337 | 1.5612 | 0.5015 | 0.5801 |
| PIQE | #3 | 0.8826 | 0.4857 | 1.2280 | 0.1455 | 0.2926 | 0.4881 | 0.3565 | 1.0772 | 973.7577 | **0.2646** | 0.3640 | 0.6095 | 1.4671 | 0.4214 | 0.8966 |
| SNR | #1 | 0.8046 | 0.4828 | 0.7719 | 0.1544 | 0.9049 | 0.4897 | 0.5410 | 1.8831 | 646.7556 | 0.3053 | **0.6673** | 0.3663 | 3.6241 | 0.3307 | 0.6971 |
| SNR | #2 | 0.5963 | 0.3791 | 0.6515 | 0.1575 | 0.6322 | 0.3244 | 0.4452 | 1.7868 | 614.0573 | 0.3208 | **0.4166** | 0.4189 | 3.3153 | 0.4234 | 0.6606 |
| SNR | #3 | 0.7072 | 0.4574 | 0.6288 | 0.1545 | 0.7060 | 0.4404 | 0.6830 | 1.8733 | 706.4569 | 0.3110 | **0.4789** | 0.4052 | 5.5970 | 0.2724 | 1.0838 |
| VIF | #1 | 0.8141 | 0.5624 | 0.8487 | 0.1511 | 0.3819 | 0.4736 | 0.4829 | 1.7588 | 825.5509 | 0.2969 | 0.3429 | **0.3922** | 3.6775 | 0.3988 | 0.7429 |
| VIF | #2 | 1.0271 | 0.7134 | 0.6675 | 0.1513 | 0.3096 | 0.4229 | 0.3672 | 2.0187 | 742.7014 | 0.3083 | 0.2967 | **0.4661** | 2.7077 | 0.2635 | 0.7258 |
| VIF | #3 | 1.3520 | 0.7812 | 0.8209 | 0.1521 | 0.8008 | 0.5328 | 0.7413 | 2.4019 | 810.6348 | 0.2729 | 0.6121 | **0.6938** | 4.6593 | 0.2381 | 1.0355 |
| FSIM | #1 | 1.0054 | 0.6856 | 0.9359 | 0.1518 | 0.4447 | 0.4508 | 0.4465 | 1.6305 | 612.9276 | 0.3022 | 0.4696 | 0.4589 | **3.1051** | 0.3695 | 0.4388 |
| FSIM | #2 | 0.9352 | 0.6139 | 1.0774 | 0.1523 | 0.4557 | 0.8041 | 0.5927 | 1.7794 | 931.7832 | 0.3089 | 0.4214 | 0.3603 | **3.3518** | 0.3681 | 1.1965 |
| FSIM | #3 | 1.3151 | 0.8944 | 1.1727 | 0.1476 | 0.3099 | 1.2815 | 0.3875 | 1.6400 | 785.3310 | 0.3052 | 0.3558 | 0.7009 | **2.4308** | 0.3537 | 1.1149 |
| GMSD | #1 | 0.8757 | 0.4979 | 1.4493 | 0.1486 | 0.2172 | 0.6247 | 0.2211 | 1.2986 | 976.8377 | 0.3311 | 0.2394 | 0.2932 | 1.6564 | **0.8508** | 0.3220 |
| GMSD | #2 | 0.8395 | 0.4549 | 1.5786 | 0.1504 | 0.3308 | 0.4376 | 0.2669 | 1.2697 | 697.0237 | 0.3488 | 0.4127 | 0.4148 | 1.8647 | **0.7208** | 0.4693 |
| GMSD | #3 | 1.1083 | 0.5363 | 0.9148 | 0.1518 | 0.3270 | 0.7395 | 0.3407 | 1.9294 | 712.0541 | 0.3738 | 0.3516 | 0.5486 | 2.2699 | **0.4046** | 0.4381 |
| NCC | #1 | 0.7940 | 0.4080 | 0.6670 | 0.1523 | 0.7091 | 0.6745 | 0.4876 | 2.0591 | 680.1015 | 0.2816 | 0.7261 | 0.3853 | 4.6163 | 0.6090 | **1.1485** |
| NCC | #2 | 0.6925 | 0.4264 | 0.5873 | 0.1610 | 0.8181 | 0.4089 | 0.4712 | 1.9700 | 623.3235 | 0.2895 | 0.5000 | 0.3780 | 5.3950 | 0.4029 | **0.4568** |
| NCC | #3 | 0.8015 | 0.4223 | 0.6986 | 0.1562 | 2.0394 | 0.7175 | 0.5491 | 1.8930 | 567.5703 | 0.2952 | 1.0613 | 0.4929 | 3.8104 | 0.5756 | **1.8228** |

Bold values indicates the ability to distinguish between different images with the same content for the corresponding Normalized (Localized)-IQM values

**Table 6** *Localized-IQM* values obtained from experiments performed using Test Image #4 from the Animal Image Dataset

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| | | 20.0000 | 200.0000 | 50.0000 | 100.0000 | 0.8000 | 45.0000 | 0.8000 | 25.0000 | – | – | 10.0000 | 0.5000 | 0.9000 | 0.3000 | 0.9000 |
| MAE | #1 | **1.2027** | 1.0291 | 0.2679 | 0.1861 | 0.6481 | 0.3255 | 0.8318 | 2.9918 | 362.5981 | 0.3984 | 0.5139 | 0.9892 | 3.3657 | 0.3130 | 8.0008 |
| MAE | #2 | **0.6159** | 0.4749 | 0.2386 | 0.1928 | 0.8197 | 0.2643 | 0.7486 | 2.1220 | 349.4742 | 0.4163 | 0.4329 | 0.7946 | 3.2448 | 0.2494 | 6.5446 |
| MAE | #3 | **0.6329** | 0.3913 | 0.2779 | 0.1962 | 0.7503 | 0.2998 | 0.9573 | 2.3506 | 338.6017 | 0.4281 | 0.4695 | 0.8766 | 4.2538 | 0.3990 | 6.1373 |
| MSE | #1 | 0.7978 | **0.5416** | 0.2761 | 0.2254 | 1.2453 | 0.2870 | 1.1150 | 3.0826 | 358.4152 | 0.4102 | 0.8039 | 2.0839 | 4.6700 | 0.3328 | 11.5949 |
| MSE | #2 | 0.8536 | **0.5684** | 0.2485 | 0.2077 | 1.3692 | 0.2611 | 1.1869 | 2.8465 | 351.9923 | 0.4094 | 0.5787 | 1.8309 | 4.3730 | 0.3376 | 16.7261 |
| MSE | #3 | 0.9898 | **0.5420** | 0.2209 | 0.2251 | 1.1542 | 0.2233 | 0.9195 | 4.0983 | 379.4554 | 0.4025 | 0.5855 | 1.9850 | 4.7273 | 0.3700 | 8.3882 |
| ERGAS | #1 | 0.8108 | 0.6151 | **0.3045** | 0.2149 | 0.4026 | 0.3751 | 0.5946 | 2.0617 | 394.4891 | 0.3975 | 0.3939 | 1.1489 | 4.4707 | 0.4011 | 4.5563 |
| ERGAS | #2 | 0.6248 | 0.4375 | **0.3575** | 0.1966 | 0.5677 | 0.2824 | 0.6375 | 1.3508 | 400.9289 | 0.4140 | 0.5365 | 0.8550 | 2.5100 | 0.3364 | 6.6494 |
| ERGAS | #3 | 0.3800 | 0.2391 | **0.7683** | 0.1912 | 0.2742 | 0.7913 | 1.5250 | 1.2117 | 358.6254 | 0.3945 | 1.3405 | 0.8505 | 4.7883 | 0.3231 | 7.1687 |
| LMSE | #1 | 0.6558 | 0.4927 | 0.3350 | **0.2111** | 0.8663 | 0.2575 | 0.9773 | 1.7152 | 365.3441 | 0.3802 | 0.6731 | 1.1240 | 3.7846 | 0.2254 | 12.3212 |
| LMSE | #2 | 0.7020 | 0.4611 | 0.3396 | **0.2063** | 0.8412 | 0.3921 | 0.8251 | 2.1611 | 374.3128 | 0.4030 | 0.5982 | 1.0316 | 3.0558 | 0.1991 | 16.9576 |
| LMSE | #3 | 0.8830 | 0.6548 | 0.2894 | **0.2261** | 0.9582 | 0.2623 | 0.8432 | 3.1221 | 376.4389 | 0.4022 | 0.5924 | 0.9455 | 3.8673 | 0.1976 | 13.6048 |
| SSIM | #1 | 0.8846 | 0.5334 | 0.2876 | 0.2006 | **0.6407** | 0.3372 | 0.6339 | 2.7136 | 351.6422 | 0.4168 | 0.4740 | 0.8392 | 2.9747 | 0.2390 | 5.1160 |
| SSIM | #2 | 0.5751 | 0.3910 | 0.3858 | 0.1904 | **0.8999** | 0.3949 | 0.7450 | 1.2926 | 377.8998 | 0.4142 | 0.5304 | 1.1159 | 3.6640 | 0.2817 | 6.9047 |
| SSIM | #3 | 0.7689 | 0.4657 | 0.3494 | 0.1900 | **0.6129** | 0.4073 | 0.7565 | 2.4282 | 403.8611 | 0.4171 | 0.5946 | 1.3518 | 3.2398 | 0.3833 | 4.5205 |
| SAM | #1 | 0.4404 | 0.2856 | 0.6293 | 0.1711 | 0.2717 | **1.6594** | 0.5729 | 0.7110 | 433.1954 | 0.3618 | 0.2988 | 1.1482 | 2.2275 | 0.8538 | 5.4458 |
| SAM | #2 | 0.4358 | 0.2854 | 0.6053 | 0.1696 | 0.2782 | **1.7862** | 0.5233 | 0.6869 | 413.9083 | 0.3881 | 0.3074 | 0.5417 | 2.9248 | 0.6140 | 4.8827 |
| SAM | #3 | 0.4682 | 0.3135 | 0.5305 | 0.1677 | 0.2893 | **1.4708** | 0.4684 | 0.6776 | 404.1548 | 0.4098 | 0.2398 | 0.5080 | 2.7170 | 0.6878 | 3.1297 |
| UIQI | #1 | 1.0837 | 0.7242 | 0.2612 | 0.2188 | 0.9068 | 0.3072 | **1.2728** | 3.8609 | 382.9183 | 0.4230 | 0.5919 | 1.1969 | 4.2817 | 0.5078 | 9.0422 |
| UIQI | #2 | 0.8132 | 0.5791 | 0.2785 | 0.2192 | 0.4301 | 0.2524 | **1.1376** | 2.9638 | 367.4712 | 0.3671 | 0.5989 | 1.4708 | 4.3407 | 0.4161 | 10.1911 |
| UIQI | #3 | 0.8086 | 0.6519 | 0.2338 | 0.1989 | 0.7980 | 0.3366 | **0.7450** | 2.4432 | 368.6480 | 0.4100 | 0.4772 | 1.0485 | 3.2033 | 0.2204 | 7.0604 |
| PSNR | #1 | 0.7483 | 0.5219 | 0.2602 | 0.2185 | 0.8891 | 0.2992 | 0.9832 | **2.7715** | 385.9102 | 0.4218 | 0.6031 | 1.6031 | 4.6069 | 0.5458 | 7.6409 |
| PSNR | #2 | 0.7696 | 0.4781 | 0.2267 | 0.2184 | 0.8007 | 0.2773 | 1.0136 | **2.8844** | 371.7332 | 0.4015 | 0.5255 | 1.4799 | 3.9377 | 0.3213 | 9.6758 |
| PSNR | #3 | 1.0496 | 0.5832 | 0.2456 | 0.2397 | 0.8080 | 0.2766 | 1.2121 | **4.7930** | 369.7465 | 0.4370 | 0.5567 | 1.5355 | 6.2055 | 0.4020 | 13.8346 |
| NIQE | #1 | 0.5777 | 0.3067 | 0.3876 | 0.1759 | 0.2334 | 0.7361 | 0.6046 | 2.2486 | **366.2902** | 0.4960 | 0.6472 | 0.7240 | 2.1130 | 0.4147 | 3.6252 |
| NIQE | #2 | 0.3655 | 0.2464 | 0.5852 | 0.1758 | 0.1557 | 0.8308 | 0.6584 | 0.8252 | **401.6965** | 0.4312 | 0.6912 | 0.6530 | 2.5295 | 0.4695 | 1.3281 |

**Table 6** (continued)

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| NIQE | #3 | 0.9607 | 0.6610 | 0.2556 | 0.1913 | 0.3134 | 0.3039 | 0.5377 | 3.0402 | **388.9481** | 0.4687 | 0.4334 | 0.5652 | 2.2192 | 0.4542 | 3.0226 |
| PIQE | #1 | 0.4201 | 0.2948 | 0.5401 | 0.1694 | 0.2053 | 0.7574 | 0.3115 | 0.5532 | 467.2847 | **0.4855** | 0.3711 | 0.4632 | 1.2762 | 0.4060 | 2.7542 |
| PIQE | #2 | 0.5757 | 0.3715 | 0.3815 | 0.1733 | 0.2460 | 0.5280 | 0.3954 | 1.4182 | 397.7901 | **0.4464** | 0.3923 | 0.3811 | 1.2663 | 0.3062 | 0.5754 |
| PIQE | #3 | 0.4036 | 0.2787 | 0.5409 | 0.1743 | 0.3861 | 0.3286 | 0.3688 | 0.5310 | 433.2134 | **0.4276** | 0.4916 | 0.5258 | 1.4103 | 0.3536 | 1.9456 |
| SNR | #1 | 0.7087 | 0.4785 | 0.3259 | 0.1815 | 0.3774 | 0.3162 | 0.4705 | 1.5168 | 435.7203 | 0.4671 | **0.3876** | 0.8688 | 2.3159 | 0.4361 | 3.4698 |
| SNR | #2 | 0.7318 | 0.4861 | 0.2687 | 0.1881 | 0.3853 | 0.3093 | 0.3685 | 1.9013 | 423.3874 | 0.4951 | **0.3124** | 0.3884 | 1.1099 | 0.3160 | 1.7912 |
| SNR | #3 | 0.6729 | 0.3997 | 0.3336 | 0.1785 | 0.4197 | 0.4165 | 0.4132 | 1.8110 | 401.8457 | 0.4928 | **0.4882** | 0.5570 | 1.9559 | 0.4952 | 3.4363 |
| VIF | #1 | 0.7157 | 0.4732 | 0.3592 | 0.1737 | 0.4366 | 0.4759 | 0.3932 | 2.0164 | 390.1299 | 0.3962 | 0.4018 | **0.3636** | 1.7053 | 0.2310 | 2.0784 |
| VIF | #2 | 0.8677 | 0.4856 | 0.3573 | 0.1817 | 0.7627 | 0.5458 | 0.5290 | 2.2450 | 387.9512 | 0.4286 | 0.4764 | **0.6436** | 1.8095 | 0.2318 | 5.0007 |
| VIF | #3 | 0.9525 | 0.7344 | 0.3056 | 0.1851 | 0.6646 | 0.3053 | 0.4432 | 2.7217 | 402.4719 | 0.3784 | 0.3982 | **0.4692** | 1.4898 | 0.3362 | 2.2531 |
| FSIM | #1 | 0.8313 | 0.5109 | 0.3154 | 0.1778 | 0.2486 | 0.5402 | 0.4617 | 2.4821 | 408.8575 | 0.4021 | 0.4062 | 0.5788 | **2.0079** | 0.3643 | 1.8667 |
| FSIM | #2 | 0.7992 | 0.5701 | 0.2853 | 0.1914 | 0.2398 | 0.3299 | 0.4350 | 2.1784 | 388.0730 | 0.4432 | 0.3720 | 0.5064 | **1.8837** | 0.3529 | 2.9511 |
| FSIM | #3 | 0.7033 | 0.4291 | 0.3242 | 0.1840 | 0.5792 | 0.3713 | 0.5715 | 1.9751 | 418.5213 | 0.4286 | 0.4689 | 0.7484 | **2.1887** | 0.2976 | 5.9820 |
| GMSD | #1 | 0.7937 | 0.5101 | 0.2495 | 0.1808 | 0.3941 | 0.3255 | 0.5063 | 2.1931 | 378.9265 | 0.4661 | 0.3554 | 0.7537 | 1.6198 | **0.4469** | 4.2987 |
| GMSD | #2 | 0.6185 | 0.3782 | 0.4422 | 0.1741 | 0.3369 | 0.6565 | 0.4136 | 1.2504 | 407.8948 | 0.4241 | 0.3451 | 0.5634 | 1.6080 | **0.3578** | 3.9047 |
| GMSD | #3 | 0.7319 | 0.5219 | 0.3330 | 0.1900 | 0.4290 | 0.4861 | 0.3403 | 1.5555 | 382.6021 | 0.4823 | 0.3426 | 0.4423 | 1.3005 | **0.3359** | 3.1017 |
| NCC | #1 | 0.4623 | 0.2842 | 0.4503 | 0.1937 | 0.1782 | 0.6357 | 0.5843 | 1.3960 | 394.3997 | 0.3950 | 0.4511 | 0.9203 | 5.6834 | 0.4733 | **4.4601** |
| NCC | #2 | 0.5109 | 0.3533 | 0.4209 | 0.1756 | 0.5976 | 0.7648 | 0.4613 | 0.8048 | 415.2549 | 0.4377 | 0.3552 | 0.7071 | 2.2629 | 0.4249 | **4.0317** |
| NCC | #3 | 0.4341 | 0.2997 | 0.4343 | 0.2055 | 0.2518 | 0.5439 | 0.6823 | 0.8449 | 406.1148 | 0.4499 | 0.4458 | 0.7994 | 3.4467 | 0.7415 | **3.9976** |

Bold values indicates the ability to distinguish between different images with the same content for the corresponding Normalized (Localized)-IQM values

**Table 7** *Localized-IQM* values obtained from experiments performed using Test Image #1 from the Human Image Dataset

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| | | 20.0000 | 200.0000 | 50.0000 | 100.0000 | 0.8000 | 45.0000 | 0.8000 | 25.0000 | – | – | 10.0000 | 0.5000 | 0.9000 | 0.3000 | 0.9000 |
| MAE | #1 | **0.4906** | 0.2433 | 0.4519 | 0.2098 | 0.3418 | 0.5304 | 0.6523 | 1.3811 | 847.5678 | 0.2968 | 0.6540 | 0.4058 | 3.4980 | 0.3052 | 1.5246 |
| MAE | #2 | **0.8224** | 0.4299 | 0.3299 | 0.2029 | 0.3793 | 0.3889 | 0.6940 | 2.2332 | 767.2379 | 0.2835 | 0.4801 | 0.2374 | 4.0084 | 0.3916 | 1.3752 |
| MAE | #3 | **1.0614** | 0.5150 | 0.2844 | 0.1959 | 0.4549 | 0.2692 | 0.6841 | 1.7193 | 809.6880 | 0.2746 | 0.4947 | 0.2696 | 3.6847 | 0.2990 | 0.8184 |
| MSE | #1 | 0.6475 | **0.3762** | 0.2183 | 0.1925 | 0.3033 | 0.2037 | 0.5791 | 1.7657 | 904.9590 | 0.2855 | 0.4613 | 0.3755 | 4.8484 | 0.2792 | 1.1243 |
| MSE | #2 | 0.5664 | **0.3442** | 0.2690 | 0.1927 | 0.7140 | 0.2694 | 0.7072 | 1.2215 | 719.1841 | 0.2490 | 0.5577 | 0.5659 | 5.1036 | 0.1905 | 2.5969 |
| MSE | #3 | 0.4329 | **0.2469** | 0.2162 | 0.2029 | 0.4532 | 0.2425 | 0.6248 | 1.2135 | 1001.1033 | 0.3640 | 0.5495 | 0.2979 | 3.6553 | 0.3073 | 1.0492 |
| ERGAS | #1 | 0.7961 | 0.3814 | **0.3734** | 0.2029 | 0.3120 | 0.2642 | 0.6648 | 1.6942 | 866.3397 | 0.2677 | 0.4582 | 0.5532 | 4.0776 | 0.2226 | 1.8986 |
| ERGAS | #2 | 0.9037 | 0.5245 | **0.2968** | 0.2003 | 0.2333 | 0.2830 | 0.4909 | 2.2801 | 883.7980 | 0.2785 | 0.3655 | 0.2915 | 3.0517 | 0.3510 | 1.8175 |
| ERGAS | #3 | 0.8730 | 0.3970 | **0.4108** | 0.2007 | 0.1750 | 0.6436 | 0.4728 | 1.6474 | 979.6727 | 0.3545 | 0.5031 | 0.3366 | 3.5140 | 0.4766 | 0.6275 |
| LMSE | #1 | 0.5146 | 0.3843 | 0.3217 | **0.1885** | 0.3925 | 0.4920 | 0.3669 | 0.9754 | 849.0387 | 0.2700 | 0.2735 | 0.2611 | 2.0698 | 0.3098 | 1.2091 |
| LMSE | #2 | 0.4391 | 0.3032 | 0.2451 | **0.1735** | 0.4122 | 0.2682 | 0.5926 | 1.2457 | 1031.1629 | 0.2948 | 0.3431 | 0.2834 | 3.0768 | 0.3018 | 1.3679 |
| LMSE | #3 | 0.4578 | 0.2814 | 0.3365 | **0.1785** | 0.5879 | 0.2550 | 0.7599 | 1.0582 | 774.2321 | 0.3529 | 0.4299 | 0.3165 | 3.5916 | 0.4795 | 0.9046 |
| SSIM | #1 | 0.7038 | 0.3712 | 0.4695 | 0.1938 | **0.5243** | 0.4434 | 0.7882 | 1.5654 | 819.7260 | 0.2821 | 0.6198 | 0.2995 | 4.8210 | 0.3373 | 5.1959 |
| SSIM | #2 | 0.5316 | 0.3041 | 0.3030 | 0.2059 | **0.4532** | 0.2783 | 0.7005 | 1.2975 | 799.8760 | 0.2832 | 0.4271 | 0.3740 | 3.9784 | 0.2938 | 1.9922 |
| SSIM | #3 | 0.7441 | 0.5011 | 0.3008 | 0.1965 | **0.3006** | 0.3624 | 0.6520 | 1.6945 | 900.7335 | 0.2771 | 0.4310 | 0.2695 | 4.7838 | 0.2525 | 1.8177 |
| SAM | #1 | 0.6321 | 0.4098 | 0.4698 | 0.1427 | 0.2830 | **1.3403** | 0.1936 | 0.6595 | 937.4774 | 0.4350 | 0.1242 | 0.3525 | 1.4528 | 0.6959 | 0.4292 |
| SAM | #2 | 0.6315 | 0.4595 | 0.4771 | 0.1575 | 0.1280 | **0.8045** | 0.3422 | 0.8231 | 841.3280 | 0.3284 | 0.1682 | 0.2740 | 1.9761 | 0.3851 | 0.8951 |
| SAM | #3 | 0.5508 | 0.3290 | 0.6644 | 0.1594 | 0.2193 | **0.9845** | 0.4365 | 0.7538 | 899.7972 | 0.2830 | 0.2646 | 0.2881 | 2.6500 | 0.5577 | 2.1657 |
| UIQI | #1 | 0.4097 | 0.2892 | 0.2828 | 0.1836 | 0.3688 | 0.3142 | **0.8274** | 0.8559 | 759.8806 | 0.2539 | 0.7210 | 0.4283 | 4.4064 | 0.1960 | 5.3412 |
| UIQI | #2 | 0.6663 | 0.4436 | 0.2450 | 0.1735 | 0.5752 | 0.2784 | **0.8033** | 1.8132 | 777.1187 | 0.2781 | 0.4909 | 0.6142 | 3.8450 | 0.2462 | 7.7945 |
| UIQI | #3 | 0.7625 | 0.3812 | 0.2904 | 0.1856 | 0.5674 | 0.2657 | **0.9002** | 1.6648 | 754.7137 | 0.2474 | 0.6844 | 0.7119 | 6.0062 | 0.1969 | 8.2798 |
| PSNR | #1 | 0.7317 | 0.4419 | 0.2212 | 0.1970 | 0.6432 | 0.2045 | 0.7075 | **1.7570** | 887.4524 | 0.2879 | 0.5454 | 0.3111 | 3.6794 | 0.2564 | 1.6003 |
| PSNR | #2 | 0.4093 | 0.2426 | 0.2732 | 0.1733 | 0.4369 | 0.3038 | 0.7740 | **1.1088** | 788.5634 | 0.2675 | 0.7652 | 0.4322 | 3.8827 | 0.1933 | 5.3331 |
| PSNR | #3 | 0.5933 | 0.3161 | 0.2115 | 0.1957 | 0.2828 | 0.1940 | 0.5987 | **1.7182** | 961.8146 | 0.3278 | 0.3957 | 0.2603 | 3.0066 | 0.3224 | 1.4453 |
| NIQE | #1 | 0.9322 | 0.4489 | 0.4473 | 0.1647 | 0.3124 | 0.4618 | 0.3405 | 1.9614 | **797.5412** | 0.2786 | 0.3159 | 0.3664 | 1.5293 | 0.2168 | 1.0203 |
| NIQE | #2 | 0.6185 | 0.3725 | 0.5490 | 0.1790 | 0.1313 | 0.3898 | 0.2757 | 0.9398 | **827.5202** | 0.2488 | 0.2648 | 0.3016 | 1.4735 | 0.3332 | 0.3666 |

**Table 7** (continued)

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| NIQE | #3 | 0.8417 | 0.5002 | 0.3589 | 0.1825 | 0.2892 | 0.2354 | 0.3474 | 1.5419 | **694.7353** | 0.2582 | 0.3126 | 0.3355 | 1.5069 | 0.2094 | 1.0064 |
| PIQE | #1 | 0.5577 | 0.3392 | 0.7633 | 0.1750 | 0.1793 | 0.7046 | 0.2295 | 0.7324 | 603.6876 | **0.2446** | 0.2314 | 0.2577 | NaN | 0.3632 | 0.3690 |
| PIQE | #2 | 0.5456 | 0.3570 | 0.7179 | 0.1623 | 0.1809 | 0.6278 | 0.2005 | 0.6829 | 802.5678 | **0.2508** | 0.2152 | 0.2318 | 1.1729 | 0.3607 | 0.2613 |
| PIQE | #3 | 0.6322 | 0.4057 | 0.5739 | 0.1680 | 0.2011 | 0.4160 | 0.2371 | 0.8993 | 746.4624 | **0.2465** | 0.2374 | 0.2650 | 1.2603 | 0.3176 | 0.3410 |
| SNR | #1 | 1.0396 | 0.6350 | 0.3045 | 0.1716 | 0.3004 | 0.2902 | 0.3904 | 2.7497 | 820.2317 | 0.2663 | **0.3204** | 0.3219 | 2.0212 | 0.1846 | 1.5428 |
| SNR | #2 | 0.8903 | 0.5489 | 0.2738 | 0.1830 | 0.3038 | 0.2379 | 0.3876 | 1.9747 | 772.7059 | 0.2552 | **0.3160** | 0.3477 | 1.8100 | 0.1983 | 1.2576 |
| SNR | #3 | 0.4170 | 0.3052 | 0.2662 | 0.1861 | 0.2773 | 0.2838 | 0.3589 | 0.9689 | 877.8056 | 0.2445 | **0.2677** | 0.2718 | 2.0489 | 0.2780 | 0.5119 |
| VIF | #1 | 0.7852 | 0.4511 | 0.4219 | 0.1774 | 0.3160 | 0.3794 | 0.3755 | 1.7843 | 1081.6998 | 0.3084 | 0.3140 | **0.2877** | 1.9937 | 0.3226 | 0.5376 |
| VIF | #2 | 0.4684 | 0.2546 | 0.5095 | 0.1741 | 0.3503 | 0.5805 | 0.3958 | 0.9467 | 1027.5821 | 0.3261 | 0.4628 | **0.2699** | 2.0713 | 0.2522 | 1.0903 |
| VIF | #3 | 0.7870 | 0.5865 | 0.3827 | 0.1848 | 0.2924 | 0.4391 | 0.3217 | 1.5273 | 941.0133 | 0.2650 | 0.2758 | **0.2587** | 1.3829 | 0.2662 | 0.6603 |
| FSIM | #1 | 0.8005 | 0.6062 | 0.4011 | 0.1894 | 0.3013 | 0.4583 | 0.2924 | 1.2848 | 971.9076 | 0.2673 | 0.2907 | 0.2431 | **1.6939** | 0.2531 | 0.3953 |
| FSIM | #2 | 0.5517 | 0.3346 | 0.2677 | 0.1889 | 0.3393 | 0.2629 | 0.3905 | 1.2944 | 1006.8656 | 0.2879 | 0.3290 | 0.2670 | **1.8460** | 0.2379 | 1.3634 |
| FSIM | #3 | 0.6437 | 0.4207 | 0.5373 | 0.1819 | 0.2845 | 0.3461 | 0.2664 | 0.8816 | 760.4450 | 0.2517 | 0.2510 | 0.3086 | **1.5172** | 0.2781 | 1.7259 |
| GMSD | #1 | 0.6655 | 0.4304 | 0.3032 | 0.1890 | 0.2703 | 0.2706 | 0.3285 | 1.1932 | 952.5403 | 0.3221 | 0.3027 | 0.2650 | 1.6718 | **0.3119** | 0.5748 |
| GMSD | #2 | 0.4773 | 0.2988 | 0.9606 | 0.1672 | 0.2645 | 0.5619 | 0.2614 | 0.5949 | 926.2608 | 0.2523 | 0.2848 | 0.3156 | 1.4761 | **0.2785** | 1.0707 |
| GMSD | #3 | 0.9596 | 0.6413 | 0.3385 | 0.1670 | 0.1349 | 0.3266 | 0.3535 | 1.4337 | 747.7249 | 0.2575 | 0.3231 | 0.3454 | 1.6666 | **0.2022** | 1.3508 |
| NCC | #1 | 0.6281 | 0.4221 | 0.4832 | 0.1787 | 0.3274 | 0.6665 | 0.2557 | 0.8250 | 990.2931 | 0.3222 | 0.2410 | 0.2663 | 1.6816 | 0.6462 | **0.8564** |
| NCC | #2 | 0.4218 | 0.2839 | 0.9696 | 0.1903 | 0.1649 | 0.5794 | 0.4803 | 0.5724 | 926.8912 | 0.2850 | 0.6218 | 0.4161 | 2.3921 | 0.5255 | **0.8779** |
| NCC | #3 | 0.8209 | 0.4182 | 0.3353 | 0.1993 | 0.1267 | 0.3473 | 0.4041 | 1.3837 | 948.7083 | 0.2857 | 0.3112 | 0.2525 | 1.8835 | 0.3346 | **0.8538** |

Bold values indicates the ability to distinguish between different images with the same content for the corresponding Normalized (Localized)-IQM values

**Table 8** *Localized-IQM* values obtained from experiments performed using Test Image #2 from the Human Image Dataset

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| | | 20.0000 | 200.0000 | 50.0000 | 100.0000 | 0.8000 | 45.0000 | 0.8000 | 25.0000 | – | – | 10.0000 | 0.5000 | 0.9000 | 0.3000 | 0.9000 |
| MAE | #1 | **0.4294** | 0.2592 | 0.2864 | 0.1999 | 0.9144 | 0.2837 | 0.6676 | 1.3306 | 545.0572 | 0.3380 | 0.5908 | 0.5376 | 3.7030 | 0.4431 | 2.8268 |
| MAE | #2 | **0.5060** | 0.3030 | 0.2588 | 0.1807 | 0.7948 | 0.2481 | 0.5135 | 1.4924 | 798.1562 | 0.3175 | 0.5427 | 0.5079 | 2.6936 | 0.5434 | 0.7595 |
| MAE | #3 | **0.8434** | 0.5703 | 0.2474 | 0.1647 | 0.8783 | 0.2277 | 0.7924 | 2.2300 | 794.7958 | 0.2803 | 0.5642 | 0.7077 | 2.6710 | 0.3086 | 1.3879 |
| MSE | #1 | 0.7302 | **0.4456** | 0.2117 | 0.1796 | 0.9496 | 0.2737 | 0.9436 | 2.0308 | 606.2240 | 0.2900 | 0.5222 | 0.7370 | 4.6596 | 0.3497 | 5.1047 |
| MSE | #2 | 1.0985 | **0.5231** | 0.2030 | 0.1788 | 0.8893 | 0.2087 | 0.8479 | 2.4183 | 680.1584 | 0.2858 | 0.5093 | 0.9654 | 5.0006 | 0.2187 | 6.5159 |
| MSE | #3 | 0.4120 | **0.2232** | 0.1792 | 0.1729 | 0.7710 | 0.1944 | 0.7637 | 1.4557 | 629.2476 | 0.2744 | 0.4015 | 0.6233 | 7.0092 | 0.2838 | 4.5732 |
| ERGAS | #1 | 0.9029 | 0.4458 | **0.3566** | 0.1879 | 0.7097 | 0.4740 | 0.7608 | 2.3601 | 568.7994 | 0.2958 | 0.6234 | 0.7369 | 7.0362 | 0.2311 | 3.2308 |
| ERGAS | #2 | 0.8392 | 0.3039 | **0.3413** | 0.1848 | 0.7131 | 0.2992 | 0.6234 | 2.0691 | 710.3534 | 0.3033 | 0.6291 | 0.5506 | 3.2111 | 0.4941 | 1.5170 |
| ERGAS | #3 | 0.5378 | 0.3748 | **0.3746** | 0.2000 | 0.2424 | 0.3077 | 0.7364 | 0.9974 | 797.5416 | 0.2918 | 0.4608 | 0.3226 | 4.3711 | 0.4102 | 4.3486 |
| LMSE | #1 | 0.8877 | 0.4191 | 0.3771 | **0.1721** | 0.6926 | 0.4080 | 0.4749 | 1.2458 | 754.8527 | 0.2930 | 0.4742 | 0.4588 | 3.9588 | 0.3683 | 3.7009 |
| LMSE | #2 | 0.7519 | 0.6734 | 0.3102 | **0.2043** | 0.6958 | 0.3362 | 0.4788 | 1.0013 | 749.8347 | 0.3138 | 0.3705 | 0.4096 | 3.8378 | 0.4447 | 2.6982 |
| LMSE | #3 | 0.4075 | 0.2964 | 0.2806 | **0.1999** | 0.1309 | 0.2817 | 0.5270 | 0.6492 | 777.6749 | 0.2979 | 0.2485 | 0.4105 | 2.9962 | 0.4135 | 1.0488 |
| SSIM | #1 | 1.0122 | 0.6266 | 0.2803 | 0.1639 | **0.3147** | 0.3018 | 0.7297 | 2.2193 | 677.3406 | 0.3208 | 0.4391 | 0.6089 | 2.5573 | 0.4201 | 1.6025 |
| SSIM | #2 | 1.0452 | 0.4171 | 0.3153 | 0.1823 | **0.7332** | 0.3734 | 0.7838 | 2.1533 | 598.5507 | 0.2993 | 0.5620 | 0.7380 | 6.6590 | 0.4194 | 3.1048 |
| SSIM | #3 | 0.7244 | 0.3996 | 0.3604 | 0.1769 | **0.6989** | 0.4119 | 0.7069 | 1.8816 | 756.8063 | 0.3115 | 0.6378 | 0.4921 | 2.2843 | 0.3449 | 2.4756 |
| SAM | #1 | 0.5350 | 0.3896 | 0.4737 | 0.1622 | 0.1999 | **0.9998** | 0.4348 | 0.5069 | 833.2634 | 0.2830 | 0.2081 | 0.5844 | 3.0828 | 0.4744 | 4.1212 |
| SAM | #2 | 0.4674 | 0.2834 | 0.8502 | 0.1754 | 0.1252 | **1.2696** | 0.6676 | 0.6339 | 658.7970 | 0.3322 | 0.4764 | 0.4532 | 3.9627 | 0.6089 | 1.3661 |
| SAM | #3 | 0.5493 | 0.3454 | 0.5573 | 0.1665 | 0.1925 | **0.9091** | 0.4796 | 0.6203 | 800.1299 | 0.2749 | 0.2617 | 0.4082 | 2.3348 | 0.6456 | 3.4292 |
| UIQI | #1 | 0.7054 | 0.3903 | 0.1931 | 0.1832 | 0.8768 | 0.2418 | **0.8598** | 2.1075 | 654.6151 | 0.2927 | 0.4770 | 0.6833 | 5.1605 | 0.1959 | 6.0593 |
| UIQI | #2 | 0.5255 | 0.2974 | 0.2182 | 0.1754 | 0.8719 | 0.2672 | **0.8873** | 1.9162 | 726.2048 | 0.3131 | 0.6649 | 0.6353 | 3.7460 | 0.2353 | 5.2273 |
| UIQI | #3 | 0.6867 | 0.3930 | 0.3480 | 0.1865 | 1.0777 | 0.3530 | **0.9570** | 1.7611 | 668.4342 | 0.2762 | 0.7669 | 0.6089 | 7.4131 | 0.2162 | 6.6262 |
| PSNR | #1 | 0.7482 | 0.5005 | 0.2138 | 0.2129 | 1.0802 | 0.2755 | 0.9093 | **3.0559** | 628.0359 | 0.3018 | 0.7326 | 0.7437 | 7.3761 | 0.3207 | 5.8112 |
| PSNR | #2 | 0.7988 | 0.4065 | 0.2192 | 0.2031 | 1.0687 | 0.2604 | 0.7545 | **2.4638** | 564.3890 | 0.2884 | 0.5461 | 0.6113 | 6.8795 | 0.2657 | 1.5431 |
| PSNR | #3 | 0.6206 | 0.2873 | 0.2396 | 0.1821 | 0.8686 | 0.2119 | 0.8111 | **2.3331** | 651.5182 | 0.2966 | 0.4542 | 0.5650 | 7.7884 | 0.2258 | 2.4286 |
| NIQE | #1 | 0.8230 | 0.6244 | 0.3682 | 0.1811 | 0.2399 | 0.3936 | 0.2706 | 0.9733 | **1134.3919** | 0.3460 | 0.2664 | 0.2666 | 1.0604 | 0.2659 | 0.6745 |
| NIQE | #2 | 0.5374 | 0.3748 | 0.2882 | 0.1758 | 0.2818 | 0.2700 | 0.2723 | 0.7175 | **1233.4706** | 0.3306 | 0.2412 | 0.2531 | 0.9499 | 0.3172 | 0.6768 |

**Table 8** (continued)

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| NIQE | #3 | 0.5643 | 0.3696 | 0.5740 | 0.1779 | 0.3228 | 0.6496 | 0.3136 | 0.6267 | **968.3787** | 0.3134 | 0.3091 | 0.2804 | 0.9717 | 0.2826 | 0.9632 |
| PIQE | #1 | 0.5057 | 0.3530 | 0.4122 | 0.1733 | 0.4200 | 0.4227 | 0.2952 | 0.6148 | 904.2346 | **0.3202** | 0.2499 | 0.2474 | 2.0155 | 0.3729 | 0.3596 |
| PIQE | #2 | 0.6587 | 0.4201 | 0.5353 | 0.1867 | 0.3504 | 0.6319 | 0.2753 | 0.7523 | 1007.5755 | **0.3021** | 0.2927 | 0.2441 | 0.9228 | 0.3095 | 0.3951 |
| PIQE | #3 | 0.5250 | 0.3469 | 0.5851 | 0.1720 | 0.3596 | 0.4312 | 0.2641 | 0.5818 | 1034.6944 | **0.2760** | 0.3233 | 0.2419 | 1.0225 | 0.3749 | 0.4065 |
| SNR | #1 | 0.8560 | 0.5362 | 0.3019 | 0.1645 | 0.4860 | 0.3009 | 0.4168 | 1.3656 | 657.1921 | 0.2912 | **0.3762** | 0.5402 | 2.9322 | 0.2288 | 0.9859 |
| SNR | #2 | 0.7381 | 0.4848 | 0.3406 | 0.1730 | 0.6191 | 0.4700 | 0.6321 | 1.4865 | 616.1170 | 0.3087 | **0.4684** | 0.5653 | 2.7695 | 0.3302 | 1.7028 |
| SNR | #3 | 0.8679 | 0.6639 | 0.3304 | 0.1696 | 0.1319 | 0.2712 | 0.4529 | 1.5036 | 803.5869 | 0.3042 | **0.3984** | 0.3643 | 3.1388 | 0.2850 | 1.3897 |
| VIF | #1 | 0.5163 | 0.2681 | 0.5593 | 0.1932 | 0.1804 | 1.1068 | 0.5631 | 1.1151 | 727.9735 | 0.3290 | 0.5986 | **0.6142** | 2.3959 | 0.2436 | 1.4011 |
| VIF | #2 | 0.6107 | 0.3771 | 0.3760 | 0.1949 | 0.1621 | 0.5952 | 0.3833 | 0.8966 | 746.5858 | 0.3157 | 0.2702 | **0.2748** | 2.1981 | 0.3389 | 1.7297 |
| VIF | #3 | 0.7396 | 0.4580 | 0.3387 | 0.1968 | 0.5708 | 0.3073 | 0.5972 | 1.3149 | 705.6724 | 0.2813 | 0.4282 | **0.4441** | 2.4394 | 0.2312 | 1.1217 |
| FSIM | #1 | 0.6396 | 0.5171 | 0.3520 | 0.1970 | 0.3143 | 0.3529 | 0.3158 | 0.7930 | 829.9344 | 0.3128 | 0.2821 | 0.2919 | **1.7206** | 0.2896 | 0.9816 |
| FSIM | #2 | 0.6381 | 0.3728 | 0.3898 | 0.1717 | 0.1275 | 0.3815 | 0.4508 | 1.1663 | 1032.8252 | 0.3273 | 0.4350 | 0.4717 | **1.6765** | 0.2608 | 1.3555 |
| FSIM | #3 | 0.9334 | 0.6167 | 0.2928 | 0.1715 | 0.3878 | 0.3806 | 0.5375 | 1.6631 | 710.1445 | 0.3538 | 0.3739 | 0.3911 | **2.0782** | 0.2834 | 1.3550 |
| GMSD | #1 | 0.7559 | 0.4966 | 0.2971 | 0.1809 | 0.3919 | 0.2299 | 0.3054 | 1.0616 | 1175.4359 | 0.3388 | 0.2797 | 0.3201 | 1.0826 | **0.2789** | 1.1928 |
| GMSD | #2 | 0.7585 | 0.5147 | 0.3374 | 0.1867 | 0.1248 | 0.3731 | 0.3583 | 1.0954 | 1031.5353 | 0.3340 | 0.2964 | 0.3122 | 1.0431 | **0.3721** | 0.7421 |
| GMSD | #3 | 0.6669 | 0.4542 | 0.4294 | 0.1742 | 0.3998 | 0.3686 | 0.3374 | 0.8755 | 950.5872 | 0.3277 | 0.3467 | 0.3499 | 1.3197 | **0.3897** | 0.9180 |
| NCC | #1 | 0.5929 | 0.3879 | 0.4502 | 0.1800 | 0.6087 | 0.3229 | 0.5957 | 0.8815 | 806.5994 | 0.2987 | 0.4628 | 0.4587 | 4.6618 | 0.4275 | **2.7255** |
| NCC | #2 | 0.4477 | 0.3152 | 0.6955 | 0.2073 | 0.2099 | 0.5074 | 0.5764 | 0.5423 | 970.7954 | 0.2884 | 0.5046 | 0.4508 | 1.8196 | 0.4928 | **1.0420** |
| NCC | #3 | 0.5032 | 0.2680 | 0.5047 | 0.1646 | 0.4810 | 0.5136 | 0.8718 | 1.1595 | 726.8067 | 0.3403 | 0.6416 | 0.5126 | 2.2449 | 0.5246 | **2.3860** |

Bold values indicates the ability to distinguish between different images with the same content for the corresponding Normalized (Localized)-IQM values

**Table 9** *Localized-IQM* values obtained from experiments performed using Test Image #3 from the Human Image Dataset

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| | | 20.0000 | 200.0000 | 50.0000 | 100.0000 | 0.8000 | 45.0000 | 0.8000 | 25.0000 | – | – | 10.0000 | 0.5000 | 0.9000 | 0.3000 | 0.9000 |
| MAE | #1 | **0.5256** | 0.3159 | 0.2857 | 0.1683 | 0.5669 | 0.1973 | 0.8169 | 1.5098 | 916.3027 | 0.2791 | 0.6325 | 0.4577 | 4.9950 | 0.2742 | 3.3322 |
| MAE | #2 | **0.7580** | 0.4572 | 0.2592 | 0.1833 | 0.8277 | 0.2388 | 0.7667 | 1.5714 | 910.9643 | 0.2419 | 0.4927 | 0.5073 | 4.8215 | 0.2719 | 2.6858 |
| MAE | #3 | **0.9916** | 0.5263 | 0.2647 | 0.1675 | 0.7149 | 0.3318 | 0.8701 | 1.4007 | 842.8436 | 0.2933 | 0.5516 | 0.3685 | 5.8084 | 0.4004 | 2.6218 |
| MSE | #1 | 0.4734 | **0.2626** | 0.2426 | 0.1670 | 1.1842 | 0.2032 | 0.9600 | 1.9078 | 676.5482 | 0.2732 | 0.7138 | 0.6320 | 3.8033 | 0.4262 | 2.8828 |
| MSE | #2 | 0.3725 | **0.2209** | 0.1824 | 0.1567 | 0.7170 | 0.1736 | 0.6248 | 1.3066 | 851.2874 | 0.2715 | 0.5082 | 0.7148 | 2.8349 | 0.3026 | 4.9490 |
| MSE | #3 | 1.0894 | **0.5621** | 0.2334 | 0.1777 | 0.8669 | 0.2073 | 0.8037 | 3.3821 | 799.3618 | 0.2886 | 0.6441 | 0.6101 | 5.1394 | 0.4026 | 2.3250 |
| ERGAS | #1 | 0.7945 | 0.3927 | **0.3292** | 0.1679 | 0.5056 | 0.2911 | 0.5205 | 1.8251 | 789.4858 | 0.3152 | 0.4811 | 0.3135 | 2.7223 | 0.6236 | 1.6505 |
| ERGAS | #2 | 1.1158 | 0.4810 | **0.3563** | 0.1735 | 0.7631 | 0.4010 | 0.6385 | 2.5635 | 890.0328 | 0.3379 | 0.5599 | 0.4586 | 5.5765 | 0.4902 | 1.0568 |
| ERGAS | #3 | 0.9373 | 0.6028 | **0.4070** | 0.1739 | 0.4945 | 0.2186 | 0.5442 | 1.7759 | 741.1989 | 0.2744 | 0.6079 | 0.5016 | 2.9057 | 0.3282 | 2.2867 |
| LMSE | #1 | 0.4558 | 0.3195 | 0.2745 | **0.1734** | 0.6338 | 0.2289 | 0.6723 | 0.9051 | 755.4803 | 0.2701 | 0.3957 | 0.3828 | 3.4108 | 0.4678 | 2.2633 |
| LMSE | #2 | 0.7447 | 0.4119 | 0.3105 | **0.1721** | 0.4619 | 0.4217 | 0.4061 | 1.4292 | 928.1319 | 0.2784 | 0.3961 | 0.3461 | 2.7701 | 0.4243 | 0.6887 |
| LMSE | #3 | 0.9074 | 0.5036 | 0.2913 | **0.1589** | 0.6959 | 0.3414 | 0.7016 | 1.7538 | 811.8237 | 0.2737 | 0.8031 | 0.5622 | 5.0514 | 0.3911 | 2.1192 |
| SSIM | #1 | 1.4415 | 0.7174 | 0.2999 | 0.1578 | **0.7871** | 0.2947 | 0.7400 | 3.2108 | 1039.1378 | 0.2897 | 0.5878 | 0.5061 | 3.1140 | 0.4344 | 4.3060 |
| SSIM | #2 | 0.9337 | 0.6907 | 0.2358 | 0.1638 | **0.6860** | 0.2054 | 0.6046 | 2.2019 | 871.0849 | 0.2982 | 0.5457 | 0.5601 | 3.5155 | 0.4642 | 3.1795 |
| SSIM | #3 | 0.9668 | 0.6993 | 0.2519 | 0.1757 | **0.7780** | 0.2811 | 0.6564 | 1.2996 | 992.7083 | 0.2498 | 0.5570 | 0.4162 | 6.1565 | 0.2864 | 3.7608 |
| SAM | #1 | 0.5232 | 0.3523 | 0.8441 | 0.1525 | 0.1562 | **1.8851** | 0.5395 | 0.5942 | 865.2746 | 0.3845 | 0.4302 | 0.6791 | 2.5135 | 0.7704 | 4.9574 |
| SAM | #2 | 0.4915 | 0.3507 | 0.7522 | 0.1545 | 0.2524 | **1.4947** | 0.8130 | 0.4927 | 1229.6354 | 0.2807 | 0.5338 | 0.6316 | 2.8063 | 0.4331 | 4.1977 |
| SAM | #3 | 0.5853 | 0.4309 | 0.4550 | 0.1512 | 0.2254 | **1.1252** | 0.6714 | 0.5878 | 1087.7723 | 0.3246 | 0.2537 | 0.6418 | 2.9910 | 0.5543 | 1.6604 |
| UIQI | #1 | 0.8596 | 0.5133 | 0.3979 | 0.1636 | 0.8134 | 0.2998 | **0.8373** | 2.4876 | 795.3173 | 0.2549 | 0.8435 | 0.5317 | 4.0089 | 0.2818 | 8.6314 |
| UIQI | #2 | 1.1171 | 0.7722 | 0.2366 | 0.1637 | 0.6705 | 0.2254 | **0.8884** | 2.2972 | 801.3173 | 0.2524 | 0.6120 | 0.5205 | 5.3331 | 0.2617 | 8.1213 |
| UIQI | #3 | 0.8941 | 0.5329 | 0.3009 | 0.1836 | 0.9840 | 0.1848 | **1.0176** | 2.4184 | 810.2729 | 0.2418 | 0.8220 | 0.6635 | 9.4276 | 0.2223 | 10.7423 |
| PSNR | #1 | 0.6572 | 0.3267 | 0.2065 | 0.1731 | 0.7340 | 0.1847 | 0.7611 | **2.1664** | 798.1157 | 0.2693 | 0.5751 | 0.6785 | 5.3109 | 0.3538 | 1.0531 |
| PSNR | #2 | 0.4876 | 0.2868 | 0.2013 | 0.1682 | 0.8665 | 0.1793 | 0.7944 | **1.4498** | 880.7314 | 0.2937 | 0.5246 | 0.7816 | 4.5004 | 0.3486 | 2.7995 |
| PSNR | #3 | 0.7108 | 0.4143 | 0.2390 | 0.1636 | 1.0132 | 0.2084 | 0.8772 | **2.5102** | 1042.7462 | 0.3021 | 0.6648 | 0.4729 | 5.4192 | 0.4638 | 2.0338 |
| NIQE | #1 | 0.6340 | 0.3981 | 0.7243 | 0.1563 | 0.4585 | 0.5055 | 0.4692 | 1.1351 | **977.4222** | 0.3107 | 0.5156 | 0.6523 | 1.9464 | 0.3807 | 0.9995 |
| NIQE | #2 | 0.5441 | 0.3647 | 0.6916 | 0.1658 | 0.3319 | 0.5309 | 0.3675 | 0.6849 | **984.6535** | 0.2640 | 0.4749 | 0.3053 | 1.4845 | 0.3648 | 2.2967 |

**Table 9** (continued)

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| NIQE | #3 | 0.7041 | 0.4070 | 0.3666 | 0.1582 | 0.4410 | 0.3524 | 0.4064 | 1.3196 | **786.7323** | 0.2566 | 0.3769 | 0.3800 | 2.4285 | 0.2240 | 1.8768 |
| PIQE | #1 | 0.5829 | 0.4207 | 0.6497 | 0.1695 | 0.1871 | 0.5631 | 0.2710 | 0.6247 | 714.6089 | **0.2347** | 0.2617 | 0.2241 | 0.9093 | 0.3250 | 1.1829 |
| PIQE | #2 | 0.6382 | 0.4378 | 0.5761 | 0.1655 | 0.2787 | 1.3996 | 0.3027 | 0.6666 | 795.6248 | **0.2268** | 0.3291 | 0.2457 | 1.0372 | 0.2824 | 1.5445 |
| PIQE | #3 | 0.5995 | 0.4297 | 0.6615 | 0.1693 | 0.2932 | 0.4198 | 0.3405 | 0.6338 | 738.3656 | **0.2290** | 0.3395 | 0.2494 | 1.1765 | 0.2809 | 0.9731 |
| SNR | #1 | 0.8711 | 0.4157 | 0.4793 | 0.1695 | 0.5041 | 0.3885 | 0.5236 | 1.6687 | 996.5483 | 0.2572 | **0.6021** | 0.4652 | 2.8361 | 0.2015 | 4.6691 |
| SNR | #2 | 1.0480 | 0.5970 | 0.3471 | 0.1643 | 0.3929 | 0.2519 | 0.3663 | 1.5567 | 832.2528 | 0.2592 | **0.4199** | 0.3840 | 1.6455 | 0.2227 | 0.9929 |
| SNR | #3 | 0.6585 | 0.3281 | 0.3919 | 0.1637 | 0.5213 | 0.4710 | 0.4732 | 1.4539 | 921.6193 | 0.2934 | **0.4394** | 0.3172 | 3.1681 | 0.3842 | 0.6231 |
| VIF | #1 | 0.5789 | 0.4351 | 0.4977 | 0.1701 | 0.3488 | 0.4133 | 0.3146 | 0.6157 | 639.2340 | 0.2760 | 0.3222 | **0.2571** | 1.3640 | 0.3779 | 0.6440 |
| VIF | #2 | 0.6870 | 0.4075 | 0.5371 | 0.1782 | 0.1264 | 0.6003 | 0.3558 | 1.2909 | 764.6747 | 0.2619 | 0.3927 | **0.2985** | 2.6518 | 0.2506 | 0.6256 |
| VIF | #3 | 0.6233 | 0.4225 | 0.5639 | 0.1651 | 0.3481 | 0.5498 | 0.3844 | 0.7514 | 972.7091 | 0.2429 | 0.3684 | **0.2765** | 1.7274 | 0.2651 | 1.9506 |
| FSIM | #1 | 1.0180 | 0.5552 | 0.3268 | 0.1668 | 0.4773 | 0.2814 | 0.3831 | 2.2870 | 1123.6727 | 0.2949 | 0.5438 | 0.4176 | **2.2046** | 0.3586 | 0.9792 |
| FSIM | #2 | 0.6770 | 0.4180 | 0.5614 | 0.1633 | 0.4835 | 0.3707 | 0.3775 | 1.5197 | 1374.3332 | 0.2688 | 0.5665 | 0.3052 | **2.3610** | 0.3275 | 0.9967 |
| FSIM | #3 | 0.6852 | 0.3594 | 0.3481 | 0.1588 | 0.4948 | 0.4371 | 0.4966 | 1.5126 | 906.2678 | 0.2628 | 0.3905 | 0.3599 | **3.2417** | 0.2687 | 3.3052 |
| GMSD | #1 | 0.6182 | 0.4186 | 0.4609 | 0.1675 | 0.3442 | 0.5576 | 0.3150 | 0.7192 | 831.7491 | 0.2725 | 0.3034 | 0.2534 | 1.1304 | **0.2712** | 0.7927 |
| GMSD | #2 | 0.6246 | 0.3639 | 0.4522 | 0.1585 | 0.4274 | 0.2126 | 0.3851 | 0.9952 | 812.1350 | 0.2608 | 0.5265 | 0.3613 | 1.3192 | **0.3055** | 1.1667 |
| GMSD | #3 | 0.4820 | 0.2737 | 0.4715 | 0.1701 | 0.5141 | 0.2108 | 0.3785 | 0.8381 | 987.4920 | 0.3624 | 0.5931 | 0.4664 | 2.0578 | **0.8262** | 0.5764 |
| NCC | #1 | 0.5995 | 0.3836 | 0.4015 | 0.1872 | 0.3747 | 0.5860 | 0.3186 | 0.6895 | 665.1962 | 0.2762 | 0.3193 | 0.2648 | 1.4227 | 0.4952 | **2.4086** |
| NCC | #2 | 0.5557 | 0.4090 | 0.7263 | 0.1867 | 0.4108 | 0.3885 | 0.7543 | 0.6832 | 858.2723 | 0.2753 | 0.5881 | 0.2427 | 2.9564 | 0.4661 | **1.0627** |
| NCC | #3 | 0.5804 | 0.3584 | 0.4082 | 0.1677 | 0.7074 | 0.4193 | 0.7426 | 1.0075 | 764.6621 | 0.2435 | 0.4302 | 0.4869 | 5.2101 | 0.3465 | **1.1835** |

Bold values indicates the ability to distinguish between different images with the same content for the corresponding Normalized (Localized)-IQM values

**Table 10** *Localized-IQM* values obtained from experiments performed using Test Image #4 from the Human Image Dataset

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| | | 20.0000 | 200.0000 | 50.0000 | 100.0000 | 0.8000 | 45.0000 | 0.8000 | 25.0000 | – | – | 10.0000 | 0.5000 | 0.9000 | 0.3000 | 0.9000 |
| MAE | #1 | **0.7810** | 0.4987 | 0.2996 | 0.2089 | 0.4215 | 0.2977 | 0.4692 | 1.6108 | 602.5531 | 0.3434 | 0.3954 | 0.7513 | 4.4558 | 0.5053 | 2.5453 |
| MAE | #2 | **0.7812** | 0.4869 | 0.2670 | 0.2146 | 0.4454 | 0.2763 | 0.5111 | 1.6743 | 664.0833 | 0.3345 | 0.4139 | 0.4663 | 4.3469 | 0.3194 | 3.8529 |
| MAE | #3 | **0.5240** | 0.3403 | 0.2611 | 0.2355 | 0.4977 | 0.2833 | 0.4610 | 1.1495 | 789.6141 | 0.3377 | 0.3142 | 0.4259 | 3.5057 | 0.2969 | 1.5375 |
| MSE | #1 | 0.7013 | **0.4512** | 0.2387 | 0.1934 | 0.7725 | 0.2275 | 0.5823 | 1.7325 | 672.8825 | 0.3124 | 0.4636 | 0.6465 | 3.5688 | 0.4060 | 3.2789 |
| MSE | #2 | 0.6544 | **0.4105** | 0.2094 | 0.1848 | 0.6892 | 0.1979 | 0.5987 | 1.6867 | 758.0967 | 0.3141 | 0.4309 | 0.7610 | 2.9745 | 0.3317 | 5.2017 |
| MSE | #3 | 0.5883 | **0.4405** | 0.2137 | 0.2075 | 0.7076 | 0.2292 | 0.5289 | 1.3515 | 668.7938 | 0.3403 | 0.4527 | 0.6133 | 7.1524 | 0.2794 | 1.0055 |
| ERGAS | #1 | 0.5424 | 0.3187 | **0.4109** | 0.2151 | 0.7658 | 0.4294 | 0.4322 | 1.3418 | 612.3508 | 0.3211 | 0.6245 | 0.6316 | 4.7076 | 0.4290 | 1.2150 |
| ERGAS | #2 | 0.8331 | 0.3213 | **0.3420** | 0.2137 | 0.2142 | 0.3791 | 0.5889 | 2.4115 | 675.6773 | 0.3204 | 0.4073 | 0.5929 | 6.6695 | 0.3831 | 3.3774 |
| ERGAS | #3 | 0.5059 | 0.2766 | **0.4139** | 0.2116 | 0.1855 | 0.4622 | 0.6462 | 2.3347 | 682.3690 | 0.3242 | 0.9076 | 0.4779 | 8.2034 | 0.2776 | 5.1357 |
| LMSE | #1 | 0.7146 | 0.4192 | 0.3059 | **0.1845** | 0.5697 | 0.3099 | 0.6013 | 1.7845 | 773.4513 | 0.3559 | 0.4495 | 0.5533 | 2.5492 | 0.6037 | 2.2848 |
| LMSE | #2 | 0.7148 | 0.3879 | 0.4789 | **0.1750** | 0.4181 | 0.5141 | 0.3445 | 1.3474 | 776.1735 | 0.3526 | 0.3929 | 0.2929 | 3.2760 | 0.4143 | 3.8381 |
| LMSE | #3 | 0.6679 | 0.4202 | 0.4455 | **0.1989** | 0.5047 | 0.3080 | 0.3820 | 1.4681 | 1041.1185 | 0.3413 | 0.4698 | 0.3555 | 2.2122 | 0.5981 | 1.1692 |
| SSIM | #1 | 0.8117 | 0.3988 | 0.2183 | 0.1960 | **0.7941** | 0.2271 | 0.6132 | 2.2674 | 679.5075 | 0.3228 | 0.4514 | 0.5929 | 3.4026 | 0.2961 | 2.9765 |
| SSIM | #2 | 0.7007 | 0.3858 | 0.2361 | 0.2579 | **0.8544** | 0.2185 | 0.4358 | 2.1235 | 597.1327 | 0.3046 | 0.6262 | 0.4777 | 8.0072 | 0.2748 | 0.6566 |
| SSIM | #3 | 0.8262 | 0.3472 | 0.2677 | 0.2148 | **0.7396** | 0.2682 | 0.6565 | 2.4564 | 651.7363 | 0.2923 | 0.5257 | 0.7340 | 8.8878 | 0.2512 | 5.2572 |
| SAM | #1 | 0.4505 | 0.3200 | 0.5634 | 0.1752 | 0.1246 | **1.4885** | 0.4528 | 0.6248 | 753.6394 | 0.4070 | 0.1998 | 0.4478 | 1.4430 | 0.5875 | 2.6128 |
| SAM | #2 | 0.4238 | 0.3045 | 0.8554 | 0.1745 | 0.3485 | **0.9310** | 0.2539 | 0.5181 | 792.3264 | 0.4551 | 0.1622 | 0.4304 | 1.5710 | 0.6680 | 1.1798 |
| SAM | #3 | 0.3850 | 0.2684 | 0.7973 | 0.1887 | 0.1902 | **1.0574** | 0.3766 | 0.4842 | 737.7929 | 0.3599 | 0.3067 | 0.3073 | 2.9744 | 0.6648 | 2.5944 |
| UIQI | #1 | 1.6755 | 0.9126 | 0.2331 | 0.2033 | 0.7926 | 0.2335 | **0.5999** | 2.4147 | 684.7592 | 0.3059 | 0.5230 | 0.8546 | 10.0727 | 0.1883 | 6.6767 |
| UIQI | #2 | 1.1503 | 0.4452 | 0.2367 | 0.2008 | 0.7685 | 0.2786 | **0.6491** | 3.2944 | 683.1144 | 0.3117 | 0.6468 | 0.8061 | 10.1630 | 0.2096 | 7.5199 |
| UIQI | #3 | 0.9164 | 0.4438 | 0.3087 | 0.2202 | 0.8204 | 0.3115 | **0.6265** | 3.0086 | 653.0606 | 0.2921 | 0.6193 | 1.0645 | 8.4481 | 0.1950 | 7.9746 |
| PSNR | #1 | 1.0814 | 0.7401 | 0.2690 | 0.2032 | 0.7152 | 0.2655 | 0.6932 | **2.2283** | 697.5760 | 0.3149 | 0.4910 | 0.8920 | 6.1425 | 0.4108 | 2.5627 |
| PSNR | #2 | 0.5092 | 0.3270 | 0.2191 | 0.2068 | 0.6719 | 0.2112 | 0.5649 | **1.5382** | 620.2764 | 0.2987 | 0.4225 | 0.7324 | 6.8914 | 0.2947 | 4.0561 |
| PSNR | #3 | 1.1565 | 0.7190 | 0.2628 | 0.2011 | 0.8207 | 0.2522 | 0.6199 | **2.2668** | 685.8117 | 0.2938 | 0.4976 | 0.5996 | 9.6388 | 0.2215 | 5.1837 |
| NIQE | #1 | 0.4597 | 0.2796 | 0.7066 | 0.1988 | 0.2958 | 0.3781 | 0.2508 | 0.8299 | **665.2233** | 0.4243 | 0.3763 | 0.3336 | 1.7307 | 0.6537 | 0.4851 |
| NIQE | #2 | 0.4317 | 0.2618 | 0.9950 | 0.1928 | 0.2975 | 0.4474 | 0.2490 | 0.8879 | **682.5875** | 0.4050 | 0.3806 | 0.3278 | 1.7795 | 0.7364 | 0.2912 |

**Table 10** (continued)

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| NIQE | #3 | 0.5130 | 0.2795 | 0.6540 | 0.1959 | 0.2868 | 0.9536 | 0.2056 | 1.0468 | **632.0519** | 0.4538 | 0.3745 | 0.3228 | 1.8124 | 0.6671 | 0.5083 |
| PIQE | #1 | 0.3441 | 0.2425 | 1.3881 | 0.1855 | 0.3755 | 0.3638 | 0.1757 | 0.3840 | 1479.6689 | **0.9761** | 0.3512 | 0.2033 | 0.6762 | 0.3200 | 0.3049 |
| PIQE | #2 | 0.3309 | 0.2409 | 1.6259 | 0.1820 | 0.2284 | 0.4848 | 0.1798 | 0.3618 | 2295.0586 | **0.9736** | 0.4721 | 0.2424 | 0.7064 | 0.3709 | 0.1605 |
| PIQE | #3 | 0.3197 | 0.2354 | 1.5907 | 0.1855 | 0.1262 | 0.5400 | 0.1374 | 0.3400 | 2162.9647 | **1.8052** | 0.2589 | 0.1663 | 0.7188 | 0.3399 | 0.1368 |
| SNR | #1 | 0.6122 | 0.3946 | 0.3352 | 0.2034 | 0.4231 | 0.3074 | 0.3139 | 1.1236 | 758.6566 | 0.3321 | **0.3233** | 0.3146 | 2.4654 | 0.3070 | 0.5280 |
| SNR | #2 | 0.8177 | 0.4972 | 0.3015 | 0.2099 | 0.3774 | 0.3656 | 0.3703 | 1.8156 | 823.6652 | 0.3922 | **0.3560** | 0.3022 | 2.5886 | 0.3658 | 1.6065 |
| SNR | #3 | 0.9582 | 0.5666 | 0.3183 | 0.1979 | 0.3268 | 0.3171 | 0.3887 | 1.7921 | 699.3512 | 0.3981 | **0.3394** | 0.4044 | 2.2595 | 0.3899 | 1.3995 |
| VIF | #1 | 0.6649 | 0.3691 | 0.3367 | 0.1854 | 0.2887 | 0.4042 | 0.3829 | 1.1858 | 695.0592 | 0.4159 | 0.2836 | **0.3792** | 1.6776 | 0.4354 | 1.2492 |
| VIF | #2 | 0.5655 | 0.4309 | 0.3031 | 0.1905 | 0.3870 | 0.3008 | 0.3327 | 0.9686 | 613.9733 | 0.3570 | 0.3223 | **0.3600** | 2.8667 | 0.2392 | 1.6766 |
| VIF | #3 | 0.9728 | 0.5890 | 0.3361 | 0.1980 | 0.3831 | 0.3038 | 0.3336 | 1.8899 | 740.6909 | 0.3551 | 0.3451 | **0.3316** | 2.4309 | 0.2231 | 1.4218 |
| FSIM | #1 | 0.4529 | 0.3017 | 0.6864 | 0.2081 | 0.3522 | 0.3046 | 0.3392 | 0.7166 | 983.7096 | 0.3605 | 0.4080 | 0.2545 | **2.3022** | 0.3747 | 0.4124 |
| FSIM | #2 | 0.5801 | 0.3220 | 0.4465 | 0.1909 | 0.1653 | 0.3866 | 0.3018 | 1.1887 | 937.6127 | 0.3555 | 0.4014 | 0.3423 | **1.5666** | 0.2712 | 0.8372 |
| FSIM | #3 | 0.7819 | 0.4615 | 0.3381 | 0.2271 | 0.3714 | 0.3545 | 0.3769 | 1.3757 | 660.7123 | 0.3918 | 0.3487 | 0.3932 | **2.3871** | 0.3143 | 1.1799 |
| GMSD | #1 | 0.4245 | 0.2915 | 0.8491 | 0.2104 | 0.2607 | 0.7892 | 0.2082 | 0.5355 | 1310.1626 | 0.3960 | 0.2871 | 0.1792 | 1.1541 | **0.6032** | 0.7387 |
| GMSD | #2 | 0.5591 | 0.3348 | 0.6202 | 0.1976 | 0.2879 | 0.3879 | 0.2372 | 1.1780 | 763.0343 | 0.3773 | 0.3150 | 0.3518 | 1.6417 | **0.4352** | 0.6943 |
| GMSD | #3 | 0.8391 | 0.5409 | 0.2752 | 0.2173 | 0.3541 | 0.2916 | 0.3294 | 1.6924 | 847.5272 | 0.3907 | 0.3262 | 0.3632 | 2.1951 | **0.2790** | 1.7103 |
| NCC | #1 | 0.5596 | 0.3642 | 0.5546 | 0.2518 | 0.2987 | 0.8866 | 0.3139 | 0.8056 | 1094.5508 | 0.3698 | 0.3084 | 0.2531 | 1.6364 | 0.8168 | **1.2621** |
| NCC | #2 | 0.4689 | 0.3016 | 0.5909 | 0.2315 | 0.1240 | 0.4471 | 0.4302 | 0.8302 | 818.1320 | 0.3476 | 0.3920 | 0.2565 | 3.4711 | 0.5293 | **1.3453** |
| NCC | #3 | 0.4509 | 0.2910 | 0.5985 | 0.2193 | 0.6019 | 0.4399 | 0.5222 | 0.7293 | 1077.7697 | 0.3656 | 0.4692 | 0.2524 | 2.9349 | 0.5229 | **3.4377** |

Bold values indicates the ability to distinguish between different images with the same content for the corresponding Normalized (Localized)-IQM values

**Table 11** *Localized-IQM* values obtained from experiments performed using Test Image #1 from the Earth Observation Image Dataset

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| | | 20.0000 | 200.0000 | 50.0000 | 100.0000 | 0.8000 | 45.0000 | 0.8000 | 25.0000 | – | – | 10.0000 | 0.5000 | 0.9000 | 0.3000 | 0.9000 |
| MAE | #1 | **2.7917** | 2.1734 | 0.8698 | 0.6741 | 4.2750 | 1.3222 | 2.7063 | 4.5444 | 1207.8675 | 0.5224 | 1.7376 | 2.1067 | 23.2743 | 0.7452 | 30.5665 |
| MAE | #2 | **1.5257** | 1.0301 | 0.6754 | 0.6354 | 3.6042 | 0.6832 | 4.0652 | 4.6625 | 1325.0061 | 0.4509 | 1.0542 | 1.8498 | 19.6483 | 0.4740 | 19.2668 |
| MAE | #3 | **2.1159** | 1.1110 | 1.1352 | 0.7543 | 2.0614 | 1.1801 | 2.1764 | 5.7571 | 1745.0586 | 0.4626 | 1.6088 | 3.5562 | 30.3816 | 0.5016 | 6.0347 |
| MSE | #1 | 2.8912 | **1.3096** | 0.8846 | 0.6913 | 6.8612 | 0.8834 | 5.5978 | 7.7873 | 1755.4426 | 0.4502 | 2.2837 | 2.1530 | 21.0463 | 0.3675 | 28.2157 |
| MSE | #2 | 2.0805 | **1.3398** | 0.7413 | 0.6829 | 6.4380 | 0.9016 | 4.8490 | 7.1591 | 1397.5560 | 0.5214 | 1.7591 | 2.6722 | 22.9853 | 0.7817 | 19.8764 |
| MSE | #3 | 1.1921 | **0.6794** | 1.4918 | 0.7595 | 3.9385 | 1.7898 | 7.2682 | 3.7171 | 1382.3189 | 0.6681 | 3.3876 | 3.5759 | 16.5057 | 0.7728 | 31.3498 |
| ERGAS | #1 | 2.6230 | 1.1898 | **1.2584** | 0.7003 | 1.5406 | 1.2457 | 2.0781 | 6.7081 | 1031.7400 | 0.4699 | 1.8442 | 5.5424 | 8.1566 | 0.6328 | 20.4260 |
| ERGAS | #2 | 3.0366 | 1.9355 | **0.9231** | 0.5816 | 0.7631 | 1.3852 | 1.6887 | 6.9031 | 814.1861 | 0.6244 | 1.2007 | 4.2721 | 7.9202 | 0.5910 | 15.6020 |
| ERGAS | #3 | 2.6635 | 1.0260 | **1.1975** | 0.6534 | 2.9527 | 1.3938 | 2.5144 | 7.1417 | 1020.6124 | 0.4993 | 1.9720 | 1.5841 | 15.6638 | 0.7045 | 29.1166 |
| LMSE | #1 | 4.9083 | 3.2379 | 0.8654 | **0.4614** | 4.2605 | 0.8668 | 4.7202 | 9.7858 | 1687.0257 | 0.4703 | 1.3236 | 4.2502 | 40.8877 | 0.3024 | 242.7731 |
| LMSE | #2 | 6.5723 | 2.8803 | 1.0518 | **0.5227** | 12.5217 | 0.5440 | 6.8635 | 18.1399 | 1539.8148 | 0.4977 | 1.9169 | 6.8457 | 47.9746 | 0.3795 | 289.2212 |
| LMSE | #3 | 13.8322 | 8.6325 | 0.8395 | **0.4148** | 12.4160 | 0.8990 | 5.8968 | 32.5829 | 1570.8539 | 0.4773 | 1.8325 | 4.1061 | 60.9578 | 0.3573 | 358.1113 |
| SSIM | #1 | 1.4007 | 0.7869 | 1.0038 | 0.4796 | **3.1886** | 1.4177 | 2.3590 | 3.0141 | 1505.2896 | 0.5247 | 0.9857 | 2.8070 | 8.7393 | 0.5537 | 6.4970 |
| SSIM | #2 | 1.5805 | 0.8385 | 1.6298 | 0.6678 | **2.4339** | 2.1904 | 3.4982 | 3.0522 | 1226.2591 | 0.4998 | 2.1664 | 1.5467 | 11.7296 | 0.6165 | 24.2567 |
| SSIM | #3 | 2.1707 | 1.0557 | 1.2548 | 0.5916 | **3.7732** | 2.0698 | 2.6430 | 9.0735 | 1578.5393 | 0.4724 | 1.5644 | 1.9547 | 18.8757 | 0.7226 | 46.1540 |
| SAM | #1 | 3.1054 | 1.9127 | 1.1492 | 0.5465 | 0.4150 | **2.1817** | 0.9301 | 4.4643 | 1044.0036 | 0.5037 | 0.4257 | 2.3674 | 4.7100 | 1.4875 | 5.5652 |
| SAM | #2 | 1.5603 | 0.9661 | 1.3833 | 0.5766 | 0.4570 | **1.8567** | 0.8517 | 2.4789 | 1387.4224 | 0.4544 | 0.6054 | 1.9738 | 10.5873 | 0.6304 | 3.8275 |
| SAM | #3 | 1.8430 | 1.0442 | 1.4881 | 0.5599 | 0.4415 | **2.0351** | 0.9488 | 2.8805 | 1442.3154 | 0.4632 | 0.6135 | 2.3212 | 21.2324 | 0.7642 | 5.9746 |
| UIQI | #1 | 1.2170 | 0.8384 | 0.5985 | 0.6650 | 2.7669 | 0.7873 | **2.0015** | 3.4704 | 1136.4542 | 0.5363 | 0.9062 | 2.3562 | 12.3143 | 1.5189 | 10.6867 |
| UIQI | #2 | 2.0302 | 0.7339 | 1.4747 | 0.4526 | 3.4102 | 1.5141 | **4.5560** | 5.4300 | 1298.4410 | 0.4610 | 2.2091 | 2.3410 | 16.3823 | 0.6346 | 31.4765 |
| UIQI | #3 | 2.2528 | 1.4728 | 0.7672 | 0.6932 | 3.5915 | 0.9460 | **1.9603** | 5.8852 | 1101.3852 | 0.5313 | 1.1800 | 2.3437 | 15.3941 | 0.8536 | 22.2083 |
| PSNR | #1 | 1.2761 | 0.7793 | 0.6309 | 0.7124 | 7.5703 | 0.8077 | 5.2326 | **4.6075** | 1622.9247 | 0.4844 | 1.2998 | 2.2298 | 30.2627 | 0.5479 | 31.8241 |
| PSNR | #2 | 3.2841 | 1.6200 | 0.7383 | 0.5743 | 7.4942 | 0.9654 | 3.4216 | **7.4959** | 1354.2536 | 0.3940 | 1.8911 | 2.5414 | 10.4975 | 0.5028 | 51.8493 |
| PSNR | #3 | 1.0113 | 0.7650 | 0.5826 | 0.6185 | 5.0047 | 0.6159 | 4.0308 | **3.8622** | 1221.2683 | 0.4759 | 1.3326 | 3.6720 | 13.5288 | 0.3856 | 25.9012 |
| NIQE | #1 | 1.9133 | 1.0525 | 1.9833 | 0.5259 | 0.5100 | 1.0271 | 0.7178 | 2.2610 | **1033.4336** | 0.4846 | 0.7583 | 2.5394 | 4.0953 | 0.6762 | 10.5740 |
| NIQE | #2 | 3.4585 | 1.1937 | 1.1162 | 0.5519 | 2.3384 | 1.7130 | 1.2825 | 5.3148 | **1216.6454** | 0.4294 | 1.1538 | 1.9088 | 9.8238 | 0.5061 | 8.6592 |

**Table 11** (continued)

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| NIQE | #3 | 2.4327 | 1.5055 | 0.7990 | 0.5475 | 0.6925 | 0.8817 | 0.9852 | 6.4057 | **897.4119** | 0.5329 | 0.6557 | 2.1285 | 8.1893 | 0.4529 | 11.2034 |
| PIQE | #1 | 1.3361 | 0.6618 | 2.7488 | 0.5909 | 1.6723 | 2.8192 | 0.8975 | 3.5507 | 1564.9631 | **0.4415** | 1.0779 | 1.8209 | 1.5596 | 1.6337 | 1.0673 |
| PIQE | #2 | 1.3112 | 0.7916 | 1.5108 | 0.6111 | 1.7558 | 2.3560 | 0.8359 | 2.1953 | 1745.4179 | **0.4684** | 0.7602 | 1.4860 | 4.3117 | 1.6084 | 2.8031 |
| PIQE | #3 | 1.3652 | 0.5933 | 3.1218 | 0.5397 | 1.3704 | 5.9693 | 1.5504 | 2.6749 | 1389.7482 | **0.3985** | 1.6382 | 2.6655 | 2.5462 | 1.6473 | 22.9693 |
| SNR | #1 | 1.4356 | 1.0601 | 0.8201 | 0.6542 | 1.7289 | 0.9499 | 1.7230 | 3.5424 | 1019.1440 | 0.5008 | **0.9363** | 1.8180 | 10.4784 | 0.6692 | 22.5513 |
| SNR | #2 | 2.1514 | 1.4608 | 1.0797 | 0.5795 | 2.3180 | 1.2112 | 1.6789 | 4.4739 | 1031.1945 | 0.5994 | **1.4383** | 2.9626 | 7.3799 | 0.8781 | 8.6274 |
| SNR | #3 | 3.6490 | 1.3212 | 1.1746 | 0.7247 | 2.7323 | 1.3497 | 1.7805 | 10.2738 | 1141.2476 | 0.5301 | **1.6090** | 3.6695 | 11.1009 | 0.6987 | 11.8924 |
| VIF | #1 | 2.2109 | 1.4419 | 0.7967 | 0.5631 | 0.7346 | 0.6633 | 1.0216 | 4.1458 | 996.8894 | 0.4968 | 0.6768 | **1.5355** | 8.7300 | 0.5450 | 21.0487 |
| VIF | #2 | 1.7544 | 1.3443 | 0.7580 | 0.6553 | 1.2635 | 0.9529 | 1.9261 | 4.3757 | 983.5865 | 0.4997 | 1.0261 | **1.4698** | 11.8503 | 0.7191 | 18.9276 |
| VIF | #3 | 3.7252 | 2.1812 | 0.7924 | 0.5670 | 3.4120 | 0.9859 | 1.6779 | 10.8273 | 1533.3929 | 0.4696 | 1.1281 | **1.6856** | 15.5570 | 0.7355 | 25.3043 |
| FSIM | #1 | 1.9805 | 1.1276 | 0.7151 | 0.6331 | 1.9383 | 0.9055 | 0.9778 | 6.3593 | 908.8211 | 0.5964 | 0.8408 | 1.3971 | **4.7178** | 1.0912 | 4.7182 |
| FSIM | #2 | 2.0797 | 1.0979 | 1.5826 | 0.4839 | 0.6021 | 2.1154 | 0.6618 | 3.3075 | 1525.8824 | 0.4453 | 0.6793 | 1.4206 | **5.7692** | 1.0206 | 4.8553 |
| FSIM | #3 | 2.2890 | 1.1646 | 1.5413 | 0.4453 | 0.7212 | 1.9659 | 0.7565 | 3.4504 | 1509.9996 | 0.4117 | 0.7223 | 1.4279 | **5.5166** | 0.8375 | 7.9219 |
| GMSD | #1 | 2.8633 | 1.6445 | 1.3120 | 0.5163 | 0.6733 | 0.7996 | 0.7410 | 3.6206 | 1107.8767 | 0.6637 | 0.6659 | 2.1917 | 4.6857 | **0.6742** | 8.0722 |
| GMSD | #2 | 1.6692 | 0.7579 | 1.9078 | 0.6333 | 2.7694 | 1.5527 | 1.5991 | 2.6409 | 1000.3264 | 0.6184 | 1.7310 | 4.3095 | 6.2621 | **1.3220** | 14.3959 |
| GMSD | #3 | 5.9690 | 3.8586 | 1.0580 | 0.5051 | 0.8580 | 1.4111 | 0.8210 | 9.6668 | 1015.0806 | 0.6098 | 0.6995 | 1.5109 | 5.6129 | **1.0030** | 15.1541 |
| NCC | #1 | 1.8643 | 1.0048 | 1.4374 | 0.4005 | 1.4662 | 1.6542 | 1.0808 | 2.8285 | 1424.3831 | 0.4028 | 0.8219 | 2.2544 | 5.2085 | 1.5752 | **5.3235** |
| NCC | #2 | 2.0364 | 1.1443 | 1.2741 | 0.3994 | 1.0867 | 1.1213 | 0.9878 | 3.0964 | 1283.5109 | 0.5633 | 0.7477 | 1.5036 | 6.3757 | 1.4466 | **8.4098** |
| NCC | #3 | 1.8580 | 0.7496 | 1.9916 | 0.6344 | 1.8871 | 2.4785 | 2.2127 | 5.2485 | 1056.3778 | 0.4895 | 2.1691 | 2.1063 | 12.3762 | 0.4795 | **8.2892** |

Bold values indicates the ability to distinguish between different images with the same content for the corresponding Normalized (Localized)-IQM values

**Table 12** *Localized-IQM* values obtained from experiments performed using Test Image #2 from the Earth Observation Image Dataset

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| | | 20.0000 | 200.0000 | 50.0000 | 100.0000 | 0.8000 | 45.0000 | 0.8000 | 25.0000 | – | – | 10.0000 | 0.5000 | 0.9000 | 0.3000 | 0.9000 |
| MAE | #1 | **2.6413** | 1.4621 | 0.7534 | 0.7632 | 3.1418 | 0.4766 | 2.4522 | 7.1455 | 681.5882 | 0.4500 | 1.6020 | 2.1019 | 6.3106 | 0.5488 | 6.0694 |
| MAE | #2 | **1.5994** | 0.8238 | 0.9547 | 0.7192 | 2.3437 | 0.8427 | 4.1804 | 6.9873 | 822.5139 | 0.4203 | 1.6462 | 2.1749 | 21.3499 | 0.6287 | 9.6428 |
| MAE | #3 | **1.7758** | 1.2616 | 0.6408 | 0.6666 | 2.8695 | 0.4945 | 3.8822 | 4.1095 | 664.7995 | 0.4316 | 1.3869 | 1.4632 | 26.5291 | 0.6175 | 13.6250 |
| MSE | #1 | 1.1771 | **0.6673** | 0.5573 | 0.6321 | 6.0368 | 0.5962 | 4.5339 | 3.3630 | 774.7652 | 0.4632 | 1.4587 | 2.4270 | 16.2478 | 0.4311 | 21.8301 |
| MSE | #2 | 1.1364 | **0.7565** | 0.7768 | 0.8604 | 2.9869 | 0.6597 | 6.5169 | 4.4105 | 845.5312 | 0.4392 | 2.0372 | 3.3958 | 19.1065 | 0.3976 | 14.3672 |
| MSE | #3 | 1.1414 | **0.7951** | 0.6010 | 0.7236 | 4.4102 | 0.5013 | 4.9077 | 3.9427 | 841.5570 | 0.4338 | 1.7224 | 3.3142 | 45.3880 | 0.5829 | 26.4790 |
| ERGAS | #1 | 4.6703 | 1.8559 | **0.9710** | 0.6890 | 1.7584 | 0.9174 | 2.6883 | 6.6611 | 914.6429 | 0.4116 | 1.5520 | 3.1128 | 8.8195 | 0.5730 | 15.4589 |
| ERGAS | #2 | 2.5709 | 1.1053 | **0.8955** | 0.7731 | 1.1495 | 0.9848 | 2.3540 | 4.8514 | 832.3557 | 0.4704 | 1.1139 | 1.5580 | 15.8667 | 0.5723 | 10.9384 |
| ERGAS | #3 | 2.1497 | 1.3367 | **1.1036** | 0.7430 | 1.0697 | 1.3866 | 1.8522 | 4.2955 | 725.0268 | 0.4108 | 1.4531 | 1.4698 | 14.9474 | 0.7962 | 11.4169 |
| LMSE | #1 | 6.2439 | 3.7486 | 1.0408 | **0.4277** | 4.3438 | 0.5310 | 5.4108 | 10.5259 | 863.0264 | 0.4257 | 1.6869 | 5.0877 | 27.1874 | 0.1757 | 266.3045 |
| LMSE | #2 | 7.6721 | 3.3812 | 0.9765 | **0.5582** | 7.3624 | 0.6246 | 5.3752 | 12.7611 | 858.8550 | 0.4375 | 2.0710 | 6.3172 | 46.1794 | 0.4377 | 204.4552 |
| LMSE | #3 | 6.1430 | 2.7256 | 0.8993 | **0.4394** | 4.5487 | 0.5450 | 6.1497 | 12.9524 | 841.1758 | 0.4058 | 1.5551 | 3.6713 | 42.9448 | 0.2903 | 182.8695 |
| SSIM | #1 | 2.2596 | 1.4684 | 0.9983 | 0.6100 | **1.4580** | 0.7234 | 1.8117 | 4.4618 | 724.9364 | 0.4001 | 1.0068 | 2.4207 | 11.4970 | 0.7694 | 15.1709 |
| SSIM | #2 | 1.4060 | 0.7275 | 1.0546 | 0.5265 | **1.7143** | 1.0444 | 2.1230 | 3.0059 | 841.0264 | 0.4309 | 0.9726 | 1.6169 | 5.9798 | 0.6593 | 4.2570 |
| SSIM | #3 | 2.2704 | 1.0873 | 0.9924 | 0.6689 | **1.6362** | 1.5756 | 2.3234 | 6.1101 | 554.5310 | 0.4742 | 1.1527 | 2.9909 | 9.8372 | 0.7475 | 9.3484 |
| SAM | #1 | 1.8909 | 0.9053 | 2.1994 | 0.4804 | 0.4170 | **3.2257** | 1.8360 | 3.0003 | 917.9053 | 0.4403 | 0.9300 | 2.2623 | 8.5549 | 0.5703 | 7.7827 |
| SAM | #2 | 1.4754 | 0.8128 | 2.0582 | 0.3987 | 0.4258 | **2.5637** | 1.3890 | 2.4036 | 928.2511 | 0.4527 | 0.8085 | 1.2514 | 9.7100 | 0.4410 | 3.5426 |
| SAM | #3 | 2.1120 | 1.2218 | 1.6202 | 0.4728 | 0.4425 | **3.1217** | 1.6057 | 2.4985 | 839.4140 | 0.4114 | 0.6084 | 2.3553 | 5.2645 | 1.3888 | 6.4365 |
| UIQI | #1 | 1.7842 | 1.0874 | 0.8654 | 0.7654 | 0.6133 | 1.1545 | **2.8332** | 4.8630 | 825.7071 | 0.4753 | 1.0061 | 2.4658 | 10.7119 | 0.7020 | 8.6335 |
| UIQI | #2 | 3.8197 | 1.4297 | 1.1688 | 0.8658 | 0.8753 | 1.7366 | **2.6008** | 5.3692 | 698.2408 | 0.4325 | 1.5378 | 2.2116 | 15.6806 | 0.5529 | 11.1837 |
| UIQI | #3 | 2.3949 | 1.5653 | 0.9691 | 0.7469 | 1.7984 | 1.3961 | **1.8354** | 4.8604 | 536.0992 | 0.4795 | 1.4154 | 2.9827 | 7.2051 | 0.6502 | 13.1086 |
| PSNR | #1 | 1.0291 | 0.4619 | 1.2165 | 0.5262 | 1.3962 | 1.3529 | 3.9236 | **7.4923** | 765.8303 | 0.4041 | 1.8980 | 3.6407 | 19.2318 | 0.4425 | 8.9227 |
| PSNR | #2 | 1.3303 | 0.6509 | 0.8060 | 0.7319 | 3.0487 | 0.5754 | 3.9730 | **3.9951** | 683.6039 | 0.4920 | 2.2809 | 3.1543 | 11.4263 | 0.5897 | 20.5511 |
| PSNR | #3 | 1.6947 | 0.9053 | 0.8658 | 0.7557 | 5.6414 | 0.5081 | 7.9850 | **4.6349** | 724.3147 | 0.4494 | 2.4252 | 2.2906 | 19.4545 | 0.4453 | 13.2802 |
| NIQE | #1 | 2.1332 | 0.9489 | 1.3961 | 0.4357 | 0.8034 | 0.8380 | 1.0852 | 3.3634 | **724.7837** | 0.4971 | 0.9844 | 1.6034 | 6.1687 | 0.6507 | 4.0056 |
| NIQE | #2 | 1.5546 | 0.9935 | 0.7041 | 0.6199 | 1.3408 | 0.8482 | 1.7670 | 2.8362 | **628.2771** | 0.4423 | 1.1843 | 1.9820 | 11.6237 | 0.5401 | 7.4798 |

**Table 12** (continued)

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| NIQE | #3 | 3.2283 | 1.4094 | 1.4148 | 0.6024 | 0.3831 | 1.8765 | 1.7303 | 6.7164 | **692.3767** | 0.4452 | 1.1427 | 2.3951 | 9.2459 | 0.3856 | 7.7302 |
| PIQE | #1 | 1.4264 | 0.8611 | 1.9668 | 0.4490 | 0.5887 | 1.1317 | 0.4727 | 1.5065 | 1038.5903 | **0.3727** | 0.5840 | 1.2415 | 1.8917 | 0.8324 | 3.1121 |
| PIQE | #2 | 2.1170 | 0.9486 | 1.6890 | 0.5956 | 0.9703 | 2.1495 | 1.5962 | 3.0628 | 688.4026 | **0.3918** | 1.5009 | 1.8432 | 4.6918 | 0.3660 | 8.5711 |
| PIQE | #3 | 2.1080 | 1.0873 | 1.7058 | 0.5840 | 1.3210 | 1.4758 | 1.5248 | 2.6473 | 731.1722 | **0.3674** | 1.3644 | 2.2860 | 3.6152 | 0.3787 | 6.9908 |
| SNR | #1 | 1.2069 | 0.7298 | 0.8586 | 0.7229 | 1.3758 | 0.5521 | 2.0360 | 2.3700 | 554.7702 | 0.4983 | **1.3454** | 2.5097 | 4.6424 | 0.6053 | 11.2206 |
| SNR | #2 | 2.0517 | 1.1943 | 0.8131 | 0.6887 | 1.1461 | 0.7767 | 1.9495 | 3.8816 | 714.6212 | 0.5014 | **1.1384** | 2.1200 | 7.3972 | 0.2904 | 7.8333 |
| SNR | #3 | 1.7641 | 0.9475 | 0.9543 | 0.4286 | 1.5358 | 0.8959 | 2.4765 | 3.3174 | 744.0162 | 0.4752 | **1.2361** | 1.9558 | 8.9087 | 0.4809 | 8.8799 |
| VIF | #1 | 2.9551 | 1.6626 | 1.4613 | 0.6562 | 0.9082 | 0.9213 | 1.2698 | 4.3392 | 720.0884 | 0.4069 | 1.0015 | **2.8480** | 8.8787 | 0.4067 | 14.6595 |
| VIF | #2 | 2.5524 | 1.2246 | 1.3390 | 0.5274 | 1.3180 | 1.7035 | 1.5901 | 4.6717 | 604.5751 | 0.4414 | 1.3423 | **1.6814** | 5.3539 | 0.5821 | 17.8181 |
| VIF | #3 | 2.7432 | 1.3221 | 1.3643 | 0.3937 | 1.1545 | 1.1408 | 2.1852 | 4.5016 | 770.8506 | 0.4104 | 1.1236 | **1.7748** | 11.3207 | 0.3528 | 11.2338 |
| FSIM | #1 | 1.4144 | 0.8327 | 0.8498 | 0.5951 | 0.7555 | 1.2295 | 1.9764 | 3.1816 | 722.5183 | 0.4311 | 0.9311 | 2.0580 | **4.1873** | 0.7429 | 5.1659 |
| FSIM | #2 | 2.2093 | 0.9081 | 1.5783 | 0.4589 | 1.0491 | 1.5148 | 1.1373 | 4.1433 | 787.4289 | 0.4350 | 1.2393 | 2.0906 | **5.0676** | 0.6926 | 6.8599 |
| FSIM | #3 | 1.2491 | 0.8989 | 0.6303 | 0.7211 | 0.7668 | 0.6111 | 1.6377 | 2.4563 | 512.6082 | 0.4447 | 0.8725 | 2.1786 | **5.4023** | 0.7061 | 6.5487 |
| GMSD | #1 | 1.6971 | 1.0188 | 1.1185 | 0.6119 | 0.5750 | 1.4473 | 0.5180 | 2.2171 | 596.3639 | 0.5237 | 0.5136 | 0.6749 | 1.9516 | **0.5728** | 3.5854 |
| GMSD | #2 | 1.3290 | 0.7111 | 2.1420 | 0.6786 | 0.6966 | 2.8450 | 1.6470 | 2.9153 | 649.6027 | 0.4847 | 3.2208 | 2.8999 | 3.3018 | **0.8027** | 6.3379 |
| GMSD | #3 | 1.4851 | 0.6293 | 1.8262 | 0.6014 | 1.1347 | 2.9283 | 1.1493 | 3.0072 | 775.0455 | 0.4857 | 2.3163 | 0.8494 | 3.8036 | **0.6432** | 2.9850 |
| NCC | #1 | 2.0018 | 1.0452 | 1.2039 | 0.4780 | 1.2190 | 1.6323 | 1.2898 | 4.7883 | 568.6601 | 0.4494 | 0.8315 | 2.3851 | 7.3664 | 0.9513 | **6.2866** |
| NCC | #2 | 1.8070 | 1.1667 | 1.1714 | 0.4870 | 1.0134 | 1.4319 | 1.2892 | 2.8202 | 619.3813 | 0.4634 | 0.8298 | 1.9205 | 5.8452 | 0.6468 | **5.6688** |
| NCC | #3 | 1.3008 | 0.7637 | 1.7304 | 0.7880 | 0.7279 | 2.2740 | 2.0350 | 2.5168 | 907.9288 | 0.4469 | 1.4647 | 1.6671 | 14.3231 | 0.6445 | **4.7394** |

Bold values indicates the ability to distinguish between different images with the same content for the corresponding Normalized (Localized)-IQM values

**Table 13** *Localized-IQM* values obtained from experiments performed using Test Image #3 from the Earth Observation Image Dataset

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| | | 20.0000 | 200.0000 | 50.0000 | 100.0000 | 0.8000 | 45.0000 | 0.8000 | 25.0000 | – | – | 10.0000 | 0.5000 | 0.9000 | 0.3000 | 0.9000 |
| MAE | #1 | **3.9040** | 2.0747 | 1.1243 | 0.7506 | 3.6229 | 1.5829 | 2.2402 | 10.4301 | 586.0033 | 0.3960 | 2.3210 | 5.1810 | 12.2688 | 0.6287 | 15.8970 |
| MAE | #2 | **2.8447** | 1.8258 | 1.0336 | 0.8093 | 4.1853 | 1.1844 | 3.9988 | 7.2905 | 630.2227 | 0.3934 | 2.0272 | 2.4060 | 11.0044 | 0.8094 | 27.4602 |
| MAE | #3 | **2.6903** | 1.7416 | 0.9564 | 0.7807 | 2.9767 | 1.1926 | 3.4746 | 11.3883 | 1059.8254 | 0.3208 | 1.7200 | 4.3838 | 42.4606 | 0.5555 | 19.9995 |
| MSE | #1 | 4.4824 | **2.5666** | 1.1163 | 0.7944 | 9.5467 | 1.4070 | 7.4810 | 15.1523 | 1008.6135 | 0.3867 | 2.7845 | 4.7864 | 13.3158 | 0.7953 | 26.1678 |
| MSE | #2 | 1.4148 | **0.8053** | 0.6168 | 0.6714 | 5.9106 | 0.8518 | 4.0686 | 5.9689 | 1007.1800 | 0.3287 | 1.7322 | 3.6582 | 46.4588 | 0.3741 | 47.4244 |
| MSE | #3 | 3.4532 | **1.8626** | 1.3964 | 0.8268 | 10.9320 | 1.2961 | 6.2026 | 10.6674 | 1188.2387 | 0.3186 | 3.6473 | 4.3005 | 13.5864 | 0.4956 | 23.1027 |
| ERGAS | #1 | 1.3918 | 0.7495 | **3.8239** | 0.7484 | 2.2722 | 3.7687 | 3.9490 | 2.9888 | 572.1881 | 0.3691 | 9.3298 | 4.7175 | 18.5677 | 0.6924 | 24.8535 |
| ERGAS | #2 | 2.9466 | 1.3606 | **2.0677** | 0.7170 | 0.8143 | 3.0428 | 3.4919 | 7.3738 | 995.0710 | 0.3581 | 2.5631 | 3.6495 | 7.9883 | 0.7743 | 16.5895 |
| ERGAS | #3 | 2.1360 | 1.2622 | **0.9556** | 0.7787 | 1.1135 | 1.0345 | 2.8741 | 8.7612 | 607.1297 | 0.3876 | 1.0690 | 4.4506 | 17.2981 | 0.8582 | 31.4040 |
| LMSE | #1 | 5.9995 | 3.0031 | 1.3851 | **0.6568** | 16.8285 | 1.2130 | 11.1421 | 16.4133 | 998.8457 | 0.3585 | 2.7520 | 7.1535 | 76.7947 | 0.6393 | 228.1424 |
| LMSE | #2 | 8.6724 | 2.2908 | 1.6064 | **0.6016** | 10.7406 | 1.7193 | 17.3208 | 22.0624 | 1135.7401 | 0.3252 | 3.5177 | 6.6726 | 52.9399 | 0.4238 | 360.3019 |
| LMSE | #3 | 7.6084 | 3.4715 | 1.6602 | **0.6891** | 8.4209 | 1.5322 | 10.2157 | 20.9915 | 1031.5699 | 0.3422 | 2.3660 | 7.1074 | 43.6013 | 0.3879 | 319.0821 |
| SSIM | #1 | 7.4638 | 4.0351 | 1.4500 | 0.7201 | **1.3824** | 1.6706 | 2.2908 | 10.8470 | 658.6641 | 0.3906 | 1.2977 | 4.0957 | 14.1873 | 0.8086 | 28.4313 |
| SSIM | #2 | 2.9115 | 1.3617 | 1.9259 | 0.6442 | **2.9380** | 2.2656 | 2.3768 | 5.6535 | 980.0129 | 0.3296 | 1.8711 | 3.3738 | 8.7717 | 0.9072 | 10.3008 |
| SSIM | #3 | 3.5114 | 2.2086 | 1.3450 | 0.6973 | **2.1299** | 2.4243 | 1.8890 | 8.4161 | 550.1779 | 0.3999 | 1.4116 | 3.7212 | 11.2621 | 0.6841 | 15.0448 |
| SAM | #1 | 3.3085 | 1.7046 | 1.5149 | 0.7159 | 0.1916 | **2.4572** | 1.1406 | 4.9143 | 1373.7635 | 0.4065 | 0.7002 | 2.1475 | 6.5094 | 1.2323 | 6.0825 |
| SAM | #2 | 2.8711 | 1.4665 | 2.1733 | 0.5425 | 0.6945 | **4.7358** | 1.5325 | 3.3973 | 1469.4721 | 0.4026 | 0.7369 | 2.0524 | 5.0196 | 1.4229 | 6.2008 |
| SAM | #3 | 3.1926 | 1.8716 | 1.2892 | 0.7018 | 0.2291 | **2.2145** | 1.0789 | 4.5498 | 1116.4254 | 0.3867 | 0.5890 | 1.8879 | 10.7135 | 0.9943 | 5.3421 |
| UIQI | #1 | 4.2140 | 2.0150 | 0.9909 | 0.7828 | 3.1442 | 1.9774 | **2.4471** | 13.0282 | 715.4075 | 0.4096 | 1.6109 | 4.2329 | 12.6740 | 1.2502 | 11.8893 |
| UIQI | #2 | 4.4869 | 1.9105 | 1.5257 | 0.7921 | 2.5851 | 3.2021 | **2.6192** | 8.9659 | 671.4962 | 0.3931 | 1.7518 | 3.7194 | 14.0607 | 0.9687 | 17.7598 |
| UIQI | #3 | 2.7849 | 1.7416 | 1.1708 | 0.8934 | 2.2069 | 2.1869 | **2.0895** | 6.6344 | 509.1455 | 0.4245 | 1.4864 | 2.9640 | 10.8501 | 0.9362 | 18.8716 |
| PSNR | #1 | 3.1821 | 1.6145 | 1.4867 | 0.7297 | 7.0606 | 1.3585 | 8.7376 | **8.5195** | 891.3819 | 0.3832 | 4.5118 | 2.2509 | 18.6426 | 0.5388 | 34.4189 |
| PSNR | #2 | 2.5971 | 1.3956 | 0.7302 | 0.8383 | 5.4218 | 1.1080 | 4.4344 | **10.4704** | 1227.8265 | 0.3613 | 2.0094 | 5.3113 | 19.4850 | 0.6095 | 30.7669 |
| PSNR | #3 | 1.9257 | 1.1498 | 0.9538 | 0.8997 | 5.3748 | 1.0187 | 4.7066 | **7.1183** | 924.6797 | 0.4104 | 1.9550 | 4.7389 | 9.7039 | 0.7378 | 20.1417 |
| NIQE | #1 | 3.1122 | 1.7168 | 2.2052 | 0.4927 | 0.6082 | 1.3261 | 0.7205 | 3.4803 | **976.2138** | 0.3942 | 0.8010 | 1.9614 | 4.5597 | 1.1928 | 9.2665 |
| NIQE | #2 | 4.2385 | 2.0970 | 1.4944 | 0.5541 | 0.8506 | 2.8270 | 0.9388 | 7.0846 | **823.4827** | 0.3640 | 0.9231 | 2.6184 | 9.8207 | 0.7495 | 4.7211 |

**Table 13** (continued)

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| NIQE | #3 | 2.4143 | 1.3183 | 1.9491 | 0.6284 | 2.5588 | 1.3841 | 1.8572 | 3.5011 | **765.6560** | 0.4149 | 1.9397 | 3.2943 | 7.6987 | 0.6498 | 17.1286 |
| PIQE | #1 | 1.6631 | 0.8120 | 5.0470 | 0.6239 | 0.9900 | 4.5280 | 2.6944 | 3.0233 | 1429.0580 | **0.3432** | 1.9927 | 2.1795 | 3.2633 | 1.6610 | 6.5675 |
| PIQE | #2 | 1.6737 | 0.9145 | 3.7195 | 0.6494 | 1.1782 | 4.2193 | 1.5351 | 2.7011 | 1401.2092 | **0.3166** | 1.7596 | 1.5609 | 3.5127 | 2.0307 | 5.1475 |
| PIQE | #3 | 1.8642 | 0.9768 | 3.2342 | 0.6396 | 1.2829 | 4.0741 | 1.3661 | 3.1213 | 1383.7548 | **0.2983** | 1.6167 | 2.0044 | 3.3188 | 2.1197 | 6.2549 |
| SNR | #1 | 2.3106 | 1.2095 | 2.0726 | 0.5691 | 3.6388 | 1.2571 | 3.9282 | 3.8488 | 876.9085 | 0.3545 | **3.1348** | 1.9187 | 18.2659 | 0.7292 | 20.8205 |
| SNR | #2 | 3.0730 | 1.3792 | 1.4375 | 0.6584 | 2.6662 | 3.1589 | 2.5976 | 8.9809 | 778.4224 | 0.3972 | **2.8545** | 1.4615 | 22.2821 | 0.9044 | 34.5142 |
| SNR | #3 | 4.9920 | 2.8364 | 1.1739 | 0.7252 | 2.0220 | 1.2632 | 2.6696 | 11.0448 | 583.3762 | 0.3811 | **1.4161** | 1.8379 | 9.0717 | 0.9623 | 17.3167 |
| VIF | #1 | 5.6547 | 3.6832 | 1.0595 | 0.6980 | 3.0112 | 1.7601 | 2.3655 | 12.9807 | 979.5107 | 0.3862 | 1.4687 | **3.7537** | 22.7457 | 0.5863 | 15.3149 |
| VIF | #2 | 3.3337 | 1.7959 | 1.4897 | 0.5817 | 4.3547 | 1.0376 | 3.2541 | 5.3632 | 983.6484 | 0.3532 | 2.5538 | **2.4564** | 26.7603 | 0.7969 | 48.4050 |
| VIF | #3 | 5.7879 | 3.4219 | 0.9775 | 0.6189 | 3.0411 | 1.2270 | 2.7098 | 20.1628 | 700.3327 | 0.3586 | 1.6831 | **1.9150** | 27.8366 | 0.9774 | 23.0222 |
| FSIM | #1 | 3.6479 | 1.9399 | 1.9755 | 0.5120 | 1.0425 | 1.6679 | 1.1672 | 4.8968 | 1096.6201 | 0.3847 | 1.0480 | 2.2360 | **6.2140** | 0.8681 | 17.2112 |
| FSIM | #2 | 2.9812 | 1.6200 | 2.1181 | 0.5827 | 0.7620 | 1.6899 | 0.9405 | 3.7500 | 865.2744 | 0.3565 | 1.0194 | 2.6817 | **7.2559** | 0.9317 | 10.4547 |
| FSIM | #3 | 6.2048 | 2.5558 | 1.7852 | 0.5729 | 1.1298 | 4.2113 | 1.1240 | 10.9999 | 788.8147 | 0.4356 | 1.0261 | 2.1987 | **6.4632** | 1.1227 | 12.2735 |
| GMSD | #1 | 3.0987 | 1.7744 | 1.7359 | 0.6704 | 0.5646 | 1.6539 | 0.6370 | 4.0543 | 772.2871 | 0.5211 | 0.7374 | 1.0770 | 3.9999 | **2.4214** | 3.6987 |
| GMSD | #2 | 2.2116 | 1.0052 | 2.5431 | 0.5718 | 1.8973 | 4.5809 | 1.0826 | 4.1823 | 564.3254 | 0.5257 | 1.7034 | 3.3832 | 6.6899 | **0.9082** | 5.7560 |
| GMSD | #3 | 3.6260 | 1.5124 | 1.7954 | 0.6843 | 1.2850 | 2.3925 | 1.0940 | 6.7235 | 467.0041 | 0.5098 | 1.3251 | 3.4504 | 6.4737 | **0.9046** | 7.1123 |
| NCC | #1 | 3.8383 | 2.2377 | 1.4976 | 0.6164 | 0.8979 | 1.6698 | 0.8194 | 5.9583 | 1119.4305 | 0.5032 | 0.8355 | 2.3105 | 4.6108 | 1.8951 | **7.6562** |
| NCC | #2 | 4.0403 | 1.9054 | 1.6015 | 0.7707 | 1.3976 | 2.7323 | 1.4619 | 8.2467 | 707.3028 | 0.4114 | 1.2559 | 1.9692 | 8.6563 | 0.9607 | **12.1733** |
| NCC | #3 | 2.0115 | 0.9693 | 3.2187 | 0.6989 | 1.1623 | 3.8422 | 2.4084 | 3.7633 | 675.6705 | 0.3885 | 2.1601 | 2.1114 | 6.8706 | 1.3472 | **9.6583** |

Bold values indicates the ability to distinguish between different images with the same content for the corresponding Normalized (Localized)-IQM values

**Table 14** *Localized-IQM* values obtained from experiments performed using Test Image #4 from the Earth Observation Image Dataset

| Tested IQM | Experiment | Normalized-IQMs | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
| | | 20.0000 | 200.0000 | 50.0000 | 100.0000 | 0.8000 | 45.0000 | 0.8000 | 25.0000 | – | – | 10.0000 | 0.5000 | 0.9000 | 0.3000 | 0.9000 |
| MAE | #1 | **2.0640** | 1.1316 | 0.9492 | 0.8767 | 1.4980 | 0.9282 | 1.3638 | 5.5256 | 902.9775 | 0.4977 | 1.2338 | 2.2797 | 8.7137 | 1.7847 | 11.1010 |
| MAE | #2 | **1.6093** | 1.0365 | 0.7051 | 0.8281 | 1.1025 | 0.8656 | 1.3385 | 4.9071 | 995.0417 | 0.4825 | 0.8751 | 1.7630 | 12.4682 | 0.4477 | 15.1264 |
| MAE | #3 | **1.9102** | 0.7475 | 1.5459 | 0.5420 | 1.7297 | 1.8491 | 2.4300 | 4.7542 | 1089.3169 | 0.5148 | 1.7095 | 2.2095 | 12.9383 | 0.3521 | 8.2998 |
| MSE | #1 | 1.6728 | **1.1726** | 0.8094 | 0.5546 | 2.2649 | 0.7662 | 2.5096 | 4.1939 | 1286.6040 | 0.4909 | 1.6040 | 2.9978 | 16.9842 | 0.3456 | 41.5288 |
| MSE | #2 | 1.5347 | **0.9271** | 0.6418 | 0.8165 | 1.9728 | 0.6717 | 2.1014 | 4.9952 | 1153.3854 | 0.4640 | 1.2364 | 2.8510 | 14.9455 | 0.2941 | 34.8688 |
| MSE | #3 | 1.2171 | **0.8145** | 0.9388 | 0.7190 | 2.3917 | 0.5424 | 2.5069 | 4.3527 | 1321.6307 | 0.5093 | 1.9342 | 2.2574 | 13.4114 | 0.4771 | 13.2946 |
| ERGAS | #1 | 3.0088 | 1.5553 | **1.0789** | 0.8016 | 1.2709 | 0.7815 | 1.7493 | 8.1396 | 1161.8319 | 0.5580 | 1.3331 | 2.9349 | 13.0074 | 0.4683 | 8.2835 |
| ERGAS | #2 | 1.9965 | 1.1064 | **0.7831** | 0.7044 | 0.9325 | 0.8340 | 1.4510 | 8.0109 | 1221.3127 | 0.4794 | 0.9447 | 2.7996 | 11.3414 | 0.3741 | 8.8302 |
| ERGAS | #3 | 2.3182 | 1.9151 | **0.8362** | 0.5846 | 1.5760 | 0.8189 | 1.5133 | 8.2944 | 1411.2922 | 0.4675 | 1.0381 | 2.6056 | 9.8186 | 0.3600 | 26.4611 |
| LMSE | #1 | 3.7006 | 1.7802 | 1.1537 | **0.4849** | 1.9566 | 0.5433 | 1.9989 | 8.9872 | 1355.3063 | 0.4373 | 1.6405 | 3.8223 | 11.6250 | 0.2695 | 110.6832 |
| LMSE | #2 | 4.7019 | 2.6878 | 0.8533 | **0.5999** | 2.0338 | 0.5166 | 2.1758 | 12.7627 | 1322.6965 | 0.4644 | 1.3918 | 4.2315 | 13.1480 | 0.2646 | 106.2717 |
| LMSE | #3 | 4.4886 | 2.7738 | 0.8655 | **0.5441** | 1.8500 | 0.4832 | 1.9616 | 14.0557 | 1303.5816 | 0.4522 | 1.3086 | 3.4192 | 12.3246 | 0.2644 | 108.3310 |
| SSIM | #1 | 2.9315 | 1.4147 | 1.2805 | 0.4495 | **1.7930** | 0.8706 | 1.8174 | 6.8049 | 1245.8771 | 0.5101 | 1.3474 | 2.3735 | 11.8964 | 0.3593 | 22.9952 |
| SSIM | #2 | 4.3434 | 1.9411 | 1.0960 | 0.7425 | **1.0109** | 1.2949 | 1.6214 | 14.1804 | 1156.8322 | 0.5531 | 1.1478 | 3.5152 | 12.7959 | 0.3844 | 11.9063 |
| SSIM | #3 | 2.8080 | 1.4126 | 1.1132 | 0.8820 | **1.3115** | 1.2077 | 1.4820 | 7.5771 | 1212.9980 | 0.5079 | 1.2466 | 2.2721 | 7.6908 | 1.3186 | 7.7561 |
| SAM | #1 | 1.1990 | 0.6163 | 2.9559 | 0.5033 | 0.7292 | **5.1851** | 1.3793 | 3.1665 | 1436.4148 | 0.4576 | 0.8703 | 3.7517 | 5.7438 | 0.8211 | 2.7016 |
| SAM | #2 | 1.4599 | 1.0406 | 1.2666 | 0.7122 | 0.5937 | **3.3037** | 0.6859 | 2.9169 | 1006.9862 | 0.5766 | 0.3313 | 2.8301 | 11.0017 | 0.5248 | 6.2645 |
| SAM | #3 | 1.2922 | 0.7151 | 2.6689 | 0.5951 | 0.8021 | **4.2685** | 1.2525 | 2.3075 | 930.2726 | 0.4956 | 0.9332 | 2.1897 | 5.6701 | 0.6576 | 6.4718 |
| UIQI | #1 | 3.9013 | 1.9334 | 1.1331 | 0.4556 | 1.9024 | 1.0624 | **2.0580** | 8.6901 | 1236.9271 | 0.5177 | 1.3920 | 2.4356 | 12.6761 | 0.3539 | 19.0319 |
| UIQI | #2 | 2.2551 | 1.2243 | 1.2231 | 0.7210 | 1.8807 | 1.2673 | **2.2070** | 6.2005 | 1251.3903 | 0.4788 | 1.4597 | 2.2107 | 10.7423 | 0.4328 | 41.0151 |
| UIQI | #3 | 1.2096 | 0.6938 | 1.7950 | 0.5657 | 2.2530 | 0.7935 | **2.4469** | 3.0044 | 1392.2349 | 0.4210 | 2.9270 | 1.8854 | 12.4131 | 0.3927 | 14.2512 |
| PSNR | #1 | 1.9766 | 1.1785 | 0.8375 | 0.7266 | 1.8509 | 0.7529 | 2.6265 | **5.4646** | 1136.5965 | 0.4815 | 1.5947 | 1.2659 | 14.6150 | 0.4357 | 13.8604 |
| PSNR | #2 | 1.6903 | 0.8801 | 0.9069 | 0.5721 | 2.3994 | 0.6772 | 2.6977 | **4.2641** | 1018.4372 | 0.4432 | 1.8079 | 1.1244 | 12.5174 | 0.4307 | 31.1728 |
| PSNR | #3 | 1.7275 | 0.9669 | 0.9364 | 0.6721 | 2.1939 | 0.7786 | 2.3559 | **4.6775** | 1246.0598 | 0.4836 | 1.8723 | 3.2792 | 13.8150 | 0.3060 | 21.0346 |
| NIQE | #1 | 1.0777 | 0.6245 | 3.7048 | 0.5168 | 0.5206 | 1.5230 | 0.3723 | 1.5473 | **1441.1724** | 0.5639 | 0.5542 | 1.0689 | 2.0095 | 1.1269 | 1.8539 |
| NIQE | #2 | 1.2011 | 0.6689 | 3.1242 | 0.3976 | 0.5423 | 2.6865 | 0.4030 | 1.8967 | **1074.8543** | 0.6643 | 0.5628 | 1.0413 | 2.1951 | 1.3412 | 1.4850 |

**Table 14** (continued)

| Tested IQM | Experiment | Normalized-IQMs MAE | MSE | ERGAS | LMSE | SSIM | SAM | UIQI | PSNR | NIQE | PIQE | SNR | VIF | FSIM | GMSD | NCC |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NIQE | #3 | 1.2339 | 0.6928 | 3.0432 | 0.4929 | 0.5561 | 2.1046 | 0.4121 | 2.0118 | **1294.2579** | 0.5579 | 0.6173 | 1.0295 | 2.3294 | 1.6098 | 3.8708 |
| PIQE | #1 | 1.2253 | 0.7919 | 1.7802 | 0.5027 | 0.4390 | 2.9024 | 0.3462 | 2.7734 | 1390.7633 | **0.4819** | 0.4446 | 0.6669 | 2.2935 | 1.4585 | 0.5483 |
| PIQE | #2 | 1.3716 | 0.8108 | 1.8601 | 0.5484 | 0.5053 | 4.2252 | 0.3922 | 3.1938 | 1395.6915 | **0.4683** | 0.4985 | 0.7804 | 2.7680 | 2.2394 | 0.7664 |
| PIQE | #3 | 1.2436 | 0.7757 | 2.0285 | 0.5179 | 0.4558 | 3.4806 | 0.3552 | 2.6241 | 1434.1915 | **0.4289** | 0.4445 | 0.6142 | 2.0001 | 1.1735 | 0.5590 |
| SNR | #1 | 1.4359 | 0.8500 | 0.6859 | 0.6808 | 0.9616 | 0.6809 | 1.0201 | 4.1379 | 815.0220 | 0.7683 | **0.8219** | 1.2802 | 6.5167 | 0.8410 | 16.2917 |
| SNR | #2 | 1.0173 | 0.6946 | 0.6150 | 0.6198 | 0.9458 | 0.4481 | 0.9354 | 3.2174 | 851.2809 | 0.7491 | **0.7238** | 1.2529 | 7.2423 | 0.7730 | 15.6264 |
| SNR | #3 | 1.9656 | 1.1942 | 0.8469 | 0.7650 | 1.1822 | 0.3664 | 1.1718 | 5.4002 | 736.4693 | 0.7970 | **1.0268** | 1.3258 | 7.4004 | 0.7984 | 18.7486 |
| VIF | #1 | 2.6619 | 1.3578 | 1.1770 | 0.6545 | 0.9379 | 2.0325 | 0.9533 | 6.7680 | 1024.1772 | 0.7296 | 1.0240 | **1.7312** | 6.7652 | 0.8118 | 19.1879 |
| VIF | #2 | 1.6893 | 1.0822 | 0.8030 | 0.6132 | 0.7859 | 0.5284 | 0.7799 | 4.3328 | 1068.0130 | 0.7261 | 0.7062 | **1.5415** | 6.4986 | 0.8202 | 10.2206 |
| VIF | #3 | 2.4413 | 1.2909 | 1.3167 | 0.5223 | 0.8750 | 1.4461 | 0.8227 | 6.4537 | 1034.3831 | 0.7234 | 0.9158 | **1.3927** | 6.2616 | 0.8252 | 13.1672 |
| FSIM | #1 | 1.4104 | 0.7065 | 2.5748 | 0.5041 | 0.8882 | 4.0697 | 0.8890 | 2.6377 | 1141.9672 | 0.7034 | 1.0512 | 2.3617 | **5.9995** | 0.9306 | 5.4235 |
| FSIM | #2 | 1.6525 | 0.7760 | 2.0590 | 0.4761 | 0.7947 | 3.4562 | 0.9426 | 3.9127 | 1136.4523 | 0.7358 | 0.9469 | 2.1789 | **6.2841** | 1.0597 | 4.6605 |
| FSIM | #3 | 1.9754 | 1.1094 | 1.5161 | 0.4916 | 0.8171 | 1.1590 | 0.7718 | 4.1539 | 1337.6908 | 0.7561 | 0.8482 | 1.8164 | **5.4246** | 0.9241 | 14.2789 |
| GMSD | #1 | 1.3516 | 0.7797 | 1.6687 | 0.4052 | 0.5114 | 1.1261 | 0.5054 | 2.2943 | 1552.1718 | 0.8055 | 0.5003 | 1.6197 | 2.5188 | **1.0777** | 3.0535 |
| GMSD | #2 | 1.2146 | 0.6895 | 2.6491 | 0.4591 | 0.5492 | 2.0490 | 0.5188 | 1.9613 | 1214.6981 | 0.7868 | 0.5816 | 1.3593 | 3.8592 | **1.3289** | 2.5090 |
| GMSD | #3 | 1.6719 | 1.1179 | 1.1944 | 0.5965 | 0.6082 | 1.0700 | 0.5810 | 4.0200 | 1478.5761 | 0.4369 | 0.6150 | 1.3235 | 3.3066 | **1.5236** | 2.3403 |
| NCC | #1 | 1.9947 | 1.2330 | 1.2514 | 0.5479 | 0.7797 | 2.7029 | 0.7173 | 4.7992 | 1382.9261 | 0.4773 | 0.7148 | 1.9417 | 4.2940 | 1.2268 | **6.6594** |
| NCC | #2 | 1.2301 | 0.7008 | 2.3919 | 0.5281 | 0.8023 | 1.8275 | 1.6784 | 2.1034 | 1261.5010 | 0.5608 | 1.3958 | 1.8903 | 6.6906 | 0.7786 | **7.2671** |
| NCC | #3 | 1.7366 | 1.0331 | 1.2570 | 0.5081 | 0.7664 | 1.7045 | 0.8344 | 3.9739 | 1253.4646 | 0.5370 | 0.6882 | 2.2523 | 4.7334 | 0.8583 | **5.2122** |

Bold values indicates the ability to distinguish between different images with the same content for the corresponding Normalized (Localized)-IQM values

**Table 15** Robustness ranking of Image Quality Metrics (IQMs) in discrimination tasks (lower → higher)

| Rank | Metric(s) | Ref. type | Rationale/Evidence |
|---|---|---|---|
| High | Localized-IQM variants (e.g., Localized-SSIM, Localized-PSNR, Localized-NIQE, etc.) | FR/NR | Variance-weighted tiling captures local differences and separates images that share identical global IQM scores; consistent discrimination across datasets (Figs. 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 14; Tables 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 14) |
| Moderate | SSIM, VIF, FSIM | FR | Perception-oriented metrics show comparatively better alignment but still exhibit cases with close or identical scores in one-to-many comparisons; partial failures observed (Figs.3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 14) |
| Low | MAE, MSE, ERGAS, LMSE, PSNR, SNR, NCC, GMSD, NIQE, PIQE, SAM, UIQI | FR/NR | Frequently yield identical or near-identical values at $10^{-2}$ precision for visually different images under numerical perturbations; insufficient discrimination in one-to-many settings (Figs. 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 14; Tables 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 14) |

FR: Full-Reference IQMs, NR: No(Blind)-Reference IQMs

ues are distinctly different from the pre-determined reference values for the corresponding IQMs (shown in Table 2) used during the experiments. Consequently, the Localized-IQM was able to successfully discriminate all 540 images under examination.

For two images of size $[M \cdot N \cdot D]$, classical global IQMs that scan pixels once have time complexity $\mathcal{O}(M \cdot N \cdot D)$. Localized-IQM partitions the image into $T = (M \cdot N)/k^2$ non-overlapping $[k \cdot k \cdot D]$ tiles, computes the same IQM per tile, and forms a variance-weighted average, which preserves the $\mathcal{O}(M \cdot N \cdot D)$ per-pixel cost and adds only $\mathcal{O}(T)$ tile-level bookkeeping with $\mathcal{O}(T)$ extra memory.

## 6 Conclusion

IQMs face several challenges and limitations. Objective metrics may not fully capture perceptual quality due to the complex nature of human vision. Subjective assessment, while accurate, requires significant resources and may be time-consuming. Deep learning-based approaches depend heavily on large, annotated datasets, limiting their generalization to diverse image types. Hybrid methods struggle with feature selection, impacting their overall performance. Image content and context can influence the accuracy of objective metrics. Furthermore, variations in image distortions and subjective perceptions add complexity to the evaluation process. Developing IQMs that effectively balance computational efficiency and human perceptual relevance remains a significant challenge. Overcoming these obstacles is crucial to ensure reliable and efficient image quality assessment in various real-world applications.

The results obtained in this paper have shown that MAE, MSE, ERGAS, LMSE, SSIM, SAM, UIQI, PSNR, NIQE, PIQE, SNR, VIF, FSIM, GMSD, and NCC are sensitive to *chromatic* changes in the relevant images up to a certain limit. The respective IQMs produce responses that are consistent with their analytical structures, but these responses do not always provide a coherent idea about the visual appearance of the images. When comparing IQMs obtained for the images, ensuring that the relevant images are captured with the same sensors under the same conditions will help researchers make more consistent interpretations. Using different analytical structures for multiple IQMs when producing accurate judgments about images can partially protect researchers from making misleading interpretations. The experiments conducted in this article have demonstrated that using multiple objective and subjective criteria and IQMs to make accurate judgments about images is an unavoidable necessity.

The findings also carry significant implications for practical application domains. In medical imaging, robust IQMs are essential to ensure that diagnostic images preserve clinically relevant details while avoiding misleading numerical scores that could obscure subtle pathological indicators. For environmental monitoring, accurate quality assessment supports the detection of land cover changes, vegetation stress, or atmospheric anomalies, where even slight chromatic or structural differences can have substantial ecological implications. In security and surveillance systems, reliable IQMs are critical to evaluate the integrity of facial recognition, license plate detection, or low-light video enhancement, where inconsistent or misleading metrics may compromise operational decision-making. The proposed Localized-IQM approach, by addressing local variations and reducing the misleading influence of global averaging, demonstrates practical robustness that enhances trustworthiness in these domains.

Classical IQMs provide computational efficiency and transparency, yet their limitations in perceptual alignment and discrimination highlight the need for refined methods. The Localized-IQM method introduced here represents a promising step toward bridging this gap by incorporating local statistical variations into quality assessment. This enables more reliable deployment of IQMs across application areas that demand both precision and interpretability. Advancing robust and perceptually consistent IQM frameworks is therefore essential not only for academic research but also for ensuring the reliability of image-based decision-making in healthcare, environmental monitoring, and security-critical systems.

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## Declarations

**Conflict of interest** The authors declare no conflict of interest or conflict of interest.

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